

AK High-Dimensional Projection Structural Theory

Appendix CR-v21

Final Claim Register

Canonical Identity, Migration, Source Binding, and Release Closure

Version 21.0.0

Atsushi Kobayashi

August 2026

Final register status

Document role: claim-governance appendix. Evidence status: operational. Invocation role: operational rule. Conformance status: conformant. CR-v21 is registered after closure of Main Chapters 1–10 and all v21 appendices. Registration governs identity and admissible use but does not strengthen mathematics. Package release and publication remain independent acts.

Contents

1	Purpose, finality, and minimality	2
2	Canonical source set and identity closure	2
3	Canonical ownership and theorem-identity precedence	2
4	Canonical v21 register schema	2
4.1	UID policy	2
5	Claim-use and result-sort kernel	3
6	Failure-route kernel	3
7	Final closure and release-critical records	3
8	Inventory policy and counts	5
9	Current v21 canonical and local registered index	5
9.1	Main Chapters 1-10 (829)	5
9.2	Foundation Appendices A-G (140)	12
9.3	Appendix DG (20)	14
9.4	Appendix HT (23)	14
9.5	Appendix UB (21)	14
9.6	Appendix CM (19)	15
9.7	Appendix TB (19)	15
9.8	Technical Extension Appendices H-N (70)	15
9.9	Problem Interface Appendices MS/NS (73)	16
9.10	Search/Platform Appendices U-Z (128)	16
9.11	Companion (78)	18
10	Preserved v20 lower-baseline epoch	18
11	Source-binding and Known-Theorem Recovery closure	18
11.1	Iwasawa track	18
11.2	Serrin track	19
12	Release-state vector and final nonconfusion	19

1 Purpose, finality, and minimality

This document is the final compact governance layer for AK-HPDST v21.0.0. It preserves the complete registered v20 lower baseline, assigns stable v21 identities to current canonical and source-local registered statements, separates canonical statements from expanded proofs, validation and governance, and records source binding, migration, target strength, failure semantics, artifact identity, and release closure.

Minimality and source-control rule

The PDF does not duplicate complete hypotheses, proofs, dependency prose, permitted-use clauses, target-strength preorders, or artifact byte hashes in every row. The immutable source artifact, exact locator, machine claim register, migration matrix, and external identity manifest jointly control those fields. If any source, register, migration record, or consumer disagrees, the weakest source-supported type-correct reading governs.

Routine remarks and examples remain source-local. Definitions are registered when they are Main-canonical or carry an explicit v21 local identity/-operational contract; proof expansions of Main statements do not receive competing theorem UUIDs.

2 Canonical source set and identity closure

Artifact	Document	Role	Pages	SHA-256
DOC-MAIN	Main Chapters 1-10	canonical_statement_owner	305	3657764a99dc37dc0b28e124d910920be0f609a3591fa94893791498b35674ba
DOC-FOUNDATION	Foundation Appendices A-G	expanded_proof_and_schema_owner	232	0c945bd4690d796c4f82d49f6a1c261c17a7f5feac684a3b1f0fdd3b76434d1e
DOC-DG	Appendix DG	expanded_proof_and_local_lemma_owner	28	24cb54fb48f0d92b0b90e43daa4eec794a55ef41a31cdcd89fcef57ccbf2014e
DOC-HT	Appendix HT	research_spine_and_expanded_proof_owner	180	18e5bb1644ff5f2fafa2808a9b6f556813fafc3ea09cc0fff39b9b6e8b27452ce
DOC-UB	Appendix UB	component_library_and_validation_owner	160	dbc2d0c2b33570691bfa5a72c91839b4c3510f2bb8fc5a42c4f3f23920478f34
DOC-CM	Appendix CM	microtheorem_and_consumer_contract_owner	42	3a79bab7b668dc4395778639ffe8a33142940ed2a687ab72ff0eb26307290426
DOC-TB	Appendix TB	toy_theorem_and_counterexample_owner	46	a9a94f47c8cae77fea6254e74ab4bae20318da49c2c99c8a44842b731d2ce27
DOC-TE	Technical Extension Appendices H-N	bounded_extension_owner	140	4c659695d86aac733a7593a03fa6dc6c27f348e5319a7e23f9aee3de8fd66a3e
DOC-PI	Problem Interface Appendices MS/NS	source_specific_interface_and_return_owner	182	ce55ed0f51a5d723146fef3f62c804f3351bdbbed69a1fc5a27490f04cb6eba5
DOC-SP	Search/Platform Appendices U-Z	infrastructure_owner	199	34e4fe586805db22947961fae55a3ba5304a9b60274e16cb95a3008216c21432
DOC-COMP	Companion	calibration_execution_handoff_owner	61	1a44e8adb3d6c172ad688edec495cf8b88233db6da521fb31e35815f47ccb2a4
DOC-CR-v20	Appendix CR-v20 Compact Final Claim Register	immutable_predecessor_claim_registry	46	56be7b1e0bbde1ef2765961361a1eeaf11cc1d3e462aaade021a72f41e6ab5b
SRC-IW-STMT	Iwasawa 1959 exact source-statement binding	authoritative_local_statement_binding	n/a	a5f0a65a73c802492167676aba024b01651cd855bbdc00bc3d3c48d46cd6f0d7
SRC-IW-REG	Iwasawa source-binding registry	registry_with_repair_ancestry	n/a	7a3c2eb157eba3ccbe1613c4ade739991a26f8639083724f1213406a04ebff6c
SRC-IW-MANIFEST	Iwasawa Known-Theorem Recovery manifest	recovery_manifest	n/a	acccc9fc2f756aa5d35e00e3889e0a9a2f99f30baa06966968cc257caabd0bf3
SRC-NS-STMT	Serrin L6 exact source-statement binding	authoritative_local_statement_binding	n/a	608216fda60bd3fcd055243c0169bdb31b53c337bb99d9b7c9cb43a215996423
SRC-NS-REG	Navier-Stokes source-binding registry	source_registry	n/a	445dcd8863d73cdfd68099ee0539fe3a4e96881f4d788d9e817076e61666de18
SRC-NS-MANIFEST	Navier-Stokes Known-Theorem Recovery manifest	recovery_manifest	n/a	da09e31137613381f42a1cba8dd50ed1ae8780bbf642a78ca8ed9ba7be9c209a

Complete identities, filenames and ancillary audit sidecars are recorded in `ARTIFACT_IDENTITY_MANIFEST.json`. The final CR PDF and TeX are hashed only after generation, in the package-level SHA-256 file, avoiding circular self-hash claims.

3 Canonical ownership and theorem-identity precedence

1. Main Chapters 1–10 own canonical generic Core-P, Core-DG, Interface, verifier, Bridge/Return, Validation, and governing statements.
2. Foundation A–G and Appendix DG own expanded proofs, schemas, local lemmas, counterexamples and regression witnesses. A Main repetition is a proof locator, not a second theorem authority.
3. Appendix HT owns the HC–PE–SS–RL–IW/Return research spine and module-local refinements; Main Chapter 9 retains generic Bridge authority.
4. Appendix UB owns reusable executable components, local derived results and validated cases, without converting validation into universal completeness.
5. CM owns compact corollaries and consumer contracts; TB owns explicitly bounded toy results, counterexamples and regression demonstrations.
6. H–N own bounded Technical-Extension results. MS/NS own source-specific Problem-Interface components and Returns. U–Z own Search/Platform infrastructure. Companion owns operational projections only.
7. CR-v21 owns identity, migration, registration and admissible-use governance. It owns no underlying mathematical proof merely by registration.

4 Canonical v21 register schema

Every registered current claim has a stable UUID, immutable source artifact, exact locator, primary layer, statement kind, canonical owner, evidence status, invocation role, migration record or new-v21 marker, target-relative strength policy, comparison policy, typed failure route, and prohibited stronger reading. Full machine rows are in `CLAIM_REGISTER_V21.json/csv`.

4.1 UID policy

- Main identities use `ak-hpdst:21.0.0:main:chN:<kind>-<number>`.
- Explicit local identifiers retain their printed token inside an artifact-qualified namespace.

- Historical v20 ClaimIDs and UIDs are preserved without renumbering in a separate immutable epoch.
- A proof locator has no competing claim UID. A repair, strengthening, retyping or genuinely new local theorem receives a distinct v21 identity.

5 Claim-use and result-sort kernel

1. Registration is not proof. Missing required evidence is `undefined`; a well-typed false obligation is `reject`; `not_invoked` requires an immutable scope exclusion.
2. Atomic verifier status, P2DG readiness, `descent_input_ready`, `bridge_input_ready`, Type IV status, Bridge result, workflow, migration, registration, package release and publication are distinct sorts.
3. Comparison modes are exactly `exact`, `metric_gap_certified`, `theorem_controlled_nonmetric`, and `not_compared`.
4. Detector stages are exactly `windowed_prethreshold`, `repaired_postthreshold`, `kernel_defect`, `cokernel_defect`, `diagram_probe`, and `descent_coherence`.
5. Target strength is a downward-closed set in a declared preorder. No greatest element, lattice operation, intersection, product object, or cross-target coherence is inferred without proof.
6. Finite observation is not finite proof; record globalization is not object globalization; realization is not Return; validation is not premise; execution and replay are not mathematical truth.

6 Failure-route kernel

Route	Registered meaning
FR - MISSING	Required evidence absent or ill-typed; atomic status undefined.
FR - T1	Certified repaired-persistence nonvanishing.
FR - T2	Certified eligible categorical/higher-edge nonvanishing.
FR - T3	Certified detector, metric, comparison, representative, coverage, Restart, reduction, coherence, execution-admission, or route failure.
FR - T4	Certified nonzero terminal Type IV diagnostic.
FR - TRANSIENT	Certified retained interior/transient defect; not automatically Type IV.
FR - RETURN - MISSING	Required Interface, reduction, rigidity, descent, Bridge component, or Return theorem missing; requested external target remains undefined or obstructed at its declared result sort.
FR - T5	Identity, authority, source binding, provenance, schema, permission, migration, promotion, or registration failure.
FR - NONVERDICT	Operational, advisory, toy, Search, Companion, spec, or validation material used beyond its declared theorem authority.

7 Final closure and release-critical records

CR-v21-FINAL-001: Final v21 source-set closure and weaker-safe claim use

UID: `ak-hpdst:21.0.0:cr:CR-v21-FINAL-001`

Typing: `operational_policy` / `operational`

Registered reading: All eleven current v21 source artifacts and the registered source-binding sidecars are identity-frozen by the external manifest. Every claim is consumed only at source-supported strength.

CR-v21-FINAL-002: Conservative inheritance of the complete v20 lower baseline

UID: `ak-hpdst:21.0.0:cr:CR-v21-FINAL-002`

Typing: `operational_policy` / `operational`

Registered reading: All historical v20 identities remain immutable. A v21 strengthening, repair, retyping, or new local result has a distinct v21 identity and never overwrites predecessor truth conditions.

CR-v21-FINAL-003: Canonical-owner uniqueness and proof-locator separation

UID: `ak-hpdst:21.0.0:cr:CR-v21-FINAL-003`

Typing: `operational_policy` / `operational`

Registered reading: Main owns generic canonical statements; Foundation and DG expanded repetitions are proof locators; Appendix-local, source-specific, validation, infrastructure and Companion results retain only their declared local authority.

CR-v21-FINAL-004: Minimal source-bound registry schema

UID: `ak-hpdst:21.0.0:cr:CR-v21-FINAL-004`

Typing: `operational_policy` / `operational`

Registered reading: The compact PDF indexes identity, owner, source locator, kind, role/status and title. Full hypotheses, proofs, strength, failure routes and prohibited readings remain controlled by the immutable source and machine register.

CR-v21-FINAL-005: Seven-axis and result-sort noncollapse

UID: `ak-hpdst:21.0.0:cr:CR-v21-FINAL-005`

Typing: `operational_policy` / `operational`

Registered reading: Document role, evidence, invocation, conformance, verifier, workflow and migration are independent. Atomic status, P2DG readiness, handoff tokens, Type IV status, Bridge result and release states are pairwise distinct.

CR-v21-FINAL-006: Target-relative downward-closed strength registration

UID: `ak-hpdst:21.0.0:cr:CR-v21-FINAL-006`

Typing: `operational_policy` / `operational`

Registered reading: Each route exports only a downward-closed supported-strength set in the declared target preorder. Incomparable maxima

are permitted; independent targets use product records rather than untyped intersections.

CR-v21-FINAL-007: Proof-DAG closure, one active descent lane and noncompensation

UID: ak-hpdst:21.0.0:cr:CR-v21-FINAL-007

Typing: operational_policy / operational

Registered reading: Every registered Return uses a dependency-complete target slice, one active descent lane when invoked, complete source/-target typing and a single-owner failure route. Metric, workflow, storage or replay success cannot fill a missing theorem edge.

CR-v21-FINAL-008: Core-P, Core-DG, record-globalization and object-globalization separation

UID: ak-hpdst:21.0.0:cr:CR-v21-FINAL-008

Typing: operational_policy / operational

Registered reading: Persistence certificates do not manufacture diagrams; record gluing does not manufacture global objects; actual morphisms, cocycles, coherence and effectivity are mandatory at their declared stages.

CR-v21-FINAL-009: Generic Bridge and source-specific Return ownership

UID: ak-hpdst:21.0.0:cr:CR-v21-FINAL-009

Typing: operational_policy / operational

Registered reading: Main Chapter 9 owns generic Bridge/Return calculus; HT expands the research spine; UB owns reusable implementations and validation; MS/NS own only source-specific components and Returns.

CR-v21-FINAL-010: Validation and benchmark no-backflow

UID: ak-hpdst:21.0.0:cr:CR-v21-FINAL-010

Typing: operational_policy / operational

Registered reading: Toy, validation, benchmark and external truth values are downstream leaves. They may confirm exact strength after route closure but may not supply premises to the route under validation.

CR-v21-FINAL-011: Exact Known-Theorem Recovery registration

UID: ak-hpdst:21.0.0:cr:CR-v21-FINAL-011

Typing: interface_governance / proved

Registered reading: IW-KTR-GROWTH and NS-KTR-SERRIN-L6 are registered at their exact source-bound strengths only: the Iwasawa characteristic-growth clause relative to its precursor package, and the one-way fixed-exponent strict-subcritical Serrin implication.

CR-v21-FINAL-012: Source-binding repair and historical-reference quarantine

UID: ak-hpdst:21.0.0:cr:CR-v21-FINAL-012

Typing: source_binding_policy / operational

Registered reading: The actual Iwasawa statement-binding hash is authoritative; the stale hash in the legacy registry is repair ancestry. The nine nonexistent CM-v20 tokens in TB-v20 remain historical errors and are consumed only through the explicit CM/TB v21 repair map.

CR-v21-FINAL-013: Infrastructure, replay and package closure are nonproof

UID: ak-hpdst:21.0.0:cr:CR-v21-FINAL-013

Typing: operational_policy / operational

Registered reading: Search coverage, agent agreement, formal wiring, Merkle closure, cache admission, successful compilation, replay and package integrity support auditability but do not establish missing mathematics.

CR-v21-FINAL-014: Claim registration, package release and publication are independent axes

UID: ak-hpdst:21.0.0:cr:CR-v21-FINAL-014

Typing: operational_policy / operational

Registered reading: This artifact closes and registers CR-v21. Package publication, DOI/Zenodo publication and public release remain separate acts with their own state fields.

CR-v21-FINAL-015: Machine inventory and migration closure

UID: ak-hpdst:21.0.0:cr:CR-v21-FINAL-015

Typing: operational_policy / operational

Registered reading: The machine register contains 1420 current source-document identities plus 16 CR governance identities, and 1950 preserved v20 registered identities, together with the full statement inventory and artifact manifest.

CR-v21-NONCLAIM-001: Global external nonclaim

UID: ak-hpdst:21.0.0:cr:CR-v21-NONCLAIM-001

Typing: non_claim / operational

Registered reading: No BSD, ABC, RH, Iwasawa main conjecture, unconditional three-dimensional Navier-Stokes regularity, general HMS/-Fukaya equivalence, unrestricted higher-categorical descent, or other named open problem is promoted by this register.

8 Inventory policy and counts

The human-readable index lists every current v21 registered identity. The complete extracted statement inventory also records proof locators, source-local unregistered definitions, remarks, examples and retained predecessor bodies. The preserved v20 compact register is incorporated machine-readably as an immutable historical epoch rather than reprinted in full.

Artifact	Document	Registered	Theorem-bearing	Definition/policy
DOC-MAIN	Main Chapters 1-10	829	293	536
DOC-FOUNDATION	Foundation Appendices A-G	140	138	2
DOC-DG	Appendix DG	20	14	6
DOC-HT	Appendix HT	23	23	0
DOC-UB	Appendix UB	21	21	0
DOC-CM	Appendix CM	19	13	5
DOC-TB	Appendix TB	19	13	6
DOC-TE	Technical Extension Appendices H-N	70	54	16
DOC-PI	Problem Interface Appendices MS/NS	73	55	18
DOC-SP	Search/Platform Appendices U-Z	128	84	44
DOC-COMP	Companion	78	37	41
DOC-CR-v21	Final CR governance records	16	1	15
Current v21 total		1436		
Preserved v20 registered epoch		1950		
Full extracted statement inventory		5966		

9 Current v21 canonical and local registered index

9.1 Main Chapters 1-10 (829)

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:main:chl:definition-1.1	DOC-MAIN p.3; Definition 1.1	Definition / operational	v21 statement-role taxonomy
ak-hpdst:21.0.0:main:chl:policy-1	DOC-MAIN p.4; Policy 1	Policy / operational	Weaker-safe interpretation
ak-hpdst:21.0.0:main:chl:definition-1.2	DOC-MAIN p.4; Definition 1.2	Definition / operational	Seven-axis statement typing
ak-hpdst:21.0.0:main:chl:definition-1.4	DOC-MAIN p.5; Definition 1.4	Definition / operational	Immutable v20 lower baseline
ak-hpdst:21.0.0:main:chl:definition-1.5	DOC-MAIN p.5; Definition 1.5	Definition / operational	Conservative v21 evolution
ak-hpdst:21.0.0:main:chl:theorem-1.6	DOC-MAIN p.5; Theorem 1.6	Theorem / proved	No retroactive theorem mutation
ak-hpdst:21.0.0:main:chl:definition-1.7	DOC-MAIN p.5; Definition 1.7	Definition / operational	Dual-Core architecture
ak-hpdst:21.0.0:main:chl:definition-1.10	DOC-MAIN p.6; Definition 1.10	Definition / operational	v21 layer taxonomy
ak-hpdst:21.0.0:main:chl:definition-1.8	DOC-MAIN p.6; Definition 1.8	Definition / operational	Core-P standing scope
ak-hpdst:21.0.0:main:chl:definition-1.9	DOC-MAIN p.6; Definition 1.9	Definition / operational	Core-DG standing scope
ak-hpdst:21.0.0:main:ch2:policy-2	DOC-MAIN p.6; Policy 2	Policy / operational	No silent Core transfer
ak-hpdst:21.0.0:main:chl:definition-1.11	DOC-MAIN p.7; Definition 1.11	Definition / operational	v21 migration-status record
ak-hpdst:21.0.0:main:chl:definition-1.12	DOC-MAIN p.7; Definition 1.12	Definition / operational	Migration record
ak-hpdst:21.0.0:main:chl:definition-1.13	DOC-MAIN p.8; Definition 1.13	Definition / operational	Theorem identity and namespace rule
ak-hpdst:21.0.0:main:ch3:policy-3	DOC-MAIN p.8; Policy 3	Policy / operational	Selective HT-UB promotion
ak-hpdst:21.0.0:main:ch4:policy-4	DOC-MAIN p.8; Policy 4	Policy / operational	No duplicate mathematical authority
ak-hpdst:21.0.0:main:chl:definition-1.14	DOC-MAIN p.9; Definition 1.14	Definition / operational	Ordinary proof dependency graph
ak-hpdst:21.0.0:main:chl:definition-1.15	DOC-MAIN p.9; Definition 1.15	Definition / operational	Admissible cyclic semantics
ak-hpdst:21.0.0:main:ch5:policy-5	DOC-MAIN p.9; Policy 5	Policy / operational	Layer direction and downstream non-interference
ak-hpdst:21.0.0:main:ch6:policy-6	DOC-MAIN p.9; Policy 6	Policy / operational	Benchmark quarantine
ak-hpdst:21.0.0:main:chl:definition-1.17	DOC-MAIN p.10; Definition 1.17	Definition / operational	Return-strength record
ak-hpdst:21.0.0:main:chl:definition-1.18	DOC-MAIN p.10; Definition 1.18	Definition / operational	Canonical v21 Return-strength families
ak-hpdst:21.0.0:main:ch7:policy-7	DOC-MAIN p.10; Policy 7	Policy / operational	Return strengths are partially ordered
ak-hpdst:21.0.0:main:chl:definition-1.19	DOC-MAIN p.11; Definition 1.19	Definition / operational	P-to-DG Interface record
ak-hpdst:21.0.0:main:chl:definition-1.20	DOC-MAIN p.11; Definition 1.20	Definition / operational	DG-to-P Interface record
ak-hpdst:21.0.0:main:ch8:policy-8	DOC-MAIN p.11; Policy 8	Policy / operational	Route-admissible Return set and conditional ceiling
ak-hpdst:21.0.0:main:chl:definition-1.21	DOC-MAIN p.12; Definition 1.21	Definition / operational	Unified v21 proof route
ak-hpdst:21.0.0:main:chl:definition-1.22	DOC-MAIN p.12; Definition 1.22	Definition / operational	Record-level globalization
ak-hpdst:21.0.0:main:chl:definition-1.23	DOC-MAIN p.12; Definition 1.23	Definition / operational	Object-level globalization
ak-hpdst:21.0.0:main:chl:theorem-1.24	DOC-MAIN p.12; Theorem 1.24	Theorem / proved	No globalization by record gluing alone
ak-hpdst:21.0.0:main:chl:definition-1.25	DOC-MAIN p.12; Definition 1.25	Definition / operational	Core-DG constitutional target theorems
ak-hpdst:21.0.0:main:chl:definition-1.26	DOC-MAIN p.13; Definition 1.26	Definition / operational	Core-P gate
ak-hpdst:21.0.0:main:chl:definition-1.27	DOC-MAIN p.13; Definition 1.27	Definition / operational	Core-DG gate
ak-hpdst:21.0.0:main:chl:definition-1.28	DOC-MAIN p.13; Definition 1.28	Definition / operational	Unified Bridge/Return gate
ak-hpdst:21.0.0:main:chl:definition-1.29	DOC-MAIN p.13; Definition 1.29	Definition / operational	v21 aggregate gate
ak-hpdst:21.0.0:main:chl:definition-1.30	DOC-MAIN p.14; Definition 1.30	Definition / operational	Chapter 1 completion condition
ak-hpdst:21.0.0:main:chl0:policy-10	DOC-MAIN p.14; Policy 10	Policy / operational	G21-2: Core-P regression
ak-hpdst:21.0.0:main:chl1:policy-11	DOC-MAIN p.14; Policy 11	Policy / operational	G21-3: Core-DG closure
ak-hpdst:21.0.0:main:chl2:policy-12	DOC-MAIN p.14; Policy 12	Policy / operational	G21-4: Core-P-to-Core-DG closure
ak-hpdst:21.0.0:main:chl3:policy-13	DOC-MAIN p.14; Policy 13	Policy / operational	G21-5: Object-level globalization
ak-hpdst:21.0.0:main:chl4:policy-14	DOC-MAIN p.14; Policy 14	Policy / operational	G21-6: Certificate complexity
ak-hpdst:21.0.0:main:chl5:policy-15	DOC-MAIN p.14; Policy 15	Policy / operational	G21-7: Unified external Return
ak-hpdst:21.0.0:main:chl6:policy-16	DOC-MAIN p.14; Policy 16	Policy / operational	G21-8: Non-duplication and namespace closure
ak-hpdst:21.0.0:main:chl7:policy-17	DOC-MAIN p.14; Policy 17	Policy / operational	G21-9: Release and artifact closure
ak-hpdst:21.0.0:main:chl8:policy-18	DOC-MAIN p.14; Policy 18	Policy / operational	v21 release acceptance
ak-hpdst:21.0.0:main:chl9:policy-9	DOC-MAIN p.14; Policy 9	Policy / operational	G21-1: Migration completeness
ak-hpdst:21.0.0:main:chl:definition-1.31	DOC-MAIN p.15; Definition 1.31	Definition / operational	Conservative Core-P re-issuance
ak-hpdst:21.0.0:main:chl:definition-1.32	DOC-MAIN p.15; Definition 1.32	Definition / operational	Inherited v20 four-pillar Core-P contract
ak-hpdst:21.0.0:main:chl:definition-1.33	DOC-MAIN p.15; Definition 1.33	Definition / operational	Finite observation and finite proof certificate
ak-hpdst:21.0.0:main:chl:definition-1.34	DOC-MAIN p.16; Definition 1.34	Definition / operational	Detector profile stage
ak-hpdst:21.0.0:main:chl:definition-1.35	DOC-MAIN p.16; Definition 1.35	Definition / operational	Finite detector certificate record
ak-hpdst:21.0.0:main:chl:definition-1.36	DOC-MAIN p.17; Definition 1.36	Definition / operational	Two-axis detector status
ak-hpdst:21.0.0:main:chl:definition-1.37	DOC-MAIN p.17; Definition 1.37	Definition / operational	Metric-safe detector transport chain
ak-hpdst:21.0.0:main:chl9:policy-19	DOC-MAIN p.17; Policy 19	Policy / operational	Detector admission and non-compensation
ak-hpdst:21.0.0:main:chl:definition-1.38	DOC-MAIN p.18; Definition 1.38	Definition / operational	Higher obstruction contract
ak-hpdst:21.0.0:main:chl:definition-1.39	DOC-MAIN p.18; Definition 1.39	Definition / operational	Natural zero-reflection upgrade
ak-hpdst:21.0.0:main:chl:definition-1.41	DOC-MAIN p.19; Definition 1.41	Definition / operational	Family-relative proof compression
ak-hpdst:21.0.0:main:chl:definition-1.42	DOC-MAIN p.19; Definition 1.42	Definition / operational	Inherited Core-P Problem-Bridge proof chain
ak-hpdst:21.0.0:main:chl:theorem-1.44	DOC-MAIN p.20; Theorem 1.44	Theorem / proved	Threshold-gap-safe stability under the standard bottleneck metric
ak-hpdst:21.0.0:main:chl:definition-1.46	DOC-MAIN p.20; Definition 1.46	Definition / operational	Core-P Unified Collapse Contract: inherited and conservatively re-issued data
ak-hpdst:21.0.0:main:ch20:policy-20	DOC-MAIN p.20; Policy 20	Policy / operational	Return-Theorem and finite-to-infinite discipline
ak-hpdst:21.0.0:main:ch21:policy-21	DOC-MAIN p.21; Policy 21	Policy / operational	Core-P Unified Collapse Contract: audit policy
ak-hpdst:21.0.0:main:chl:definition-1.47	DOC-MAIN p.23; Definition 1.47	Definition / operational	v21 Core-P UCC-Contract
ak-hpdst:21.0.0:main:chl:definition-1.48	DOC-MAIN p.23; Definition 1.48	Definition / operational	Right-open windows, partition covers, and overlap covers
ak-hpdst:21.0.0:main:chl:definition-1.49	DOC-MAIN p.24; Definition 1.49	Definition / operational	Window restriction
ak-hpdst:21.0.0:main:chl:definition-1.50	DOC-MAIN p.24; Definition 1.50	Definition / operational	Windowed pre-threshold profile
ak-hpdst:21.0.0:main:chl:definition-1.51	DOC-MAIN p.24; Definition 1.51	Definition / operational	Windowed repaired profile
ak-hpdst:21.0.0:main:chl:definition-1.52	DOC-MAIN p.24; Definition 1.52	Definition / operational	Canonical comparison mode
ak-hpdst:21.0.0:main:chl:definition-1.53	DOC-MAIN p.24; Definition 1.53	Definition / operational	Exact comparison record
ak-hpdst:21.0.0:main:chl:definition-1.54	DOC-MAIN p.25; Definition 1.54	Definition / operational	Threshold clearance
ak-hpdst:21.0.0:main:chl:definition-1.55	DOC-MAIN p.25; Definition 1.55	Definition / operational	Threshold safety margin
ak-hpdst:21.0.0:main:chl:definition-1.56	DOC-MAIN p.25; Definition 1.56	Definition / operational	Window-edge clearance and crop budget
ak-hpdst:21.0.0:main:chl:definition-1.57	DOC-MAIN p.25; Definition 1.57	Definition / operational	Metric threshold-gap record
ak-hpdst:21.0.0:main:chl:definition-1.61	DOC-MAIN p.26; Definition 1.61	Definition / operational	Typed pipeline ledger and scalarized budget
ak-hpdst:21.0.0:main:chl:definition-1.64	DOC-MAIN p.27; Definition 1.64	Definition / operational	Overlap Gate
ak-hpdst:21.0.0:main:chl:definition-1.66	DOC-MAIN p.28; Definition 1.66	Definition / operational	B-Gate+ (inherited Core-P finite-certificate gate)
ak-hpdst:21.0.0:main:chl:definition-1.70	DOC-MAIN p.29; Definition 1.70	Definition / operational	Failure landscape
ak-hpdst:21.0.0:main:chl:definition-1.71	DOC-MAIN p.30; Definition 1.71	Definition / operational	Type IV scope status at the Chapter 1 contract level
ak-hpdst:21.0.0:main:chl:definition-1.72	DOC-MAIN p.30; Definition 1.72	Definition / operational	Tower comparison artifact
ak-hpdst:21.0.0:main:chl:definition-1.73	DOC-MAIN p.30; Definition 1.73	Definition / operational	Terminal-generic Type IV diagnostics at contract level

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:main:ch1:definition-1.76	DOC-MAIN p.31; Definition 1.76	Definition / operational	Transient-defect detector contract
ak-hpdst:21.0.0:main:ch1:definition-1.78	DOC-MAIN p.31; Definition 1.78	Definition / operational	Monitored Windowed Collapse Zone
ak-hpdst:21.0.0:main:ch1:axiom-1.1	DOC-MAIN p.32; Axiom 1.1	Axiom / proved	Minimal Manifest Axiom
ak-hpdst:21.0.0:main:ch1:definition-1.84	DOC-MAIN p.33; Definition 1.84	Definition / operational	Manifest-complete proof-object candidate
ak-hpdst:21.0.0:main:ch1:definition-1.85	DOC-MAIN p.33; Definition 1.85	Definition / operational	Certified proof object
ak-hpdst:21.0.0:main:ch22:policy-22	DOC-MAIN p.34; Policy 22	Policy / operational	Countable atlas forward contract
ak-hpdst:21.0.0:main:ch23:policy-23	DOC-MAIN p.41; Policy 23	Policy / operational	Immutable v20 metric baseline
ak-hpdst:21.0.0:main:ch2:definition-2.5	DOC-MAIN p.42; Definition 2.5	Definition / operational	Constructible persistence module
ak-hpdst:21.0.0:main:ch2:definition-2.7	DOC-MAIN p.42; Definition 2.7	Definition / operational	Finite-type filtered chain complex
ak-hpdst:21.0.0:main:ch2:definition-2.10	DOC-MAIN p.43; Definition 2.10	Definition / operational	Bar and barcode
ak-hpdst:21.0.0:main:ch2:theorem-2.11	DOC-MAIN p.43; Theorem 2.11	Theorem / proved	Constructible interval decomposition
ak-hpdst:21.0.0:main:ch2:definition-2.12	DOC-MAIN p.43; Definition 2.12	Definition / operational	Length of a bar
ak-hpdst:21.0.0:main:ch2:definition-2.14	DOC-MAIN p.43; Definition 2.14	Definition / operational	Endpoint-discipline record
ak-hpdst:21.0.0:main:ch2:definition-2.9	DOC-MAIN p.43; Definition 2.9	Definition / operational	Interval module
ak-hpdst:21.0.0:main:ch2:definition-2.15	DOC-MAIN p.44; Definition 2.15	Definition / operational	Window-cofinal bar
ak-hpdst:21.0.0:main:ch2:definition-2.17	DOC-MAIN p.44; Definition 2.17	Definition / operational	Standard bottleneck distance with infinite bars
ak-hpdst:21.0.0:main:ch2:lemma-2.18	DOC-MAIN p.44; Lemma 2.18	Lemma / proved	Infinite-bar threshold invariance
ak-hpdst:21.0.0:main:ch2:definition-2.20	DOC-MAIN p.45; Definition 2.20	Definition / operational	Right-open window
ak-hpdst:21.0.0:main:ch2:definition-2.21	DOC-MAIN p.45; Definition 2.21	Definition / operational	Window restriction
ak-hpdst:21.0.0:main:ch2:definition-2.23	DOC-MAIN p.45; Definition 2.23	Definition / operational	Windowed collapsed persistence evaluation
ak-hpdst:21.0.0:main:ch2:definition-2.25	DOC-MAIN p.45; Definition 2.25	Definition / operational	Repaired evaluation record
ak-hpdst:21.0.0:main:ch2:definition-2.26	DOC-MAIN p.46; Definition 2.26	Definition / operational	Threshold-gap-certified repaired evaluation comparison
ak-hpdst:21.0.0:main:ch2:definition-2.28	DOC-MAIN p.46; Definition 2.28	Definition / operational	Hard barcode threshold deletion
ak-hpdst:21.0.0:main:ch2:proposition-2.30	DOC-MAIN p.46; Proposition 2.30	Proposition / proved	Basic properties of hard threshold deletion
ak-hpdst:21.0.0:main:ch2:proposition-2.31	DOC-MAIN p.47; Proposition 2.31	Proposition / proved	Threshold composition
ak-hpdst:21.0.0:main:ch2:proposition-2.32	DOC-MAIN p.47; Proposition 2.32	Proposition / proved	Positive-threshold short-bar objects are not Serre in general
ak-hpdst:21.0.0:main:ch2:corollary-2.33	DOC-MAIN p.47; Corollary 2.33	Corollary / proved	No global exact reflector claim
ak-hpdst:21.0.0:main:ch2:corollary-2.37	DOC-MAIN p.48; Corollary 2.37	Corollary / proved	Deprecated v18 metric claim
ak-hpdst:21.0.0:main:ch2:definition-2.38	DOC-MAIN p.48; Definition 2.38	Definition / operational	Threshold clearance
ak-hpdst:21.0.0:main:ch2:definition-2.39	DOC-MAIN p.49; Definition 2.39	Definition / operational	Threshold gap as ledger alias
ak-hpdst:21.0.0:main:ch2:definition-2.40	DOC-MAIN p.49; Definition 2.40	Definition / operational	Bottleneck scalarization and typed metric budget
ak-hpdst:21.0.0:main:ch2:definition-2.42	DOC-MAIN p.50; Definition 2.42	Definition / operational	Windowed pre-threshold profile
ak-hpdst:21.0.0:main:ch2:definition-2.44	DOC-MAIN p.50; Definition 2.44	Definition / operational	Threshold-gap comparison record
ak-hpdst:21.0.0:main:ch2:theorem-2.46	DOC-MAIN p.51; Theorem 2.46	Theorem / proved	Metric-repair theorem: sharp threshold-gap-safe stability
ak-hpdst:21.0.0:main:ch2:corollary-2.49	DOC-MAIN p.53; Corollary 2.49	Corollary / proved	Scalarized-ledger-safe threshold classification
ak-hpdst:21.0.0:main:ch2:definition-2.50	DOC-MAIN p.53; Definition 2.50	Definition / operational	◆-reduced barcode profile
ak-hpdst:21.0.0:main:ch2:definition-2.51	DOC-MAIN p.53; Definition 2.51	Definition / operational	Zero radius of a finite barcode profile
ak-hpdst:21.0.0:main:ch2:lemma-2.52	DOC-MAIN p.53; Lemma 2.52	Lemma / proved	Distance-to-zero formula
ak-hpdst:21.0.0:main:ch2:theorem-2.53	DOC-MAIN p.54; Theorem 2.53	Theorem / proved	Sharp zero separation for ◆◆-reduced profiles
ak-hpdst:21.0.0:main:ch2:proposition-2.55	DOC-MAIN p.54; Proposition 2.55	Proposition / proved	Exact zero transport
ak-hpdst:21.0.0:main:ch2:theorem-2.56	DOC-MAIN p.54; Theorem 2.56	Theorem / proved	Metric-safe repaired-zero transport
ak-hpdst:21.0.0:main:ch2:corollary-2.57	DOC-MAIN p.55; Corollary 2.57	Corollary / proved	Scalarized metric-safe repaired-zero transport
ak-hpdst:21.0.0:main:ch2:definition-2.58	DOC-MAIN p.55; Definition 2.58	Definition / operational	Post-threshold zero-separation record
ak-hpdst:21.0.0:main:ch2:proposition-2.59	DOC-MAIN p.55; Proposition 2.59	Proposition / proved	No unconditional crop-collapse commutation
ak-hpdst:21.0.0:main:ch2:definition-2.60	DOC-MAIN p.56; Definition 2.60	Definition / operational	Window-edge clearance
ak-hpdst:21.0.0:main:ch2:definition-2.61	DOC-MAIN p.56; Definition 2.61	Definition / operational	Crop budget
ak-hpdst:21.0.0:main:ch2:definition-2.64	DOC-MAIN p.56; Definition 2.64	Definition / operational	◆◆-window-adapted condition
ak-hpdst:21.0.0:main:ch2:proposition-2.65	DOC-MAIN p.57; Proposition 2.65	Proposition / proved	Exact crop-collapse commutation criterion
ak-hpdst:21.0.0:main:ch2:definition-2.67	DOC-MAIN p.57; Definition 2.67	Definition / operational	Canonical v21 comparison modes
ak-hpdst:21.0.0:main:ch2:definition-2.68	DOC-MAIN p.58; Definition 2.68	Definition / operational	Theorem-controlled nonmetric comparison record
ak-hpdst:21.0.0:main:ch2:theorem-2.69	DOC-MAIN p.58; Theorem 2.69	Theorem / proved	Comparison-mode separation
ak-hpdst:21.0.0:main:ch2:definition-2.70	DOC-MAIN p.58; Definition 2.70	Definition / operational	Typed discrepancy level
ak-hpdst:21.0.0:main:ch2:definition-2.71	DOC-MAIN p.58; Definition 2.71	Definition / operational	Barcode discrepancy
ak-hpdst:21.0.0:main:ch2:definition-2.72	DOC-MAIN p.58; Definition 2.72	Definition / operational	Persistence-module discrepancy
ak-hpdst:21.0.0:main:ch2:definition-2.73	DOC-MAIN p.59; Definition 2.73	Definition / operational	Higher-level discrepancy placeholder
ak-hpdst:21.0.0:main:ch2:theorem-2.74	DOC-MAIN p.59; Theorem 2.74	Theorem / proved	No automatic upward discrepancy transport
ak-hpdst:21.0.0:main:ch2:definition-2.76	DOC-MAIN p.59; Definition 2.76	Definition / operational	Finite critical-grid datum
ak-hpdst:21.0.0:main:ch2:definition-2.77	DOC-MAIN p.59; Definition 2.77	Definition / operational	Core-P to Core-DG input record
ak-hpdst:21.0.0:main:ch2:definition-2.78	DOC-MAIN p.60; Definition 2.78	Definition / operational	P2DG readiness status
ak-hpdst:21.0.0:main:ch2:theorem-2.79	DOC-MAIN p.60; Theorem 2.79	Theorem / proved	P2DG handoff boundary
ak-hpdst:21.0.0:main:ch2:definition-2.81	DOC-MAIN p.61; Definition 2.81	Definition / operational	Metric-safe detector-zero transport record
ak-hpdst:21.0.0:main:ch2:corollary-2.82	DOC-MAIN p.61; Corollary 2.82	Corollary / proved	Detector-certified metric zero transport
ak-hpdst:21.0.0:main:ch2:definition-2.83	DOC-MAIN p.61; Definition 2.83	Definition / operational	Pointwise span-rank observation
ak-hpdst:21.0.0:main:ch2:proposition-2.85	DOC-MAIN p.62; Proposition 2.85	Proposition / proved	No bottleneck stability theorem for raw pointwise detector ranks
ak-hpdst:21.0.0:main:ch2:definition-2.87	DOC-MAIN p.62; Definition 2.87	Definition / operational	Overlap pre-threshold profiles
ak-hpdst:21.0.0:main:ch2:proposition-2.89	DOC-MAIN p.63; Proposition 2.89	Proposition / proved	Overlap transfer under a scalarized threshold-gap certificate
ak-hpdst:21.0.0:main:ch2:corollary-2.90	DOC-MAIN p.63; Corollary 2.90	Corollary / proved	Overlap repaired-zero equivalence
ak-hpdst:21.0.0:main:ch2:definition-2.91	DOC-MAIN p.63; Definition 2.91	Definition / operational	◆◆-split object
ak-hpdst:21.0.0:main:ch2:definition-2.92	DOC-MAIN p.63; Definition 2.92	Definition / operational	◆◆-split morphism
ak-hpdst:21.0.0:main:ch2:definition-2.93	DOC-MAIN p.64; Definition 2.93	Definition / operational	◆◆-split regime
ak-hpdst:21.0.0:main:ch2:proposition-2.94	DOC-MAIN p.64; Proposition 2.94	Proposition / proved	Functional deletion in a ◆◆-split regime
ak-hpdst:21.0.0:main:ch2:definition-2.95	DOC-MAIN p.64; Definition 2.95	Definition / operational	Exact-admissible collapse regime
ak-hpdst:21.0.0:main:ch2:definition-2.97	DOC-MAIN p.64; Definition 2.97	Definition / operational	Windowed collapsed persistence profile
ak-hpdst:21.0.0:main:ch2:definition-2.98	DOC-MAIN p.64; Definition 2.98	Definition / operational	Admissible filtered representative
ak-hpdst:21.0.0:main:ch2:definition-2.100	DOC-MAIN p.65; Definition 2.100	Definition / operational	Representative equivalence
ak-hpdst:21.0.0:main:ch2:definition-2.102	DOC-MAIN p.65; Definition 2.102	Definition / operational	Windowed ◆◆-side input
ak-hpdst:21.0.0:main:ch2:definition-2.103	DOC-MAIN p.65; Definition 2.103	Definition / operational	Windowed topological clause
ak-hpdst:21.0.0:main:ch2:definition-2.104	DOC-MAIN p.65; Definition 2.104	Definition / operational	Windowed categorical clause
ak-hpdst:21.0.0:main:ch2:definition-2.105	DOC-MAIN p.65; Definition 2.105	Definition / operational	Comparison-typed ◆◆-Gate+ input
ak-hpdst:21.0.0:main:ch2:definition-2.107	DOC-MAIN p.66; Definition 2.107	Definition / operational	Windowed collapsed tower objects
ak-hpdst:21.0.0:main:ch2:definition-2.108	DOC-MAIN p.66; Definition 2.108	Definition / operational	Tower comparison artifact
ak-hpdst:21.0.0:main:ch2:definition-2.110	DOC-MAIN p.67; Definition 2.110	Definition / operational	Typed Type IV scope status
ak-hpdst:21.0.0:main:ch2:definition-2.111	DOC-MAIN p.67; Definition 2.111	Definition / operational	Collapse diagnostics notation contract
ak-hpdst:21.0.0:main:ch2:definition-2.114	DOC-MAIN p.68; Definition 2.114	Definition / operational	Collapse policy manifest entry
ak-hpdst:21.0.0:main:ch2:definition-2.115	DOC-MAIN p.68; Definition 2.115	Definition / operational	Metric and crop repair record
ak-hpdst:21.0.0:main:ch3:definition-3.2	DOC-MAIN p.75; Definition 3.2	Definition / operational	Pre-threshold windowed profile
ak-hpdst:21.0.0:main:ch3:definition-3.3	DOC-MAIN p.75; Definition 3.3	Definition / operational	Repaired windowed evaluation
ak-hpdst:21.0.0:main:ch3:definition-3.6	DOC-MAIN p.76; Definition 3.6	Definition / operational	Threshold clearance on a windowed profile
ak-hpdst:21.0.0:main:ch3:definition-3.7	DOC-MAIN p.76; Definition 3.7	Definition / operational	Ledger gap
ak-hpdst:21.0.0:main:ch3:definition-3.10	DOC-MAIN p.77; Definition 3.10	Definition / operational	Bottleneck scalarization and metric ledger
ak-hpdst:21.0.0:main:ch3:definition-3.11	DOC-MAIN p.77; Definition 3.11	Definition / operational	Non-scalar record obligations
ak-hpdst:21.0.0:main:ch3:definition-3.13	DOC-MAIN p.77; Definition 3.13	Definition / operational	Repaired evaluation record
ak-hpdst:21.0.0:main:ch3:definition-3.15	DOC-MAIN p.78; Definition 3.15	Definition / operational	Single-profile threshold-gap readiness record
ak-hpdst:21.0.0:main:ch3:definition-3.16	DOC-MAIN p.78; Definition 3.16	Definition / operational	Comparison record and two-sided metric certificate
ak-hpdst:21.0.0:main:ch3:proposition-3.17	DOC-MAIN p.79; Proposition 3.17	Proposition / proved	Gap-certified stability transfer
ak-hpdst:21.0.0:main:ch3:definition-3.19	DOC-MAIN p.80; Definition 3.19	Definition / operational	Window-edge clearance record
ak-hpdst:21.0.0:main:ch3:definition-3.21	DOC-MAIN p.80; Definition 3.21	Definition / operational	Profile-level metric-ledger decomposition
ak-hpdst:21.0.0:main:ch3:definition-3.23	DOC-MAIN p.81; Definition 3.23	Definition / operational	Infinite-bar policy for windowed evaluation
ak-hpdst:21.0.0:main:ch3:definition-3.25	DOC-MAIN p.81; Definition 3.25	Definition / operational	Admissible filtered representative
ak-hpdst:21.0.0:main:ch3:definition-3.26	DOC-MAIN p.81; Definition 3.26	Definition / operational	Representative record
ak-hpdst:21.0.0:main:ch3:definition-3.28	DOC-MAIN p.82; Definition 3.28	Definition / operational	Realization-compatible representative
ak-hpdst:21.0.0:main:ch3:proposition-3.30	DOC-MAIN p.82; Proposition 3.30	Proposition / proved	Profile-level representative invariance
ak-hpdst:21.0.0:main:ch3:definition-3.32	DOC-MAIN p.83; Definition 3.32	Definition / operational	Topological clause input
ak-hpdst:21.0.0:main:ch3:definition-3.33	DOC-MAIN p.83; Definition 3.33	Definition / operational	Scalarized gap-ready topological input
ak-hpdst:21.0.0:main:ch3:definition-3.36	DOC-MAIN p.84; Definition 3.36	Definition / operational	Overlap pre-threshold profile
ak-hpdst:21.0.0:main:ch3:definition-3.37	DOC-MAIN p.84; Definition 3.37	Definition / operational	Overlap repaired evaluation
ak-hpdst:21.0.0:main:ch3:proposition-3.39	DOC-MAIN p.84; Proposition 3.39	Proposition / proved	Scalarized overlap threshold safety
ak-hpdst:21.0.0:main:ch3:definition-3.40	DOC-MAIN p.85; Definition 3.40	Definition / operational	Detector profile stage
ak-hpdst:21.0.0:main:ch3:definition-3.41	DOC-MAIN p.85; Definition 3.41	Definition / operational	Certificate information model
ak-hpdst:21.0.0:main:ch3:definition-3.43	DOC-MAIN p.85; Definition 3.43	Definition / operational	Certificate query model
ak-hpdst:21.0.0:main:ch24:policy-24	DOC-MAIN p.86; Policy 24	Policy / operational	Exact-zero use of randomized or approximate queries
ak-hpdst:21.0.0:main:ch3:definition-3.44	DOC-MAIN p.86; Definition 3.44	Definition / operational	Certificate encoding model
ak-hpdst:21.0.0:main:ch3:definition-3.46	DOC-MAIN p.86; Definition 3.46	Definition / operational	Coordinatewise cost order
ak-hpdst:21.0.0:main:ch3:definition-3.47	DOC-MAIN p.86; Definition 3.47	Definition / operational	Certificate verifier model
ak-hpdst:21.0.0:main:ch3:definition-3.48	DOC-MAIN p.87; Definition 3.48	Definition / operational	Certificate model
ak-hpdst:21.0.0:main:ch3:definition-3.49	DOC-MAIN p.87; Definition 3.49	Definition / operational	Family-relative certificate scheme and family cost
ak-hpdst:21.0.0:main:ch3:definition-3.50	DOC-MAIN p.87; Definition 3.50	Definition / operational	Oracle-closed certificate model
ak-hpdst:21.0.0:main:ch3:definition-3.51	DOC-MAIN p.88; Definition 3.51	Definition / operational	Typed finite probe family
ak-hpdst:21.0.0:main:ch3:theorem-3.52	DOC-MAIN p.88; Theorem 3.52	Theorem / proved	Certified finite-probe upper bound
ak-hpdst:21.0.0:main:ch3:definition-3.54	DOC-MAIN p.88; Definition 3.54	Definition / operational	Effective detector domain
ak-hpdst:21.0.0:main:ch3:definition-3.56	DOC-MAIN p.89; Definition 3.56	Definition / operational	Exact finite stencil detector
ak-hpdst:21.0.0:main:ch3:definition-3.57	DOC-MAIN p.89; Definition 3.57	Definition / operational	Retained bars and active-start region
ak-hpdst:21.0.0:main:ch3:proposition-3.58	DOC-MAIN p.89; Proposition 3.58	Proposition / proved	Interval decomposition of a stencil image
ak-hpdst:21.0.0:main:ch3:theorem-3.59	DOC-MAIN p.89; Theorem 3.59	Theorem / proved	Finite-stencil soundness
ak-hpdst:21.0.0:main:ch3:definition-3.61	DOC-MAIN p.90; Definition 3.61	Definition / operational	Exhaustive birth set

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:main:ch3:theorem-3.62	DOC-MAIN p.90; Theorem 3.62	Theorem / proved	Birth detector characterization
ak-hpdst:21.0.0:main:ch3:definition-3.64	DOC-MAIN p.90; Definition 3.64	Definition / operational	Right germ and shifted germ map
ak-hpdst:21.0.0:main:ch3:definition-3.65	DOC-MAIN p.91; Definition 3.65	Definition / operational	Germ detector
ak-hpdst:21.0.0:main:ch3:theorem-3.66	DOC-MAIN p.91; Theorem 3.66	Theorem / proved	Decoration-invariant germ characterization
ak-hpdst:21.0.0:main:ch3:definition-3.67	DOC-MAIN p.91; Definition 3.67	Definition / operational	Stencil coverage certificate
ak-hpdst:21.0.0:main:ch3:theorem-3.68	DOC-MAIN p.91; Theorem 3.68	Theorem / proved	Certified stencil completeness
ak-hpdst:21.0.0:main:ch3:definition-3.70	DOC-MAIN p.92; Definition 3.70	Definition / operational	Window-relative threshold excess
ak-hpdst:21.0.0:main:ch3:definition-3.72	DOC-MAIN p.92; Definition 3.72	Definition / operational	Augmented mesh
ak-hpdst:21.0.0:main:ch3:lemma-3.73	DOC-MAIN p.92; Lemma 3.73	Lemma / proved	Augmented-mesh hitting lemma
ak-hpdst:21.0.0:main:ch3:theorem-3.74	DOC-MAIN p.93; Theorem 3.74	Theorem / proved	Family-mesh completeness
ak-hpdst:21.0.0:main:ch3:theorem-3.75	DOC-MAIN p.93; Theorem 3.75	Theorem / proved	Strict mesh sharpness and equality boundary
ak-hpdst:21.0.0:main:ch3:definition-3.76	DOC-MAIN p.94; Definition 3.76	Definition / operational	Finite detector certificate record
ak-hpdst:21.0.0:main:ch3:definition-3.77	DOC-MAIN p.94; Definition 3.77	Definition / operational	Two-axis detector status
ak-hpdst:21.0.0:main:ch3:proposition-3.78	DOC-MAIN p.94; Proposition 3.78	Proposition / proved	Detector invariance under profile isomorphism
ak-hpdst:21.0.0:main:ch3:theorem-3.79	DOC-MAIN p.95; Theorem 3.79	Theorem / proved	Metric zero-verdict composition
ak-hpdst:21.0.0:main:ch3:definition-3.81	DOC-MAIN p.95; Definition 3.81	Definition / operational	Boundary-safe robust stencil
ak-hpdst:21.0.0:main:ch3:proposition-3.82	DOC-MAIN p.95; Proposition 3.82	Proposition / proved	Robust-to-exact sandwich
ak-hpdst:21.0.0:main:ch3:definition-3.84	DOC-MAIN p.96; Definition 3.84	Definition / operational	Detector information class
ak-hpdst:21.0.0:main:ch3:definition-3.85	DOC-MAIN p.96; Definition 3.85	Definition / operational	Achievable scheme-cost set and family-relative minimal certificate
ak-hpdst:21.0.0:main:ch3:theorem-3.86	DOC-MAIN p.96; Theorem 3.86	Theorem / proved	Well-posedness of the finite Pareto frontier
ak-hpdst:21.0.0:main:ch3:proposition-3.88	DOC-MAIN p.97; Proposition 3.88	Proposition / proved	Birth-set upper bound
ak-hpdst:21.0.0:main:ch3:definition-3.89	DOC-MAIN p.97; Definition 3.89	Definition / operational	Minimal augmented-mesh stencil number
ak-hpdst:21.0.0:main:ch3:theorem-3.90	DOC-MAIN p.97; Theorem 3.90	Theorem / proved	Exact query count for an augmented-mesh target
ak-hpdst:21.0.0:main:ch3:corollary-3.91	DOC-MAIN p.97; Corollary 3.91	Corollary / proved	Family mesh upper bound for vector-valued MinCert
ak-hpdst:21.0.0:main:ch3:definition-3.92	DOC-MAIN p.98; Definition 3.92	Definition / operational	Private witness cells
ak-hpdst:21.0.0:main:ch3:theorem-3.93	DOC-MAIN p.98; Theorem 3.93	Theorem / proved	Indistinguishability lower bound
ak-hpdst:21.0.0:main:ch3:corollary-3.94	DOC-MAIN p.98; Corollary 3.94	Corollary / proved	Omitted-region, omitted-generator, and omitted-cell lower bounds
ak-hpdst:21.0.0:main:ch3:definition-3.100	DOC-MAIN p.99; Definition 3.100	Definition / operational	Composition overhead and reuse discipline
ak-hpdst:21.0.0:main:ch3:theorem-3.97	DOC-MAIN p.99; Theorem 3.97	Theorem / proved	Uniform finite-detector no-go on an unbounded birth family
ak-hpdst:21.0.0:main:ch3:definition-3.99	DOC-MAIN p.99; Definition 3.99	Definition / operational	Composable certificate schemes
ak-hpdst:21.0.0:main:ch3:theorem-3.101	DOC-MAIN p.100; Theorem 3.101	Theorem / proved	Finite certificate-scheme composition inequality
ak-hpdst:21.0.0:main:ch3:definition-3.103	DOC-MAIN p.100; Definition 3.103	Definition / operational	Certificate-to-representative handoff
ak-hpdst:21.0.0:main:ch3:theorem-3.104	DOC-MAIN p.100; Theorem 3.104	Theorem / proved	Representative handoff boundary
ak-hpdst:21.0.0:main:ch3:definition-3.105	DOC-MAIN p.101; Definition 3.105	Definition / operational	Residual degree-one bar mass
ak-hpdst:21.0.0:main:ch3:proposition-3.106	DOC-MAIN p.101; Proposition 3.106	Proposition / proved	Sound $\diamond\Diamond 1$ -vanishing detector
ak-hpdst:21.0.0:main:ch3:corollary-3.107	DOC-MAIN p.101; Corollary 3.107	Corollary / proved	Whole-barcode $\diamond\Diamond 1$ characterization
ak-hpdst:21.0.0:main:ch3:definition-3.109	DOC-MAIN p.101; Definition 3.109	Definition / operational	Chapter 3 evaluation and representative manifest extension
ak-hpdst:21.0.0:main:ch4:definition-4.3	DOC-MAIN p.109; Definition 4.3	Definition / operational	Two-axis realization classification
ak-hpdst:21.0.0:main:ch25:policy-25	DOC-MAIN p.110; Policy 25	Policy / operational	No verdict-conditioned realization
ak-hpdst:21.0.0:main:ch4:definition-4.5	DOC-MAIN p.110; Definition 4.5	Definition / operational	Finite endpoint set
ak-hpdst:21.0.0:main:ch4:definition-4.6	DOC-MAIN p.110; Definition 4.6	Definition / operational	Atom decomposition and enriched grid
ak-hpdst:21.0.0:main:ch4:definition-4.10	DOC-MAIN p.111; Definition 4.10	Definition / operational	Grid completeness record
ak-hpdst:21.0.0:main:ch4:definition-4.8	DOC-MAIN p.111; Definition 4.8	Definition / operational	Terminal sentinel policy
ak-hpdst:21.0.0:main:ch4:definition-4.9	DOC-MAIN p.111; Definition 4.9	Definition / operational	Γ -constructible profile
ak-hpdst:21.0.0:main:ch4:definition-4.12	DOC-MAIN p.112; Definition 4.12	Definition / operational	Finite linear-quiver category
ak-hpdst:21.0.0:main:ch4:definition-4.13	DOC-MAIN p.112; Definition 4.13	Definition / operational	Grid restriction and step realization
ak-hpdst:21.0.0:main:ch4:lemma-4.14	DOC-MAIN p.112; Lemma 4.14	Lemma / proved	Canonical atom transport
ak-hpdst:21.0.0:main:ch4:theorem-4.15	DOC-MAIN p.112; Theorem 4.15	Theorem / proved	Finite-grid encoding equivalence
ak-hpdst:21.0.0:main:ch4:proposition-4.16	DOC-MAIN p.113; Proposition 4.16	Proposition / proved	Probe-choice invariance on a fixed atom decomposition
ak-hpdst:21.0.0:main:ch4:definition-4.20	DOC-MAIN p.114; Definition 4.20	Definition / operational	Objective-barcode encoding
ak-hpdst:21.0.0:main:ch4:definition-4.21	DOC-MAIN p.114; Definition 4.21	Definition / operational	Grid encoding strength record
ak-hpdst:21.0.0:main:ch4:theorem-4.22	DOC-MAIN p.114; Theorem 4.22	Theorem / proved	Endpoint-complete reconstruction
ak-hpdst:21.0.0:main:ch4:definition-4.24	DOC-MAIN p.115; Definition 4.24	Definition / operational	Finite rank table
ak-hpdst:21.0.0:main:ch4:theorem-4.25	DOC-MAIN p.115; Theorem 4.25	Theorem / proved	Finite Möbius multiplicity formula
ak-hpdst:21.0.0:main:ch4:corollary-4.26	DOC-MAIN p.115; Corollary 4.26	Corollary / proved	Endpoint-complete barcode Return
ak-hpdst:21.0.0:main:ch4:definition-4.28	DOC-MAIN p.115; Definition 4.28	Definition / operational	Projective and simple probes
ak-hpdst:21.0.0:main:ch4:definition-4.29	DOC-MAIN p.116; Definition 4.29	Definition / operational	Enriched finite projective/Ext signature
ak-hpdst:21.0.0:main:ch4:proposition-4.30	DOC-MAIN p.116; Proposition 4.30	Proposition / proved	Local projective/Ext exact sequence
ak-hpdst:21.0.0:main:ch4:theorem-4.31	DOC-MAIN p.116; Theorem 4.31	Theorem / proved	Finite enriched-signature reconstruction
ak-hpdst:21.0.0:main:ch4:definition-4.33	DOC-MAIN p.117; Definition 4.33	Definition / operational	Degree-one repaired profile
ak-hpdst:21.0.0:main:ch4:definition-4.34	DOC-MAIN p.117; Definition 4.34	Definition / operational	Degree-one admissible representative
ak-hpdst:21.0.0:main:ch4:definition-4.36	DOC-MAIN p.117; Definition 4.36	Definition / operational	Realization datum
ak-hpdst:21.0.0:main:ch4:definition-4.37	DOC-MAIN p.117; Definition 4.37	Definition / operational	Residual edge extractor
ak-hpdst:21.0.0:main:ch4:definition-4.38	DOC-MAIN p.118; Definition 4.38	Definition / operational	Edge-realization certificate
ak-hpdst:21.0.0:main:ch4:definition-4.39	DOC-MAIN p.118; Definition 4.39	Definition / operational	Eligible realization record
ak-hpdst:21.0.0:main:ch4:definition-4.42	DOC-MAIN p.119; Definition 4.42	Definition / operational	Degree- $\diamond$ repaired profile
ak-hpdst:21.0.0:main:ch4:definition-4.43	DOC-MAIN p.119; Definition 4.43	Definition / operational	Degree- $\diamond\Diamond$ compatible representative
ak-hpdst:21.0.0:main:ch4:definition-4.44	DOC-MAIN p.119; Definition 4.44	Definition / operational	Degree-indexed realization datum
ak-hpdst:21.0.0:main:ch4:definition-4.45	DOC-MAIN p.120; Definition 4.45	Definition / operational	Higher residual edge extractor
ak-hpdst:21.0.0:main:ch4:definition-4.46	DOC-MAIN p.120; Definition 4.46	Definition / operational	Higher edge-realization certificate
ak-hpdst:21.0.0:main:ch4:definition-4.47	DOC-MAIN p.120; Definition 4.47	Definition / operational	Standard $\diamond\Diamond$ -eligible realization record
ak-hpdst:21.0.0:main:ch4:theorem-4.50	DOC-MAIN p.121; Theorem 4.50	Theorem / proved	Canonical higher Ext-edge over a field
ak-hpdst:21.0.0:main:ch4:corollary-4.51	DOC-MAIN p.121; Corollary 4.51	Corollary / proved	Higher edge vanishing and reflection
ak-hpdst:21.0.0:main:ch4:lemma-4.52	DOC-MAIN p.121; Lemma 4.52	Lemma / proved	Formality over a field; noncanonical splitting
ak-hpdst:21.0.0:main:ch4:theorem-4.53	DOC-MAIN p.121; Theorem 4.53	Theorem / proved	Canonical two-term Ext1-edge identification
ak-hpdst:21.0.0:main:ch4:corollary-4.54	DOC-MAIN p.122; Corollary 4.54	Corollary / proved	Naturality square
ak-hpdst:21.0.0:main:ch4:corollary-4.55	DOC-MAIN p.122; Corollary 4.55	Corollary / proved	Edge vanishing criterion
ak-hpdst:21.0.0:main:ch4:theorem-4.58	DOC-MAIN p.122; Theorem 4.58	Theorem / proved	Eligible higher Ext-edge identification
ak-hpdst:21.0.0:main:ch4:theorem-4.59	DOC-MAIN p.123; Theorem 4.59	Theorem / proved	Higher one-way repaired discharge
ak-hpdst:21.0.0:main:ch4:definition-4.60	DOC-MAIN p.123; Definition 4.60	Definition / operational	Natural higher zero-reflection package
ak-hpdst:21.0.0:main:ch4:theorem-4.61	DOC-MAIN p.123; Theorem 4.61	Theorem / proved	Zero-reflecting higher vanishing equivalence
ak-hpdst:21.0.0:main:ch4:proposition-4.62	DOC-MAIN p.123; Proposition 4.62	Proposition / proved	Functoriality of a fixed stencil detector
ak-hpdst:21.0.0:main:ch4:corollary-4.63	DOC-MAIN p.124; Corollary 4.63	Corollary / proved	Complete fixed-stencil higher vanishing equivalence
ak-hpdst:21.0.0:main:ch4:theorem-4.66	DOC-MAIN p.124; Theorem 4.66	Theorem / proved	Conditional one-way repaired bridge
ak-hpdst:21.0.0:main:ch4:definition-4.69	DOC-MAIN p.125; Definition 4.69	Definition / operational	Higher Ext-edge discharge record
ak-hpdst:21.0.0:main:ch4:definition-4.70	DOC-MAIN p.126; Definition 4.70	Definition / operational	Typed degree- $\diamond\Diamond$ Ext-clause status
ak-hpdst:21.0.0:main:ch4:proposition-4.71	DOC-MAIN p.126; Proposition 4.71	Proposition / proved	Higher B-Gate+ categorical discharge
ak-hpdst:21.0.0:main:ch4:definition-4.72	DOC-MAIN p.126; Definition 4.72	Definition / operational	Aggregate higher categorical gate status
ak-hpdst:21.0.0:main:ch4:definition-4.73	DOC-MAIN p.126; Definition 4.73	Definition / operational	Ext-edge discharge record
ak-hpdst:21.0.0:main:ch4:definition-4.74	DOC-MAIN p.126; Definition 4.74	Definition / operational	Typed Ext-edge clause status
ak-hpdst:21.0.0:main:ch4:proposition-4.75	DOC-MAIN p.127; Proposition 4.75	Proposition / proved	B-Gate+ categorical discharge
ak-hpdst:21.0.0:main:ch4:definition-4.77	DOC-MAIN p.127; Definition 4.77	Definition / operational	Degree- $\diamond$ realization-equivalent representatives
ak-hpdst:21.0.0:main:ch4:proposition-4.78	DOC-MAIN p.127; Proposition 4.78	Proposition / proved	Higher representative invariance
ak-hpdst:21.0.0:main:ch4:definition-4.79	DOC-MAIN p.128; Definition 4.79	Definition / operational	Realization-equivalent representatives
ak-hpdst:21.0.0:main:ch4:proposition-4.80	DOC-MAIN p.128; Proposition 4.80	Proposition / proved	Representative invariance of Ext discharge
ak-hpdst:21.0.0:main:ch4:proposition-4.82	DOC-MAIN p.128; Proposition 4.82	Proposition / proved	Complete detector higher Ext discharge
ak-hpdst:21.0.0:main:ch4:proposition-4.83	DOC-MAIN p.129; Proposition 4.83	Proposition / proved	Operational $\diamond\Diamond 1$ -detector discharge
ak-hpdst:21.0.0:main:ch4:proposition-4.85	DOC-MAIN p.129; Proposition 4.85	Proposition / proved	Higher reverse inference requires zero-reflection
ak-hpdst:21.0.0:main:ch4:definition-4.86	DOC-MAIN p.129; Definition 4.86	Definition / operational	Synthetic detector object
ak-hpdst:21.0.0:main:ch4:proposition-4.87	DOC-MAIN p.129; Proposition 4.87	Proposition / proved	Encoded detector duality
ak-hpdst:21.0.0:main:ch4:proposition-4.90	DOC-MAIN p.130; Proposition 4.90	Proposition / proved	Failure of the reverse bridge for a natural nonfaithful edge
ak-hpdst:21.0.0:main:ch4:definition-4.92	DOC-MAIN p.131; Definition 4.92	Definition / operational	Actual finite-diagram morphism record
ak-hpdst:21.0.0:main:ch4:definition-4.93	DOC-MAIN p.131; Definition 4.93	Definition / operational	Linear cone handoff record
ak-hpdst:21.0.0:main:ch4:proposition-4.94	DOC-MAIN p.131; Proposition 4.94	Proposition / proved	Cone zero criterion in the finite linear derived category
ak-hpdst:21.0.0:main:ch4:definition-4.96	DOC-MAIN p.131; Definition 4.96	Definition / operational	Finite linear P2DG Γ interface
ak-hpdst:21.0.0:main:ch4:definition-4.97	DOC-MAIN p.131; Definition 4.97	Definition / operational	Finite linear P2DG record
ak-hpdst:21.0.0:main:ch4:theorem-4.98	DOC-MAIN p.132; Theorem 4.98	Theorem / proved	P-to-DG finite linear handoff
ak-hpdst:21.0.0:main:ch4:theorem-4.99	DOC-MAIN p.132; Theorem 4.99	Theorem / proved	Finite profile-to-quiver-to-Return closure
ak-hpdst:21.0.0:main:ch4:definition-4.101	DOC-MAIN p.133; Definition 4.101	Definition / operational	Eligibility manifest extension
ak-hpdst:21.0.0:main:ch4:theorem-4.103	DOC-MAIN p.134; Theorem 4.103	Theorem / proved	Eligible Ext-edge bridge package
ak-hpdst:21.0.0:main:ch5:definition-5.3	DOC-MAIN p.141; Definition 5.3	Definition / operational	Typed tower variance
ak-hpdst:21.0.0:main:ch5:definition-5.4	DOC-MAIN p.141; Definition 5.4	Definition / operational	Tower semantic role
ak-hpdst:21.0.0:main:ch5:definition-5.5	DOC-MAIN p.141; Definition 5.5	Definition / operational	Actual-transient requirement
ak-hpdst:21.0.0:main:ch26:policy-26	DOC-MAIN p.142; Policy 26	Policy / operational	No variance laundering
ak-hpdst:21.0.0:main:ch5:definition-5.6	DOC-MAIN p.142; Definition 5.6	Definition / operational	Evaluation-lift record
ak-hpdst:21.0.0:main:ch5:definition-5.7	DOC-MAIN p.142; Definition 5.7	Definition / operational	v21 inherited failure types and transient-defect refinement
ak-hpdst:21.0.0:main:ch5:definition-5.12	DOC-MAIN p.143; Definition 5.12	Definition / operational	Type IV scope status
ak-hpdst:21.0.0:main:ch5:definition-5.9	DOC-MAIN p.143; Definition 5.9	Definition / operational	Failure-check order
ak-hpdst:21.0.0:main:ch5:definition-5.13	DOC-MAIN p.144; Definition 5.13	Definition / operational	Structural collapse admissibility
ak-hpdst:21.0.0:main:ch5:definition-5.14	DOC-MAIN p.144; Definition 5.14	Definition / operational	Audited collapse admissibility
ak-hpdst:21.0.0:main:ch5:proposition-5.15	DOC-MAIN p.145; Proposition 5.15	Proposition / proved	Eligible monitored vanishing discharges the invoked Type II clauses
ak-hpdst:21.0.0:main:ch5:definition-5.16	DOC-MAIN p.145; Definition 5.16	Definition / operational	Direct-system source object
ak-hpdst:21.0.0:main:ch5:definition-5.17	DOC-MAIN p.145; Definition 5.17	Definition / operational	Inverse-system source object
ak-hpdst:21.0.0:main:ch5:definition-5.18	DOC-MAIN p.145; Definition 5.18	Definition / operational	Declared apex and comparison direction
ak-hpdst:21.0.0:main:ch5:definition-5.19	DOC-MAIN p.146; Definition 5.19	Definition / operational	Windowed repaired tower objects

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:main:ch5:definition-5.20	DOC-MAIN p.146; Definition 5.20	Definition / operational	Repaired tower-profile system
ak-hpdst:21.0.0:main:ch5:definition-5.21	DOC-MAIN p.146; Definition 5.21	Definition / operational	Collapsed source profile
ak-hpdst:21.0.0:main:ch5:definition-5.22	DOC-MAIN p.146; Definition 5.22	Definition / operational	Collapsed apex profile
ak-hpdst:21.0.0:main:ch5:definition-5.24	DOC-MAIN p.147; Definition 5.24	Definition / operational	Direct Type IV tower comparison artifact
ak-hpdst:21.0.0:main:ch5:definition-5.25	DOC-MAIN p.147; Definition 5.25	Definition / operational	Inverse apex comparison record
ak-hpdst:21.0.0:main:ch5:definition-5.27	DOC-MAIN p.148; Definition 5.27	Definition / operational	Exact-admissible tower artifact
ak-hpdst:21.0.0:main:ch5:definition-5.29	DOC-MAIN p.148; Definition 5.29	Definition / operational	Countable inverse-system record
ak-hpdst:21.0.0:main:ch5:definition-5.30	DOC-MAIN p.148; Definition 5.30	Definition / operational	Roos operator and first derived-limit defect
ak-hpdst:21.0.0:main:ch5:definition-5.31	DOC-MAIN p.149; Definition 5.31	Definition / operational	Mittag–Leffler condition
ak-hpdst:21.0.0:main:ch5:theorem-5.32	DOC-MAIN p.149; Theorem 5.32	Theorem / proved	Countable Mittag–Leffler vanishing
ak-hpdst:21.0.0:main:ch5:corollary-5.33	DOC-MAIN p.149; Corollary 5.33	Corollary / proved	Finite-dimensional and finite-length ML branches
ak-hpdst:21.0.0:main:ch5:definition-5.35	DOC-MAIN p.149; Definition 5.35	Definition / operational	Inverse apex defect
ak-hpdst:21.0.0:main:ch5:definition-5.36	DOC-MAIN p.149; Definition 5.36	Definition / operational	Inverse-limit audit profile
ak-hpdst:21.0.0:main:ch5:theorem-5.37	DOC-MAIN p.150; Theorem 5.37	Theorem / proved	Derived inverse-limit and apex separation
ak-hpdst:21.0.0:main:ch5:definition-5.39	DOC-MAIN p.150; Definition 5.39	Definition / operational	Terminal-tame profile
ak-hpdst:21.0.0:main:ch5:definition-5.40	DOC-MAIN p.150; Definition 5.40	Definition / operational	Terminal-generic dimension
ak-hpdst:21.0.0:main:ch5:proposition-5.41	DOC-MAIN p.151; Proposition 5.41	Proposition / proved	Well-definedness of terminal-generic dimension
ak-hpdst:21.0.0:main:ch5:definition-5.42	DOC-MAIN p.151; Definition 5.42	Definition / operational	Cofinal bar in a window
ak-hpdst:21.0.0:main:ch5:proposition-5.44	DOC-MAIN p.151; Proposition 5.44	Proposition / proved	Terminal dimension counts cofinal bars
ak-hpdst:21.0.0:main:ch5:proposition-5.46	DOC-MAIN p.151; Proposition 5.46	Proposition / proved	Terminal tameness passes to defects
ak-hpdst:21.0.0:main:ch5:definition-5.47	DOC-MAIN p.151; Definition 5.47	Definition / operational	Defect profiles
ak-hpdst:21.0.0:main:ch5:definition-5.48	DOC-MAIN p.152; Definition 5.48	Definition / operational	Degreewise Type IV diagnostics
ak-hpdst:21.0.0:main:ch5:definition-5.49	DOC-MAIN p.152; Definition 5.49	Definition / operational	Total collapse diagnostics
ak-hpdst:21.0.0:main:ch5:definition-5.50	DOC-MAIN p.152; Definition 5.50	Definition / operational	Type IV diagnostic record
ak-hpdst:21.0.0:main:ch5:definition-5.51	DOC-MAIN p.153; Definition 5.51	Definition / operational	Type IV obstruction and Type IV-free status
ak-hpdst:21.0.0:main:ch5:proposition-5.52	DOC-MAIN p.153; Proposition 5.52	Proposition / proved	Characterization of certified Type IV-free status
ak-hpdst:21.0.0:main:ch5:definition-5.57	DOC-MAIN p.154; Definition 5.57	Definition / operational	Cofinal and interior barcode profiles
ak-hpdst:21.0.0:main:ch5:proposition-5.58	DOC-MAIN p.154; Proposition 5.58	Proposition / proved	Terminal/interior separation
ak-hpdst:21.0.0:main:ch5:definition-5.59	DOC-MAIN p.154; Definition 5.59	Definition / operational	Defect threshold and detector component
ak-hpdst:21.0.0:main:ch5:definition-5.60	DOC-MAIN p.155; Definition 5.60	Definition / operational	Kernel and cokernel defect detectors
ak-hpdst:21.0.0:main:ch5:theorem-5.61	DOC-MAIN p.155; Theorem 5.61	Theorem / proved	Complete transient-defect characterization
ak-hpdst:21.0.0:main:ch5:corollary-5.63	DOC-MAIN p.155; Corollary 5.63	Corollary / proved	Full and interior detectors coincide under terminal zero
ak-hpdst:21.0.0:main:ch5:definition-5.64	DOC-MAIN p.155; Definition 5.64	Definition / operational	Aggregate long-transient verdict
ak-hpdst:21.0.0:main:ch5:theorem-5.65	DOC-MAIN p.156; Theorem 5.65	Theorem / proved	Joint terminal/transient defect classification
ak-hpdst:21.0.0:main:ch5:definition-5.66	DOC-MAIN p.156; Definition 5.66	Definition / operational	Short-defect exclusion certificate
ak-hpdst:21.0.0:main:ch5:theorem-5.67	DOC-MAIN p.156; Theorem 5.67	Theorem / proved	Conditional full-isomorphism upgrade
ak-hpdst:21.0.0:main:ch5:definition-5.70	DOC-MAIN p.157; Definition 5.70	Definition / operational	Eventual-isomorphism certificate
ak-hpdst:21.0.0:main:ch5:theorem-5.71	DOC-MAIN p.157; Theorem 5.71	Theorem / proved	Eventually constant direct-system Return
ak-hpdst:21.0.0:main:ch5:theorem-5.72	DOC-MAIN p.157; Theorem 5.72	Theorem / proved	Eventually constant inverse-system Return
ak-hpdst:21.0.0:main:ch5:theorem-5.73	DOC-MAIN p.157; Theorem 5.73	Theorem / proved	Finite-length chain stabilization
ak-hpdst:21.0.0:main:ch5:corollary-5.74	DOC-MAIN p.157; Corollary 5.74	Corollary / proved	Stable image of a finite-length endomorphism
ak-hpdst:21.0.0:main:ch5:theorem-5.75	DOC-MAIN p.158; Theorem 5.75	Theorem / proved	Noetherian ascending-image reduction
ak-hpdst:21.0.0:main:ch5:theorem-5.77	DOC-MAIN p.158; Theorem 5.77	Theorem / proved	Finite-generation finite-window zero reduction
ak-hpdst:21.0.0:main:ch5:definition-5.79	DOC-MAIN p.158; Definition 5.79	Definition / operational	Deterministic finite-state tail system
ak-hpdst:21.0.0:main:ch5:theorem-5.80	DOC-MAIN p.158; Theorem 5.80	Theorem / proved	Finite-state terminality theorem
ak-hpdst:21.0.0:main:ch5:corollary-5.81	DOC-MAIN p.159; Corollary 5.81	Corollary / proved	Finite-state eventual-constancy lane
ak-hpdst:21.0.0:main:ch5:definition-5.82	DOC-MAIN p.159; Definition 5.82	Definition / operational	Typed finite-state linear inverse tower
ak-hpdst:21.0.0:main:ch5:definition-5.83	DOC-MAIN p.159; Definition 5.83	Definition / operational	Periodic monodromy and stable image
ak-hpdst:21.0.0:main:ch5:theorem-5.84	DOC-MAIN p.159; Theorem 5.84	Theorem / proved	Exact finite-state inverse-limit Return
ak-hpdst:21.0.0:main:ch5:theorem-5.85	DOC-MAIN p.160; Theorem 5.85	Theorem / proved	Finite-state apex-agreement criterion
ak-hpdst:21.0.0:main:ch5:definition-5.87	DOC-MAIN p.160; Definition 5.87	Definition / operational	Finite-to-infinite reduction record
ak-hpdst:21.0.0:main:ch5:theorem-5.88	DOC-MAIN p.160; Theorem 5.88	Theorem / proved	Finite-to-infinite soundness theorem
ak-hpdst:21.0.0:main:ch27:policy-27	DOC-MAIN p.161; Policy 27	Policy / operational	External Return separation
ak-hpdst:21.0.0:main:ch5:definition-5.89	DOC-MAIN p.161; Definition 5.89	Definition / operational	Internal apex Return capability profile
ak-hpdst:21.0.0:main:ch5:definition-5.90	DOC-MAIN p.161; Definition 5.90	Definition / operational	Artifact-coherent Type IV stable band
ak-hpdst:21.0.0:main:ch5:proposition-5.92	DOC-MAIN p.162; Proposition 5.92	Proposition / proved	No terminal-generic Type IV obstruction on a stable band
ak-hpdst:21.0.0:main:ch5:definition-5.93	DOC-MAIN p.162; Definition 5.93	Definition / operational	Artifact-coherent joint defect-clean band
ak-hpdst:21.0.0:main:ch5:proposition-5.94	DOC-MAIN p.162; Proposition 5.94	Proposition / proved	Uniform terminal and long-transient cleanliness
ak-hpdst:21.0.0:main:ch5:definition-5.95	DOC-MAIN p.162; Definition 5.95	Definition / operational	Type IV and transient-defect gate discharge
ak-hpdst:21.0.0:main:ch5:proposition-5.97	DOC-MAIN p.163; Proposition 5.97	Proposition / proved	Audited admissibility implies the structural and safety clauses of B-Gate+
ak-hpdst:21.0.0:main:ch5:definition-5.98	DOC-MAIN p.163; Definition 5.98	Definition / operational	Failure routing record
ak-hpdst:21.0.0:main:ch5:definition-5.99	DOC-MAIN p.164; Definition 5.99	Definition / operational	v21 tower, failure, and finite-to-infinite contract
ak-hpdst:21.0.0:main:ch5:definition-5.102	DOC-MAIN p.165; Definition 5.102	Definition / operational	Type IV manifest entry
ak-hpdst:21.0.0:main:ch5:definition-5.103	DOC-MAIN p.165; Definition 5.103	Definition / operational	Failure manifest entry
ak-hpdst:21.0.0:main:ch28:policy-28	DOC-MAIN p.168; Policy 28	Policy / operational	Chapter 5 completion gate
ak-hpdst:21.0.0:main:ch6:definition-6.3	DOC-MAIN p.171; Definition 6.3	Definition / operational	Local zero family
ak-hpdst:21.0.0:main:ch6:definition-6.4	DOC-MAIN p.172; Definition 6.4	Definition / operational	Global record verdict and global record zero
ak-hpdst:21.0.0:main:ch6:definition-6.5	DOC-MAIN p.172; Definition 6.5	Definition / operational	Global object zero
ak-hpdst:21.0.0:main:ch6:theorem-6.6	DOC-MAIN p.172; Theorem 6.6	Theorem / proved	No object descent from record coherence
ak-hpdst:21.0.0:main:ch6:definition-6.7	DOC-MAIN p.172; Definition 6.7	Definition / operational	Right-open Core window
ak-hpdst:21.0.0:main:ch6:definition-6.8	DOC-MAIN p.172; Definition 6.8	Definition / operational	Right-open partition cover
ak-hpdst:21.0.0:main:ch6:definition-6.11	DOC-MAIN p.173; Definition 6.11	Definition / operational	Local certificate record
ak-hpdst:21.0.0:main:ch6:definition-6.12	DOC-MAIN p.173; Definition 6.12	Definition / operational	Local record status
ak-hpdst:21.0.0:main:ch6:definition-6.9	DOC-MAIN p.173; Definition 6.9	Definition / operational	Overlap-admitting cover
ak-hpdst:21.0.0:main:ch6:definition-6.13	DOC-MAIN p.174; Definition 6.13	Definition / operational	Direct common-window profile
ak-hpdst:21.0.0:main:ch6:definition-6.15	DOC-MAIN p.174; Definition 6.15	Definition / operational	Bottleneck scalarization
ak-hpdst:21.0.0:main:ch6:definition-6.16	DOC-MAIN p.174; Definition 6.16	Definition / operational	Scalarized metric ledger
ak-hpdst:21.0.0:main:ch6:definition-6.17	DOC-MAIN p.175; Definition 6.17	Definition / operational	Non-scalar continuation and coherence obligations
ak-hpdst:21.0.0:main:ch6:definition-6.20	DOC-MAIN p.175; Definition 6.20	Definition / operational	Gate-scoped threshold reserve
ak-hpdst:21.0.0:main:ch6:definition-6.22	DOC-MAIN p.176; Definition 6.22	Definition / operational	Threshold reconciliation record
ak-hpdst:21.0.0:main:ch6:definition-6.23	DOC-MAIN p.176; Definition 6.23	Definition / operational	Overlap comparison mode
ak-hpdst:21.0.0:main:ch6:definition-6.24	DOC-MAIN p.176; Definition 6.24	Definition / operational	Exact overlap certificate
ak-hpdst:21.0.0:main:ch6:definition-6.25	DOC-MAIN p.177; Definition 6.25	Definition / operational	Two-sided overlap metric certificate
ak-hpdst:21.0.0:main:ch6:definition-6.26	DOC-MAIN p.177; Definition 6.26	Definition / operational	Theorem-controlled nonmetric overlap certificate
ak-hpdst:21.0.0:main:ch6:theorem-6.27	DOC-MAIN p.177; Theorem 6.27	Theorem / proved	Sharp typed overlap theorem
ak-hpdst:21.0.0:main:ch6:proposition-6.28	DOC-MAIN p.177; Proposition 6.28	Proposition / proved	Sharp overlap transfer
ak-hpdst:21.0.0:main:ch6:definition-6.29	DOC-MAIN p.178; Definition 6.29	Definition / operational	Pairwise Overlap Gate
ak-hpdst:21.0.0:main:ch6:theorem-6.32	DOC-MAIN p.179; Theorem 6.32	Theorem / proved	No higher coherence from pairwise overlap
ak-hpdst:21.0.0:main:ch6:definition-6.33	DOC-MAIN p.179; Definition 6.33	Definition / operational	Čech depth
ak-hpdst:21.0.0:main:ch6:definition-6.34	DOC-MAIN p.179; Definition 6.34	Definition / operational	Simplex coherence mode
ak-hpdst:21.0.0:main:ch6:definition-6.35	DOC-MAIN p.179; Definition 6.35	Definition / operational	Exact simplex cocycle record
ak-hpdst:21.0.0:main:ch6:definition-6.36	DOC-MAIN p.179; Definition 6.36	Definition / operational	Model-certified homotopy-coherent simplex-cell record
ak-hpdst:21.0.0:main:ch6:proposition-6.37	DOC-MAIN p.180; Proposition 6.37	Proposition / proved	Homotopy-coherent certification is model-relative
ak-hpdst:21.0.0:main:ch6:definition-6.38	DOC-MAIN p.180; Definition 6.38	Definition / operational	Theorem-controlled nonmetric simplex record
ak-hpdst:21.0.0:main:ch6:definition-6.39	DOC-MAIN p.180; Definition 6.39	Definition / operational	Metric-reference simplex record
ak-hpdst:21.0.0:main:ch6:proposition-6.40	DOC-MAIN p.180; Proposition 6.40	Proposition / proved	Pairwise bound from a metric simplex reference
ak-hpdst:21.0.0:main:ch6:definition-6.42	DOC-MAIN p.181; Definition 6.42	Definition / operational	Simplex compatibility record
ak-hpdst:21.0.0:main:ch6:definition-6.43	DOC-MAIN p.181; Definition 6.43	Definition / operational	Čech-coherent certificate atlas
ak-hpdst:21.0.0:main:ch6:definition-6.44	DOC-MAIN p.181; Definition 6.44	Definition / operational	Certificate atlas
ak-hpdst:21.0.0:main:ch6:definition-6.45	DOC-MAIN p.182; Definition 6.45	Definition / operational	Atlas verdict
ak-hpdst:21.0.0:main:ch6:proposition-6.46	DOC-MAIN p.182; Proposition 6.46	Proposition / proved	Finite-depth verification bound
ak-hpdst:21.0.0:main:ch6:theorem-6.47	DOC-MAIN p.182; Theorem 6.47	Theorem / proved	Finite-cover record-level gluing
ak-hpdst:21.0.0:main:ch6:theorem-6.48	DOC-MAIN p.182; Theorem 6.48	Theorem / proved	Finite certificate-atlas soundness
ak-hpdst:21.0.0:main:ch6:definition-6.50	DOC-MAIN p.183; Definition 6.50	Definition / operational	Global profile and local restriction interface
ak-hpdst:21.0.0:main:ch6:definition-6.51	DOC-MAIN p.183; Definition 6.51	Definition / operational	Certified local detector atlas
ak-hpdst:21.0.0:main:ch6:definition-6.52	DOC-MAIN p.184; Definition 6.52	Definition / operational	Global retained-bar coverage
ak-hpdst:21.0.0:main:ch6:theorem-6.53	DOC-MAIN p.184; Theorem 6.53	Theorem / proved	Finite retained-bar coverage theorem
ak-hpdst:21.0.0:main:ch6:corollary-6.54	DOC-MAIN p.184; Corollary 6.54	Corollary / proved	Countable detector-family zero criterion
ak-hpdst:21.0.0:main:ch6:theorem-6.55	DOC-MAIN p.185; Theorem 6.55	Theorem / proved	Detector-atlas local-to-global zero theorem
ak-hpdst:21.0.0:main:ch6:definition-6.58	DOC-MAIN p.186; Definition 6.58	Definition / operational	Restart datum
ak-hpdst:21.0.0:main:ch6:proposition-6.59	DOC-MAIN p.186; Proposition 6.59	Proposition / proved	Restart evidence monotonicity
ak-hpdst:21.0.0:main:ch6:definition-6.60	DOC-MAIN p.186; Definition 6.60	Definition / operational	Continuation artifact
ak-hpdst:21.0.0:main:ch6:definition-6.61	DOC-MAIN p.186; Definition 6.61	Definition / operational	◆◆◆◆-admissible continuation
ak-hpdst:21.0.0:main:ch6:definition-6.64	DOC-MAIN p.187; Definition 6.64	Definition / operational	Restart transition status
ak-hpdst:21.0.0:main:ch6:theorem-6.65	DOC-MAIN p.187; Theorem 6.65	Theorem / proved	Finite-horizon Restart bound
ak-hpdst:21.0.0:main:ch6:corollary-6.66	DOC-MAIN p.188; Corollary 6.66	Corollary / proved	Non-degrading continuation
ak-hpdst:21.0.0:main:ch6:definition-6.67	DOC-MAIN p.188; Definition 6.67	Definition / operational	Normalized Restart loss
ak-hpdst:21.0.0:main:ch6:corollary-6.68	DOC-MAIN p.188; Corollary 6.68	Corollary / proved	Normalized reserve criterion
ak-hpdst:21.0.0:main:ch6:definition-6.72	DOC-MAIN p.189; Definition 6.72	Definition / operational	◆◆-normalized summability
ak-hpdst:21.0.0:main:ch6:definition-6.73	DOC-MAIN p.189; Definition 6.73	Definition / operational	Countable Restart specification and hidden-tail prevention
ak-hpdst:21.0.0:main:ch6:theorem-6.74	DOC-MAIN p.189; Theorem 6.74	Theorem / proved	Conditional countable Restart principle
ak-hpdst:21.0.0:main:ch6:corollary-6.75	DOC-MAIN p.190; Corollary 6.75	Corollary / proved	Pointwise reserve and uniform reserve are distinct
ak-hpdst:21.0.0:main:ch6:definition-6.77	DOC-MAIN p.190; Definition 6.77	Definition / operational	Detector-aware Restart handoff
ak-hpdst:21.0.0:main:ch6:theorem-6.78	DOC-MAIN p.190; Theorem 6.78	Theorem / proved	Restart zero-transport theorem
ak-hpdst:21.0.0:main:ch6:theorem-6.79	DOC-MAIN p.191; Theorem 6.79	Theorem / proved	Restart record-transport theorem

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:main:ch6:definition-6.80	DOC-MAIN p.191; Definition 6.80	Definition / operational	Boundary-bridge coverage for a partition
ak-hpdst:21.0.0:main:ch6:corollary-6.81	DOC-MAIN p.191; Corollary 6.81	Corollary / proved	Detector-safe partition globalization
ak-hpdst:21.0.0:main:ch6:definition-6.83	DOC-MAIN p.191; Definition 6.83	Definition / operational	Audit-locally finite cover
ak-hpdst:21.0.0:main:ch6:definition-6.84	DOC-MAIN p.191; Definition 6.84	Definition / operational	Countable atlas enumeration and hidden-tail record
ak-hpdst:21.0.0:main:ch6:theorem-6.85	DOC-MAIN p.192; Theorem 6.85	Theorem / proved	Countable semantic record-coherence principle
ak-hpdst:21.0.0:main:ch6:definition-6.86	DOC-MAIN p.192; Definition 6.86	Definition / operational	Summable simplex exposure
ak-hpdst:21.0.0:main:ch6:theorem-6.87	DOC-MAIN p.192; Theorem 6.87	Theorem / proved	Budgeted countable record-atlas principle
ak-hpdst:21.0.0:main:ch6:definition-6.91	DOC-MAIN p.193; Definition 6.91	Definition / operational	Object-level descent interface
ak-hpdst:21.0.0:main:ch6:definition-6.92	DOC-MAIN p.193; Definition 6.92	Definition / operational	Object-descent eligibility record
ak-hpdst:21.0.0:main:ch6:theorem-6.93	DOC-MAIN p.194; Theorem 6.93	Theorem / proved	Object-descent handoff theorem
ak-hpdst:21.0.0:main:ch6:definition-6.95	DOC-MAIN p.194; Definition 6.95	Definition / operational	Local certificate status
ak-hpdst:21.0.0:main:ch6:proposition-6.96	DOC-MAIN p.194; Proposition 6.96	Proposition / proved	Compatibility does not repair a local failure
ak-hpdst:21.0.0:main:ch6:definition-6.100	DOC-MAIN p.195; Definition 6.100	Definition / operational	Chapter 6 failure routing
ak-hpdst:21.0.0:main:ch6:proposition-6.97	DOC-MAIN p.195; Proposition 6.97	Proposition / proved	Typed Type IV participation
ak-hpdst:21.0.0:main:ch6:proposition-6.98	DOC-MAIN p.195; Proposition 6.98	Proposition / proved	Inverse-audit participation
ak-hpdst:21.0.0:main:ch6:proposition-6.99	DOC-MAIN p.195; Proposition 6.99	Proposition / proved	Transient-defect participation and nontransport
ak-hpdst:21.0.0:main:ch6:proposition-6.102	DOC-MAIN p.196; Proposition 6.102	Proposition / proved	Window/overlap/Restart manifest extension
ak-hpdst:21.0.0:main:ch6:theorem-6.104	DOC-MAIN p.197; Theorem 6.104	Theorem / proved	Conditional window/overlap/Restart closure package
ak-hpdst:21.0.0:main:ch7:definition-7.1	DOC-MAIN p.202; Definition 7.1	Definition / operational	Core-DG admissible input
ak-hpdst:21.0.0:main:ch7:definition-7.3	DOC-MAIN p.203; Definition 7.3	Definition / operational	Finite category presentation
ak-hpdst:21.0.0:main:ch7:definition-7.4	DOC-MAIN p.203; Definition 7.4	Definition / operational	Finite-poset category
ak-hpdst:21.0.0:main:ch7:definition-7.5	DOC-MAIN p.203; Definition 7.5	Definition / operational	Upper Alexandrov convention
ak-hpdst:21.0.0:main:ch7:definition-7.6	DOC-MAIN p.203; Definition 7.6	Definition / operational	Finite diagram category
ak-hpdst:21.0.0:main:ch7:theorem-7.7	DOC-MAIN p.203; Theorem 7.7	Theorem / proved	Abelian structure of finite diagrams
ak-hpdst:21.0.0:main:ch7:definition-7.10	DOC-MAIN p.204; Definition 7.10	Definition / operational	Linearized representable
ak-hpdst:21.0.0:main:ch7:proposition-7.11	DOC-MAIN p.204; Proposition 7.11	Proposition / proved	Finite Yoneda evaluation
ak-hpdst:21.0.0:main:ch7:definition-7.12	DOC-MAIN p.204; Definition 7.12	Definition / operational	Finite projective generator
ak-hpdst:21.0.0:main:ch7:proposition-7.13	DOC-MAIN p.204; Proposition 7.13	Proposition / proved	Generator property
ak-hpdst:21.0.0:main:ch7:theorem-7.8	DOC-MAIN p.204; Theorem 7.8	Theorem / proved	Complexes of diagrams are diagrams of complexes
ak-hpdst:21.0.0:main:ch7:definition-7.9	DOC-MAIN p.204; Definition 7.9	Definition / operational	Bounded derived finite-diagram category
ak-hpdst:21.0.0:main:ch7:theorem-7.14	DOC-MAIN p.205; Theorem 7.14	Theorem / proved	DG-1: generator zero-reflection
ak-hpdst:21.0.0:main:ch7:theorem-7.16	DOC-MAIN p.205; Theorem 7.16	Theorem / proved	DG-2: cone-isomorphism criterion
ak-hpdst:21.0.0:main:ch7:corollary-7.17	DOC-MAIN p.205; Corollary 7.17	Corollary / proved	Generator detection of morphism isomorphism
ak-hpdst:21.0.0:main:ch7:definition-7.18	DOC-MAIN p.206; Definition 7.18	Definition / operational	Finite category algebra and handedness
ak-hpdst:21.0.0:main:ch7:definition-7.19	DOC-MAIN p.206; Definition 7.19	Definition / operational	Enriched projective probe
ak-hpdst:21.0.0:main:ch7:definition-7.20	DOC-MAIN p.206; Definition 7.20	Definition / operational	Scalar probe
ak-hpdst:21.0.0:main:ch7:theorem-7.21	DOC-MAIN p.206; Theorem 7.21	Theorem / proved	DG-3: finite Morita reconstruction
ak-hpdst:21.0.0:main:ch7:theorem-7.23	DOC-MAIN p.207; Theorem 7.23	Theorem / proved	Exact enriched-probe reconstruction
ak-hpdst:21.0.0:main:ch7:definition-7.26	DOC-MAIN p.207; Definition 7.26	Definition / operational	Finite-dimensional Alexandrov sheaf category
ak-hpdst:21.0.0:main:ch7:theorem-7.27	DOC-MAIN p.207; Theorem 7.27	Theorem / proved	DG-4: finite Alexandrov realization
ak-hpdst:21.0.0:main:ch7:corollary-7.28	DOC-MAIN p.208; Corollary 7.28	Corollary / proved	Derived Alexandrov realization
ak-hpdst:21.0.0:main:ch7:definition-7.30	DOC-MAIN p.208; Definition 7.30	Definition / operational	Finite open-cover descent datum
ak-hpdst:21.0.0:main:ch7:theorem-7.31	DOC-MAIN p.208; Theorem 7.31	Theorem / proved	DG-5: effective finite Alexandrov descent
ak-hpdst:21.0.0:main:ch7:definition-7.33	DOC-MAIN p.209; Definition 7.33	Definition / operational	Strict bounded-complex descent datum
ak-hpdst:21.0.0:main:ch7:theorem-7.34	DOC-MAIN p.209; Theorem 7.34	Theorem / proved	Strict bounded-complex descent
ak-hpdst:21.0.0:main:ch7:definition-7.35	DOC-MAIN p.209; Definition 7.35	Definition / operational	Model-certified derived descent lane
ak-hpdst:21.0.0:main:ch7:theorem-7.36	DOC-MAIN p.209; Theorem 7.36	Theorem / proved	Chapter 6 to Chapter 7 descent application
ak-hpdst:21.0.0:main:ch7:definition-7.37	DOC-MAIN p.210; Definition 7.37	Definition / operational	Finite Čech computation datum
ak-hpdst:21.0.0:main:ch7:theorem-7.38	DOC-MAIN p.210; Theorem 7.38	Theorem / proved	Finite acyclic-cover totalization
ak-hpdst:21.0.0:main:ch7:proposition-7.41	DOC-MAIN p.210; Proposition 7.41	Proposition / proved	Joint conservativity of finite Alexandrov open restriction
ak-hpdst:21.0.0:main:ch7:theorem-7.42	DOC-MAIN p.211; Theorem 7.42	Theorem / proved	DG-6a: global zero certificate
ak-hpdst:21.0.0:main:ch7:theorem-7.43	DOC-MAIN p.211; Theorem 7.43	Theorem / proved	DG-6b: global isomorphism certificate
ak-hpdst:21.0.0:main:ch7:proposition-7.45	DOC-MAIN p.211; Proposition 7.45	Proposition / proved	No local-zero twisting in the admitted descent lane
ak-hpdst:21.0.0:main:ch7:definition-7.46	DOC-MAIN p.211; Definition 7.46	Definition / operational	Family-relative global certificate cost model
ak-hpdst:21.0.0:main:ch7:theorem-7.47	DOC-MAIN p.212; Theorem 7.47	Theorem / proved	DG-7: model-relative global certificate upper bound
ak-hpdst:21.0.0:main:ch7:definition-7.53	DOC-MAIN p.213; Definition 7.53	Definition / operational	General finite-diagram handoff record
ak-hpdst:21.0.0:main:ch7:theorem-7.54	DOC-MAIN p.213; Theorem 7.54	Theorem / proved	Finite Diagram and Globalization Core package
ak-hpdst:21.0.0:main:ch7:definition-7.57	DOC-MAIN p.216; Definition 7.57	Definition / operational	Canonical token-sort registry
ak-hpdst:21.0.0:main:ch7:proposition-7.58	DOC-MAIN p.218; Proposition 7.58	Proposition / proved	No cross-sort token substitution
ak-hpdst:21.0.0:main:ch7:definition-7.59	DOC-MAIN p.218; Definition 7.59	Definition / operational	Audit clause identity
ak-hpdst:21.0.0:main:ch7:definition-7.60	DOC-MAIN p.218; Definition 7.60	Definition / operational	Typed evidence record
ak-hpdst:21.0.0:main:ch7:definition-7.61	DOC-MAIN p.218; Definition 7.61	Definition / operational	Well-founded audit dependency graph
ak-hpdst:21.0.0:main:ch7:proposition-7.62	DOC-MAIN p.218; Proposition 7.62	Proposition / proved	Determinacy on a complete well-founded graph
ak-hpdst:21.0.0:main:ch7:definition-7.63	DOC-MAIN p.219; Definition 7.63	Definition / operational	Claim supersession and conditional override record
ak-hpdst:21.0.0:main:ch7:proposition-7.64	DOC-MAIN p.219; Proposition 7.64	Proposition / proved	No silent claim supersession
ak-hpdst:21.0.0:main:ch7:definition-7.65	DOC-MAIN p.219; Definition 7.65	Definition / operational	Audit run identity
ak-hpdst:21.0.0:main:ch7:definition-7.66	DOC-MAIN p.219; Definition 7.66	Definition / operational	Immutable audit scope
ak-hpdst:21.0.0:main:ch7:definition-7.68	DOC-MAIN p.220; Definition 7.68	Definition / operational	Declared ledger algebra
ak-hpdst:21.0.0:main:ch7:definition-7.69	DOC-MAIN p.220; Definition 7.69	Definition / operational	Metric-bearing ledger components
ak-hpdst:21.0.0:main:ch7:definition-7.70	DOC-MAIN p.220; Definition 7.70	Definition / operational	Bottleneck scalarization
ak-hpdst:21.0.0:main:ch7:definition-7.72	DOC-MAIN p.220; Definition 7.72	Definition / operational	Non-scalarizable proof obligations
ak-hpdst:21.0.0:main:ch7:proposition-7.73	DOC-MAIN p.221; Proposition 7.73	Proposition / proved	No scalar repair of missing evidence
ak-hpdst:21.0.0:main:ch7:definition-7.74	DOC-MAIN p.221; Definition 7.74	Definition / operational	Gate-scoped threshold reserve
ak-hpdst:21.0.0:main:ch7:definition-7.76	DOC-MAIN p.221; Definition 7.76	Definition / operational	Ledger completeness
ak-hpdst:21.0.0:main:ch7:definition-7.77	DOC-MAIN p.222; Definition 7.77	Definition / operational	Claim-use status
ak-hpdst:21.0.0:main:ch7:definition-7.78	DOC-MAIN p.222; Definition 7.78	Definition / operational	Claim invocation role
ak-hpdst:21.0.0:main:ch7:definition-7.79	DOC-MAIN p.222; Definition 7.79	Definition / operational	Audit preflight
ak-hpdst:21.0.0:main:ch7:definition-7.80	DOC-MAIN p.223; Definition 7.80	Definition / operational	Typed preflight verdict
ak-hpdst:21.0.0:main:ch7:proposition-7.82	DOC-MAIN p.223; Proposition 7.82	Proposition / proved	Preflight failure blocks pass
ak-hpdst:21.0.0:main:ch7:definition-7.83	DOC-MAIN p.223; Definition 7.83	Definition / operational	Atomic audit status
ak-hpdst:21.0.0:main:ch7:definition-7.85	DOC-MAIN p.223; Definition 7.85	Definition / operational	Canonical specialization maps
ak-hpdst:21.0.0:main:ch7:definition-7.86	DOC-MAIN p.224; Definition 7.86	Definition / operational	Aggregate audit verdict
ak-hpdst:21.0.0:main:ch7:proposition-7.87	DOC-MAIN p.225; Proposition 7.87	Proposition / proved	Determinacy of complete records
ak-hpdst:21.0.0:main:ch7:proposition-7.88	DOC-MAIN p.225; Proposition 7.88	Proposition / proved	Undefined is neither success nor counterexample
ak-hpdst:21.0.0:main:ch7:definition-7.89	DOC-MAIN p.225; Definition 7.89	Definition / operational	Comparison-mode audit record
ak-hpdst:21.0.0:main:ch7:definition-7.90	DOC-MAIN p.225; Definition 7.90	Definition / operational	Detector-clause audit verdict
ak-hpdst:21.0.0:main:ch7:definition-7.91	DOC-MAIN p.226; Definition 7.91	Definition / operational	Degreewise higher-obstruction audit family
ak-hpdst:21.0.0:main:ch7:definition-7.92	DOC-MAIN p.226; Definition 7.92	Definition / operational	Terminal and transient defect audit
ak-hpdst:21.0.0:main:ch7:definition-7.93	DOC-MAIN p.226; Definition 7.93	Definition / operational	Global detector-atlas verdict
ak-hpdst:21.0.0:main:ch7:definition-7.94	DOC-MAIN p.226; Definition 7.94	Definition / operational	Finite-to-infinite reduction verdict
ak-hpdst:21.0.0:main:ch7:definition-7.95	DOC-MAIN p.226; Definition 7.95	Definition / operational	Object-descent and Return-Theorem verdicts
ak-hpdst:21.0.0:main:ch7:proposition-7.96	DOC-MAIN p.226; Proposition 7.96	Proposition / proved	Separation of audit conclusion levels
ak-hpdst:21.0.0:main:ch7:definition-7.97	DOC-MAIN p.227; Definition 7.97	Definition / operational	B-Gate+ audit clause set
ak-hpdst:21.0.0:main:ch7:definition-7.100	DOC-MAIN p.228; Definition 7.100	Definition / operational	Core audit reject
ak-hpdst:21.0.0:main:ch7:definition-7.101	DOC-MAIN p.228; Definition 7.101	Definition / operational	Core audit undefined
ak-hpdst:21.0.0:main:ch7:proposition-7.102	DOC-MAIN p.228; Proposition 7.102	Proposition / proved	Soundness of the pass reading
ak-hpdst:21.0.0:main:ch7:proposition-7.103	DOC-MAIN p.228; Proposition 7.103	Proposition / proved	No pass by cancellation
ak-hpdst:21.0.0:main:ch7:definition-7.104	DOC-MAIN p.228; Definition 7.104	Definition / operational	Local certificate verdict
ak-hpdst:21.0.0:main:ch7:definition-7.105	DOC-MAIN p.228; Definition 7.105	Definition / operational	Compatibility verdict
ak-hpdst:21.0.0:main:ch7:definition-7.99	DOC-MAIN p.228; Definition 7.99	Definition / operational	Core audit pass
ak-hpdst:21.0.0:main:ch7:definition-7.106	DOC-MAIN p.228; Definition 7.106	Definition / operational	Restart verdict
ak-hpdst:21.0.0:main:ch7:proposition-7.107	DOC-MAIN p.229; Proposition 7.107	Proposition / proved	Atlas composition rule
ak-hpdst:21.0.0:main:ch7:definition-7.111	DOC-MAIN p.229; Definition 7.111	Definition / operational	Manifest status
ak-hpdst:21.0.0:main:ch7:definition-7.112	DOC-MAIN p.230; Definition 7.112	Definition / operational	Canonical minimal v20 audit manifest
ak-hpdst:21.0.0:main:ch7:definition-7.113	DOC-MAIN p.231; Definition 7.113	Definition / operational	Artifact closure
ak-hpdst:21.0.0:main:ch7:definition-7.114	DOC-MAIN p.231; Definition 7.114	Definition / operational	Semantic schema-migration record
ak-hpdst:21.0.0:main:ch7:proposition-7.115	DOC-MAIN p.231; Proposition 7.115	Proposition / proved	No lossy migration of theorem evidence
ak-hpdst:21.0.0:main:ch7:proposition-7.116	DOC-MAIN p.231; Proposition 7.116	Proposition / proved	Canonical manifest conformance
ak-hpdst:21.0.0:main:ch7:definition-7.117	DOC-MAIN p.231; Definition 7.117	Definition / operational	Computational and semantic replayability
ak-hpdst:21.0.0:main:ch7:proposition-7.118	DOC-MAIN p.232; Proposition 7.118	Proposition / proved	Replay invariance under fixed inputs
ak-hpdst:21.0.0:main:ch7:proposition-7.120	DOC-MAIN p.232; Proposition 7.120	Proposition / proved	Manifest effect on the headline verdict
ak-hpdst:21.0.0:main:ch7:definition-7.121	DOC-MAIN p.232; Definition 7.121	Definition / operational	Audit failure-routing record
ak-hpdst:21.0.0:main:ch7:proposition-7.123	DOC-MAIN p.232; Proposition 7.123	Proposition / proved	Repair creates a new verdict record
ak-hpdst:21.0.0:main:ch7:definition-7.124	DOC-MAIN p.233; Definition 7.124	Definition / operational	Core sovereignty
ak-hpdst:21.0.0:main:ch7:definition-7.125	DOC-MAIN p.233; Definition 7.125	Definition / operational	Proposer and verifier
ak-hpdst:21.0.0:main:ch7:definition-7.126	DOC-MAIN p.233; Definition 7.126	Definition / operational	Re-entry protocol
ak-hpdst:21.0.0:main:ch7:proposition-7.127	DOC-MAIN p.233; Proposition 7.127	Proposition / proved	Search and AI outputs do not update verdicts
ak-hpdst:21.0.0:main:ch7:proposition-7.128	DOC-MAIN p.233; Proposition 7.128	Proposition / proved	Verifier independence
ak-hpdst:21.0.0:main:ch7:definition-7.129	DOC-MAIN p.233; Definition 7.129	Definition / operational	v20 release-acceptance record
ak-hpdst:21.0.0:main:ch7:definition-7.130	DOC-MAIN p.233; Definition 7.130	Definition / operational	Release headline status
ak-hpdst:21.0.0:main:ch7:proposition-7.131	DOC-MAIN p.234; Proposition 7.131	Proposition / proved	No release completion by local success
ak-hpdst:21.0.0:main:ch7:proposition-7.132	DOC-MAIN p.234; Proposition 7.132	Proposition / proved	Typed audit-and-sovereignty contract consequences
ak-hpdst:21.0.0:main:ch8:definition-8.3	DOC-MAIN p.239; Definition 8.3	Definition / operational	Verifier-side datum
ak-hpdst:21.0.0:main:ch8:definition-8.4	DOC-MAIN p.239; Definition 8.4	Definition / operational	Verifier input admissibility
ak-hpdst:21.0.0:main:ch8:definition-8.5	DOC-MAIN p.239; Definition 8.5	Definition / operational	Independent statement-status axes

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:main:ch8:definition-8.10	DOC-MAIN p.240; Definition 8.10	Definition / operational	Declared obligation registry
ak-hpdst:21.0.0:main:ch8:definition-8.11	DOC-MAIN p.240; Definition 8.11	Definition / operational	Certificate-readiness bundle
ak-hpdst:21.0.0:main:ch8:proposition-8.6	DOC-MAIN p.240; Proposition 8.6	Proposition / proved	No cross-axis promotion
ak-hpdst:21.0.0:main:ch8:definition-8.7	DOC-MAIN p.240; Definition 8.7	Definition / operational	Claim-use preflight
ak-hpdst:21.0.0:main:ch8:proposition-8.8	DOC-MAIN p.240; Proposition 8.8	Proposition / proved	Preflight blocks unsupported theorem use
ak-hpdst:21.0.0:main:ch8:definition-8.9	DOC-MAIN p.240; Definition 8.9	Definition / operational	Typed obligation atom
ak-hpdst:21.0.0:main:ch8:definition-8.13	DOC-MAIN p.241; Definition 8.13	Definition / operational	Canonical atomic status
ak-hpdst:21.0.0:main:ch8:definition-8.14	DOC-MAIN p.241; Definition 8.14	Definition / operational	Scope-exclusion record
ak-hpdst:21.0.0:main:ch8:definition-8.16	DOC-MAIN p.241; Definition 8.16	Definition / operational	Aggregate verifier verdict
ak-hpdst:21.0.0:main:ch8:theorem-8.17	DOC-MAIN p.241; Theorem 8.17	Theorem / proved	Non-compensation theorem
ak-hpdst:21.0.0:main:ch8:corollary-8.18	DOC-MAIN p.241; Corollary 8.18	Corollary / proved	No proof by majority or consensus
ak-hpdst:21.0.0:main:ch8:definition-8.19	DOC-MAIN p.242; Definition 8.19	Definition / operational	Canonical comparison modes
ak-hpdst:21.0.0:main:ch8:definition-8.20	DOC-MAIN p.242; Definition 8.20	Definition / operational	Claim-mode compatibility profile
ak-hpdst:21.0.0:main:ch8:definition-8.21	DOC-MAIN p.242; Definition 8.21	Definition / operational	Comparison-mode compatibility clause
ak-hpdst:21.0.0:main:ch8:theorem-8.22	DOC-MAIN p.242; Theorem 8.22	Theorem / proved	No comparison-mode substitution
ak-hpdst:21.0.0:main:ch8:definition-8.23	DOC-MAIN p.242; Definition 8.23	Definition / operational	Canonical non-scalar obligations
ak-hpdst:21.0.0:main:ch8:definition-8.24	DOC-MAIN p.243; Definition 8.24	Definition / operational	Declared metric ledger
ak-hpdst:21.0.0:main:ch8:definition-8.25	DOC-MAIN p.243; Definition 8.25	Definition / operational	Gate-scoped metric budget
ak-hpdst:21.0.0:main:ch8:theorem-8.26	DOC-MAIN p.243; Theorem 8.26	Theorem / proved	No scalar manufacture of non-scalar evidence
ak-hpdst:21.0.0:main:ch8:definition-8.27	DOC-MAIN p.243; Definition 8.27	Definition / operational	No-double-counting invariant
ak-hpdst:21.0.0:main:ch8:definition-8.28	DOC-MAIN p.243; Definition 8.28	Definition / operational	Proof-DAG
ak-hpdst:21.0.0:main:ch8:definition-8.29	DOC-MAIN p.244; Definition 8.29	Definition / operational	Canonical proof-bearing node types
ak-hpdst:21.0.0:main:ch8:definition-8.30	DOC-MAIN p.244; Definition 8.30	Definition / operational	Node semantic record
ak-hpdst:21.0.0:main:ch8:definition-8.31	DOC-MAIN p.244; Definition 8.31	Definition / operational	Primitive and derived nodes
ak-hpdst:21.0.0:main:ch8:definition-8.32	DOC-MAIN p.244; Definition 8.32	Definition / operational	Assumption context and conditionality propagation
ak-hpdst:21.0.0:main:ch8:proposition-8.33	DOC-MAIN p.244; Proposition 8.33	Proposition / proved	No silent erasure of assumptions
ak-hpdst:21.0.0:main:ch8:definition-8.34	DOC-MAIN p.245; Definition 8.34	Definition / operational	Necessary target sub-DAG
ak-hpdst:21.0.0:main:ch8:definition-8.36	DOC-MAIN p.245; Definition 8.36	Definition / operational	Node discharge certificate
ak-hpdst:21.0.0:main:ch8:proposition-8.37	DOC-MAIN p.245; Proposition 8.37	Proposition / proved	Node existence does not discharge the node
ak-hpdst:21.0.0:main:ch8:definition-8.38	DOC-MAIN p.245; Definition 8.38	Definition / operational	Canonical edge semantics
ak-hpdst:21.0.0:main:ch8:definition-8.39	DOC-MAIN p.246; Definition 8.39	Definition / operational	Edge semantic record
ak-hpdst:21.0.0:main:ch8:definition-8.40	DOC-MAIN p.246; Definition 8.40	Definition / operational	Edge compatibility
ak-hpdst:21.0.0:main:ch8:theorem-8.41	DOC-MAIN p.246; Theorem 8.41	Theorem / proved	Misrouted edges do not discharge dependencies
ak-hpdst:21.0.0:main:ch8:definition-8.42	DOC-MAIN p.246; Definition 8.42	Definition / operational	Weakening edge
ak-hpdst:21.0.0:main:ch8:proposition-8.43	DOC-MAIN p.246; Proposition 8.43	Proposition / proved	No reversal of weakening
ak-hpdst:21.0.0:main:ch8:definition-8.44	DOC-MAIN p.246; Definition 8.44	Definition / operational	Edge composition policy
ak-hpdst:21.0.0:main:ch8:proposition-8.45	DOC-MAIN p.246; Proposition 8.45	Proposition / proved	Graph reachability is not theorem composition
ak-hpdst:21.0.0:main:ch8:definition-8.46	DOC-MAIN p.246; Definition 8.46	Definition / operational	Ordinary dependency acyclicity
ak-hpdst:21.0.0:main:ch8:definition-8.47	DOC-MAIN p.247; Definition 8.47	Definition / operational	Well-founded dependency rank
ak-hpdst:21.0.0:main:ch8:definition-8.48	DOC-MAIN p.247; Definition 8.48	Definition / operational	Controlled recursive theorem node
ak-hpdst:21.0.0:main:ch8:proposition-8.49	DOC-MAIN p.247; Proposition 8.49	Proposition / proved	Unlicensed cycle blocks certification
ak-hpdst:21.0.0:main:ch8:definition-8.50	DOC-MAIN p.247; Definition 8.50	Definition / operational	Countable dependency certificate
ak-hpdst:21.0.0:main:ch8:theorem-8.51	DOC-MAIN p.247; Theorem 8.51	Theorem / proved	Determinacy on a well-founded typed DAG
ak-hpdst:21.0.0:main:ch8:definition-8.52	DOC-MAIN p.247; Definition 8.52	Definition / operational	Diagnostic failure precedence
ak-hpdst:21.0.0:main:ch8:definition-8.53	DOC-MAIN p.248; Definition 8.53	Definition / operational	Failure-routing record
ak-hpdst:21.0.0:main:ch8:proposition-8.54	DOC-MAIN p.248; Proposition 8.54	Proposition / proved	Typing precedes obstruction
ak-hpdst:21.0.0:main:ch8:proposition-8.55	DOC-MAIN p.248; Proposition 8.55	Proposition / proved	Failure precedence is not status cancellation
ak-hpdst:21.0.0:main:ch8:definition-8.56	DOC-MAIN p.248; Definition 8.56	Definition / operational	Source-statement binding
ak-hpdst:21.0.0:main:ch8:definition-8.57	DOC-MAIN p.248; Definition 8.57	Definition / operational	Precursor package
ak-hpdst:21.0.0:main:ch8:definition-8.58	DOC-MAIN p.248; Definition 8.58	Definition / operational	Benchmark conclusion
ak-hpdst:21.0.0:main:ch8:definition-8.59	DOC-MAIN p.249; Definition 8.59	Definition / operational	Benchmark-quarantine cut
ak-hpdst:21.0.0:main:ch8:definition-8.60	DOC-MAIN p.249; Definition 8.60	Definition / operational	Permitted quarantine crossings
ak-hpdst:21.0.0:main:ch8:theorem-8.61	DOC-MAIN p.249; Theorem 8.61	Theorem / proved	Benchmark leakage invalidates independent recovery
ak-hpdst:21.0.0:main:ch8:proposition-8.62	DOC-MAIN p.249; Proposition 8.62	Proposition / proved	Source storage is not theorem recovery
ak-hpdst:21.0.0:main:ch8:definition-8.63	DOC-MAIN p.249; Definition 8.63	Definition / operational	Recovery sub-DAG
ak-hpdst:21.0.0:main:ch8:definition-8.64	DOC-MAIN p.249; Definition 8.64	Definition / operational	Immutable artifact identity
ak-hpdst:21.0.0:main:ch8:proposition-8.65	DOC-MAIN p.249; Proposition 8.65	Proposition / proved	Hash identity supports identity, not truth
ak-hpdst:21.0.0:main:ch8:definition-8.66	DOC-MAIN p.250; Definition 8.66	Definition / operational	Cross-format semantic-equivalence record
ak-hpdst:21.0.0:main:ch8:definition-8.67	DOC-MAIN p.250; Definition 8.67	Definition / operational	Artifact closure
ak-hpdst:21.0.0:main:ch8:definition-8.68	DOC-MAIN p.250; Definition 8.68	Definition / operational	Canonical v21 verifier manifest
ak-hpdst:21.0.0:main:ch8:definition-8.69	DOC-MAIN p.250; Definition 8.69	Definition / operational	Manifest status
ak-hpdst:21.0.0:main:ch8:proposition-8.70	DOC-MAIN p.250; Proposition 8.70	Proposition / proved	Manifest status specialization
ak-hpdst:21.0.0:main:ch8:definition-8.71	DOC-MAIN p.251; Definition 8.71	Definition / operational	Semantic-preserving schema migration
ak-hpdst:21.0.0:main:ch8:proposition-8.72	DOC-MAIN p.251; Proposition 8.72	Proposition / proved	No lossy migration of proof evidence
ak-hpdst:21.0.0:main:ch8:definition-8.73	DOC-MAIN p.251; Definition 8.73	Definition / operational	Agent roles
ak-hpdst:21.0.0:main:ch8:definition-8.74	DOC-MAIN p.251; Definition 8.74	Definition / operational	Authority matrix
ak-hpdst:21.0.0:main:ch8:definition-8.75	DOC-MAIN p.251; Definition 8.75	Definition / operational	Proposer-to-verifier conversion certificate
ak-hpdst:21.0.0:main:ch8:proposition-8.76	DOC-MAIN p.252; Proposition 8.76	Proposition / proved	Agent assertion does not establish a Core node
ak-hpdst:21.0.0:main:ch8:proposition-8.77	DOC-MAIN p.252; Proposition 8.77	Proposition / proved	Agent consensus is non-compensating
ak-hpdst:21.0.0:main:ch8:definition-8.78	DOC-MAIN p.252; Definition 8.78	Definition / operational	Formalization and interpretation pair
ak-hpdst:21.0.0:main:ch8:proposition-8.79	DOC-MAIN p.252; Proposition 8.79	Proposition / proved	Formal verification does not silently prove the intended claim
ak-hpdst:21.0.0:main:ch8:definition-8.80	DOC-MAIN p.252; Definition 8.80	Definition / operational	Agent contamination event
ak-hpdst:21.0.0:main:ch8:proposition-8.81	DOC-MAIN p.252; Proposition 8.81	Proposition / proved	Contamination forces status downgrade
ak-hpdst:21.0.0:main:ch8:definition-8.82	DOC-MAIN p.252; Definition 8.82	Definition / operational	Replay boundary
ak-hpdst:21.0.0:main:ch8:definition-8.83	DOC-MAIN p.252; Definition 8.83	Definition / operational	Replay hierarchy
ak-hpdst:21.0.0:main:ch8:theorem-8.84	DOC-MAIN p.253; Theorem 8.84	Theorem / proved	Replay hierarchy is strict
ak-hpdst:21.0.0:main:ch8:definition-8.86	DOC-MAIN p.253; Definition 8.86	Definition / operational	Semantic replay certificate
ak-hpdst:21.0.0:main:ch8:proposition-8.87	DOC-MAIN p.253; Proposition 8.87	Proposition / proved	Replay disagreement blocks silent reuse
ak-hpdst:21.0.0:main:ch8:definition-8.88	DOC-MAIN p.253; Definition 8.88	Definition / operational	Immutable revision record
ak-hpdst:21.0.0:main:ch8:definition-8.89	DOC-MAIN p.253; Definition 8.89	Definition / operational	Impact closure
ak-hpdst:21.0.0:main:ch8:proposition-8.90	DOC-MAIN p.254; Proposition 8.90	Proposition / proved	Status does not automatically inherit across revisions
ak-hpdst:21.0.0:main:ch8:definition-8.91	DOC-MAIN p.254; Definition 8.91	Definition / operational	Repair relation
ak-hpdst:21.0.0:main:ch8:theorem-8.92	DOC-MAIN p.254; Theorem 8.92	Theorem / proved	Immutable-history theorem
ak-hpdst:21.0.0:main:ch8:definition-8.93	DOC-MAIN p.254; Definition 8.93	Definition / operational	Proper predecessor closure
ak-hpdst:21.0.0:main:ch8:definition-8.94	DOC-MAIN p.254; Definition 8.94	Definition / operational	Target derivation readiness
ak-hpdst:21.0.0:main:ch8:theorem-8.95	DOC-MAIN p.255; Theorem 8.95	Theorem / proved	Typed DAG closure theorem
ak-hpdst:21.0.0:main:ch8:corollary-8.96	DOC-MAIN p.255; Corollary 8.96	Corollary / proved	Graph existence is not typed closure
ak-hpdst:21.0.0:main:ch8:definition-8.97	DOC-MAIN p.255; Definition 8.97	Definition / operational	Route-admissible conclusion set
ak-hpdst:21.0.0:main:ch8:definition-8.98	DOC-MAIN p.255; Definition 8.98	Definition / operational	Route-admissible conclusion record
ak-hpdst:21.0.0:main:ch8:proposition-8.99	DOC-MAIN p.255; Proposition 8.99	Proposition / proved	No stronger conclusion than the route intersection
ak-hpdst:21.0.0:main:ch8:definition-8.100	DOC-MAIN p.256; Definition 8.100	Definition / operational	Unified Bridge verifier handoff
ak-hpdst:21.0.0:main:ch8:theorem-8.101	DOC-MAIN p.256; Theorem 8.101	Theorem / proved	Verifier handoff soundness
ak-hpdst:21.0.0:main:ch8:theorem-8.103	DOC-MAIN p.256; Theorem 8.103	Theorem / proved	v21 typed verification and governance kernel
ak-hpdst:21.0.0:main:ch9:definition-9.1	DOC-MAIN p.260; Definition 9.1	Definition / operational	Partial internalization policy
ak-hpdst:21.0.0:main:ch9:proposition-9.2	DOC-MAIN p.261; Proposition 9.2	Proposition / proved	Conservative promotion criterion
ak-hpdst:21.0.0:main:ch9:definition-9.4	DOC-MAIN p.261; Definition 9.4	Definition / operational	Universal v21 Bridge datum
ak-hpdst:21.0.0:main:ch9:definition-9.5	DOC-MAIN p.261; Definition 9.5	Definition / operational	Route classifier
ak-hpdst:21.0.0:main:ch9:definition-9.10	DOC-MAIN p.262; Definition 9.10	Definition / operational	Component compatibility
ak-hpdst:21.0.0:main:ch9:theorem-9.6	DOC-MAIN p.262; Theorem 9.6	Theorem / proved	Route classification and ownership
ak-hpdst:21.0.0:main:ch9:definition-9.7	DOC-MAIN p.262; Definition 9.7	Definition / operational	Target-relative Bridge strength preorder
ak-hpdst:21.0.0:main:ch9:proposition-9.8	DOC-MAIN p.262; Proposition 9.8	Proposition / proved	No comparison of unrelated partial Returns
ak-hpdst:21.0.0:main:ch9:definition-9.9	DOC-MAIN p.262; Definition 9.9	Definition / operational	Universal Bridge component
ak-hpdst:21.0.0:main:ch9:theorem-9.11	DOC-MAIN p.263; Theorem 9.11	Theorem / proved	Component compatibility theorem
ak-hpdst:21.0.0:main:ch9:definition-9.12	DOC-MAIN p.263; Definition 9.12	Definition / operational	Source realization component
ak-hpdst:21.0.0:main:ch9:theorem-9.13	DOC-MAIN p.263; Theorem 9.13	Theorem / proved	Faithfully flat zero and isomorphism reflection
ak-hpdst:21.0.0:main:ch9:theorem-9.14	DOC-MAIN p.263; Theorem 9.14	Theorem / proved	Derived Nakayama component
ak-hpdst:21.0.0:main:ch9:corollary-9.15	DOC-MAIN p.264; Corollary 9.15	Corollary / proved	Finite semikolal residue atlas
ak-hpdst:21.0.0:main:ch9:corollary-9.16	DOC-MAIN p.264; Corollary 9.16	Corollary / proved	Noetherian completion base-change component
ak-hpdst:21.0.0:main:ch9:definition-9.18	DOC-MAIN p.264; Definition 9.18	Definition / operational	Bockstein and universal-coefficient boundary component
ak-hpdst:21.0.0:main:ch9:definition-9.19	DOC-MAIN p.264; Definition 9.19	Definition / operational	Core-P component
ak-hpdst:21.0.0:main:ch9:definition-9.20	DOC-MAIN p.264; Definition 9.20	Definition / operational	Core-DG component
ak-hpdst:21.0.0:main:ch9:theorem-9.21	DOC-MAIN p.264; Theorem 9.21	Theorem / proved	P-to-DG Bridge component
ak-hpdst:21.0.0:main:ch9:definition-9.22	DOC-MAIN p.265; Definition 9.22	Definition / operational	HT sovereign components
ak-hpdst:21.0.0:main:ch9:theorem-9.23	DOC-MAIN p.265; Theorem 9.23	Theorem / proved	HC protected persistence component
ak-hpdst:21.0.0:main:ch9:definition-9.24	DOC-MAIN p.265; Definition 9.24	Definition / operational	PE component ladder
ak-hpdst:21.0.0:main:ch9:definition-9.25	DOC-MAIN p.265; Definition 9.25	Definition / operational	SS component
ak-hpdst:21.0.0:main:ch9:definition-9.26	DOC-MAIN p.265; Definition 9.26	Definition / operational	RL component
ak-hpdst:21.0.0:main:ch9:theorem-9.27	DOC-MAIN p.265; Theorem 9.27	Theorem / proved	HT transfer-spine composition
ak-hpdst:21.0.0:main:ch9:definition-9.28	DOC-MAIN p.266; Definition 9.28	Definition / operational	Tail component selector
ak-hpdst:21.0.0:main:ch9:theorem-9.29	DOC-MAIN p.266; Theorem 9.29	Theorem / proved	Tail component soundness
ak-hpdst:21.0.0:main:ch9:definition-9.30	DOC-MAIN p.266; Definition 9.30	Definition / operational	Descent/globalization component
ak-hpdst:21.0.0:main:ch9:theorem-9.31	DOC-MAIN p.266; Theorem 9.31	Theorem / proved	Finite descent Bridge
ak-hpdst:21.0.0:main:ch9:theorem-9.32	DOC-MAIN p.266; Theorem 9.32	Theorem / proved	Field Kunnet component
ak-hpdst:21.0.0:main:ch9:theorem-9.33	DOC-MAIN p.266; Theorem 9.33	Theorem / proved	Perfect duality and invertible tensor reflection
ak-hpdst:21.0.0:main:ch9:definition-9.34	DOC-MAIN p.267; Definition 9.34	Definition / operational	Determinant-level partial Return

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:main:ch9:definition-9.35	DOC-MAIN p.267; Definition 9.35	Definition / operational	Bridge coherence record
ak-hpdst:21.0.0:main:ch9:theorem-9.36	DOC-MAIN p.267; Theorem 9.36	Theorem / proved	Finite projective-presentation rigidity
ak-hpdst:21.0.0:main:ch9:definition-9.37	DOC-MAIN p.267; Definition 9.37	Definition / operational	Rigidity package
ak-hpdst:21.0.0:main:ch9:proposition-9.38	DOC-MAIN p.267; Proposition 9.38	Proposition / proved	No promotion without rigidity
ak-hpdst:21.0.0:main:ch9:definition-9.39	DOC-MAIN p.267; Definition 9.39	Definition / operational	Return target record
ak-hpdst:21.0.0:main:ch9:definition-9.40	DOC-MAIN p.268; Definition 9.40	Definition / operational	Partial Return
ak-hpdst:21.0.0:main:ch9:theorem-9.41	DOC-MAIN p.268; Theorem 9.41	Theorem / proved	Partial Return theorem
ak-hpdst:21.0.0:main:ch9:definition-9.42	DOC-MAIN p.268; Definition 9.42	Definition / operational	Replacement or regularized target datum
ak-hpdst:21.0.0:main:ch9:theorem-9.43	DOC-MAIN p.268; Theorem 9.43	Theorem / proved	Regularized-target soundness
ak-hpdst:21.0.0:main:ch9:definition-9.44	DOC-MAIN p.268; Definition 9.44	Definition / operational	Composable unified Bridge DAG
ak-hpdst:21.0.0:main:ch9:definition-9.45	DOC-MAIN p.268; Definition 9.45	Definition / operational	Unified Bridge defect stack
ak-hpdst:21.0.0:main:ch9:definition-9.46	DOC-MAIN p.269; Definition 9.46	Definition / operational	Bridge primary result
ak-hpdst:21.0.0:main:ch9:theorem-9.47	DOC-MAIN p.269; Theorem 9.47	Theorem / proved	Unified HT-UB Bridge Closure
ak-hpdst:21.0.0:main:ch9:corollary-9.48	DOC-MAIN p.270; Corollary 9.48	Corollary / proved	Typed Bridge-DAG theorem
ak-hpdst:21.0.0:main:ch9:theorem-9.49	DOC-MAIN p.270; Theorem 9.49	Theorem / proved	Non-composition theorem
ak-hpdst:21.0.0:main:ch9:theorem-9.50	DOC-MAIN p.270; Theorem 9.50	Theorem / proved	Return-ceiling and failure-leaf theorem
ak-hpdst:21.0.0:main:ch9:definition-9.51	DOC-MAIN p.270; Definition 9.51	Definition / operational	Finite filtered-chain source class
ak-hpdst:21.0.0:main:ch9:proposition-9.52	DOC-MAIN p.271; Proposition 9.52	Proposition / proved	Common endpoint-complete atom grid from source breakpoints
ak-hpdst:21.0.0:main:ch9:lemma-9.53	DOC-MAIN p.271; Lemma 9.53	Lemma / proved	Long-span criterion
ak-hpdst:21.0.0:main:ch9:theorem-9.54	DOC-MAIN p.271; Theorem 9.54	Theorem / proved	Complete HC-Core-P-Core-DG-HT/UB Return route
ak-hpdst:21.0.0:main:ch9:corollary-9.55	DOC-MAIN p.272; Corollary 9.55	Corollary / proved	Finite replay certificate for the complete route
ak-hpdst:21.0.0:main:ch9:definition-9.61	DOC-MAIN p.273; Definition 9.61	Definition / operational	Release-level claim classes
ak-hpdst:21.0.0:main:ch9:definition-9.62	DOC-MAIN p.274; Definition 9.62	Definition / operational	Claim invocation role
ak-hpdst:21.0.0:main:ch9:definition-9.63	DOC-MAIN p.274; Definition 9.63	Definition / operational	Safe-use order
ak-hpdst:21.0.0:main:ch9:definition-9.64	DOC-MAIN p.274; Definition 9.64	Definition / operational	Release token-sort conformance
ak-hpdst:21.0.0:main:ch9:proposition-9.65	DOC-MAIN p.274; Proposition 9.65	Proposition / proved	Status cannot be promoted by prose
ak-hpdst:21.0.0:main:ch9:definition-9.66	DOC-MAIN p.274; Definition 9.66	Definition / operational	Proved Core nucleus
ak-hpdst:21.0.0:main:ch9:proposition-9.67	DOC-MAIN p.274; Proposition 9.67	Proposition / proved	Contents of the proved Core nucleus
ak-hpdst:21.0.0:main:ch9:definition-9.69	DOC-MAIN p.276; Definition 9.69	Definition / operational	Conditional Core layer
ak-hpdst:21.0.0:main:ch9:definition-9.70	DOC-MAIN p.277; Definition 9.70	Definition / operational	Operational Core layer
ak-hpdst:21.0.0:main:ch9:definition-9.71	DOC-MAIN p.277; Definition 9.71	Definition / operational	Canonical comparison mode
ak-hpdst:21.0.0:main:ch9:proposition-9.72	DOC-MAIN p.277; Proposition 9.72	Proposition / proved	Conditional and operational layers do not self-promote
ak-hpdst:21.0.0:main:ch9:definition-9.74	DOC-MAIN p.278; Definition 9.74	Definition / operational	Eventually constant reduction package
ak-hpdst:21.0.0:main:ch9:theorem-9.75	DOC-MAIN p.278; Theorem 9.75	Theorem / proved	Bridge specialization: eventually constant tower with compatible apex agreement
ak-hpdst:21.0.0:main:ch9:corollary-9.76	DOC-MAIN p.278; Corollary 9.76	Corollary / proved	Vanishing of full comparison defects under reduction
ak-hpdst:21.0.0:main:ch9:definition-9.78	DOC-MAIN p.279; Definition 9.78	Definition / operational	Metric-repair and architecture rejection set
ak-hpdst:21.0.0:main:ch9:proposition-9.79	DOC-MAIN p.279; Proposition 9.79	Proposition / proved	Rejected claims cannot be repaired locally by a ledger
ak-hpdst:21.0.0:main:ch9:definition-9.80	DOC-MAIN p.280; Definition 9.80	Definition / operational	v21 Core closure contract
ak-hpdst:21.0.0:main:ch9:proposition-9.81	DOC-MAIN p.280; Proposition 9.81	Proposition / proved	Main-body Core closure
ak-hpdst:21.0.0:main:ch9:proposition-9.82	DOC-MAIN p.280; Proposition 9.82	Proposition / proved	Final-audit closure criterion
ak-hpdst:21.0.0:main:ch9:definition-9.83	DOC-MAIN p.280; Definition 9.83	Definition / operational	Main-body and release acceptance boundary
ak-hpdst:21.0.0:main:ch9:definition-9.85	DOC-MAIN p.281; Definition 9.85	Definition / operational	External source class
ak-hpdst:21.0.0:main:ch9:definition-9.86	DOC-MAIN p.281; Definition 9.86	Definition / operational	AK projection datum
ak-hpdst:21.0.0:main:ch9:definition-9.87	DOC-MAIN p.281; Definition 9.87	Definition / operational	Core-readable projected object
ak-hpdst:21.0.0:main:ch9:proposition-9.88	DOC-MAIN p.281; Proposition 9.88	Proposition / proved	Projection supplies input, not verdict
ak-hpdst:21.0.0:main:ch9:definition-9.90	DOC-MAIN p.281; Definition 9.90	Definition / operational	Purpose-faithful realization
ak-hpdst:21.0.0:main:ch9:definition-9.91	DOC-MAIN p.281; Definition 9.91	Definition / operational	Interface entry record
ak-hpdst:21.0.0:main:ch9:proposition-9.92	DOC-MAIN p.282; Proposition 9.92	Proposition / proved	No admissible Interface without typed entry
ak-hpdst:21.0.0:main:ch9:definition-9.94	DOC-MAIN p.282; Definition 9.94	Definition / operational	Detector realization theorem interface
ak-hpdst:21.0.0:main:ch9:definition-9.95	DOC-MAIN p.282; Definition 9.95	Definition / operational	Detector soundness Bridge
ak-hpdst:21.0.0:main:ch9:definition-9.96	DOC-MAIN p.282; Definition 9.96	Definition / operational	Detector completeness Bridge
ak-hpdst:21.0.0:main:ch9:definition-9.97	DOC-MAIN p.282; Definition 9.97	Definition / operational	Higher Ext realization theorem interface
ak-hpdst:21.0.0:main:ch9:definition-9.98	DOC-MAIN p.282; Definition 9.98	Definition / operational	Finite-to-infinite Interface record
ak-hpdst:21.0.0:main:ch9:proposition-9.100	DOC-MAIN p.283; Proposition 9.100	Proposition / proved	External origin does not relax Core hypotheses
ak-hpdst:21.0.0:main:ch9:definition-9.102	DOC-MAIN p.283; Definition 9.102	Definition / operational	Interface outcome
ak-hpdst:21.0.0:main:ch9:definition-9.103	DOC-MAIN p.283; Definition 9.103	Definition / operational	Core outcome for projected data
ak-hpdst:21.0.0:main:ch9:definition-9.104	DOC-MAIN p.283; Definition 9.104	Definition / operational	Bridge outcome
ak-hpdst:21.0.0:main:ch9:definition-9.105	DOC-MAIN p.283; Definition 9.105	Definition / operational	External-domain outcome
ak-hpdst:21.0.0:main:ch9:definition-9.106	DOC-MAIN p.283; Definition 9.106	Definition / operational	Finite-certificate outcome
ak-hpdst:21.0.0:main:ch9:definition-9.99	DOC-MAIN p.283; Definition 9.99	Definition / operational	Interface evaluation path
ak-hpdst:21.0.0:main:ch9:definition-9.107	DOC-MAIN p.284; Definition 9.107	Definition / operational	Reduction and Return outcomes
ak-hpdst:21.0.0:main:ch9:proposition-9.108	DOC-MAIN p.284; Proposition 9.108	Proposition / proved	No outcome-layer collapse
ak-hpdst:21.0.0:main:ch9:definition-9.109	DOC-MAIN p.284; Definition 9.109	Definition / operational	Interface-to-Core transport statement
ak-hpdst:21.0.0:main:ch9:definition-9.110	DOC-MAIN p.284; Definition 9.110	Definition / operational	Core-to-Interface transport statement
ak-hpdst:21.0.0:main:ch9:definition-9.111	DOC-MAIN p.284; Definition 9.111	Definition / operational	Transport theorem
ak-hpdst:21.0.0:main:ch9:definition-9.112	DOC-MAIN p.284; Definition 9.112	Definition / operational	External Return Theorem
ak-hpdst:21.0.0:main:ch9:definition-9.113	DOC-MAIN p.284; Definition 9.113	Definition / operational	Default status of transport statements
ak-hpdst:21.0.0:main:ch9:proposition-9.114	DOC-MAIN p.284; Proposition 9.114	Proposition / proved	No silent Bridge promotion
ak-hpdst:21.0.0:main:ch9:theorem-9.115	DOC-MAIN p.285; Theorem 9.115	Theorem / proved	Composed AK Return principle
ak-hpdst:21.0.0:main:ch9:definition-9.117	DOC-MAIN p.285; Definition 9.117	Definition / operational	Permitted adjacent support
ak-hpdst:21.0.0:main:ch9:proposition-9.118	DOC-MAIN p.285; Proposition 9.118	Proposition / proved	Adjacent material does not enlarge the Core silently
ak-hpdst:21.0.0:main:ch9:definition-9.120	DOC-MAIN p.285; Definition 9.120	Definition / operational	Known-theorem-recovery record
ak-hpdst:21.0.0:main:ch9:proposition-9.121	DOC-MAIN p.285; Proposition 9.121	Proposition / proved	Calibration is not promotion
ak-hpdst:21.0.0:main:ch9:definition-9.122	DOC-MAIN p.286; Definition 9.122	Definition / operational	v21 release-closure record
ak-hpdst:21.0.0:main:ch9:proposition-9.123	DOC-MAIN p.286; Proposition 9.123	Proposition / proved	Main closure does not predeclare release pass
ak-hpdst:21.0.0:main:ch9:definition-9.124	DOC-MAIN p.286; Definition 9.124	Definition / operational	v21 release-gate reflection matrix
ak-hpdst:21.0.0:main:ch9:proposition-9.125	DOC-MAIN p.287; Proposition 9.125	Proposition / proved	Mandatory and recommended release gates do not compensate
ak-hpdst:21.0.0:main:ch9:proposition-9.127	DOC-MAIN p.289; Proposition 9.127	Proposition / proved	Final architecture of the v21 main body
ak-hpdst:21.0.0:main:ch9:theorem-9.128	DOC-MAIN p.289; Theorem 9.128	Theorem / proved	Final AK projection-and-return principle
ak-hpdst:21.0.0:main:ch9:theorem-9.131	DOC-MAIN p.290; Theorem 9.131	Theorem / proved	v21 Unified Bridge and Return package
ak-hpdst:21.0.0:main:ch10:definition-10.1	DOC-MAIN p.291; Definition 10.1	Definition / operational	Validation taxonomy
ak-hpdst:21.0.0:main:ch10:definition-10.2	DOC-MAIN p.291; Definition 10.2	Definition / operational	Validation case record
ak-hpdst:21.0.0:main:ch10:definition-10.3	DOC-MAIN p.292; Definition 10.3	Definition / operational	Portfolio status
ak-hpdst:21.0.0:main:ch10:proposition-10.4	DOC-MAIN p.292; Proposition 10.4	Proposition / proved	Validation is strength preserving
ak-hpdst:21.0.0:main:ch10:definition-10.5	DOC-MAIN p.292; Definition 10.5	Definition / operational	Replay hierarchy used by this chapter
ak-hpdst:21.0.0:main:ch10:definition-10.6	DOC-MAIN p.292; Definition 10.6	Definition / operational	v20 regression suite
ak-hpdst:21.0.0:main:ch10:definition-10.7	DOC-MAIN p.292; Definition 10.7	Definition / operational	Regression evidence axes
ak-hpdst:21.0.0:main:ch10:proposition-10.10	DOC-MAIN p.293; Proposition 10.10	Proposition / proved	Regression R2: fixed-exponent Serrin recovery
ak-hpdst:21.0.0:main:ch10:proposition-10.9	DOC-MAIN p.293; Proposition 10.9	Proposition / proved	Regression R1: Iwasawa characteristic-growth recovery
ak-hpdst:21.0.0:main:ch10:proposition-10.11	DOC-MAIN p.294; Proposition 10.11	Proposition / proved	Regression R3: detector and Type IV separation
ak-hpdst:21.0.0:main:ch10:proposition-10.12	DOC-MAIN p.294; Proposition 10.12	Proposition / proved	Regression R4: governance and G16 closure
ak-hpdst:21.0.0:main:ch10:theorem-10.13	DOC-MAIN p.294; Theorem 10.13	Theorem / proved	Complete declared v20 regression slice
ak-hpdst:21.0.0:main:ch10:proposition-10.14	DOC-MAIN p.295; Proposition 10.14	Proposition / proved	S1 rank table
ak-hpdst:21.0.0:main:ch10:theorem-10.15	DOC-MAIN p.295; Theorem 10.15	Theorem / proved	S1 exact reconstruction
ak-hpdst:21.0.0:main:ch10:definition-10.16	DOC-MAIN p.295; Definition 10.16	Definition / operational	S1 validation verdict
ak-hpdst:21.0.0:main:ch10:proposition-10.17	DOC-MAIN p.295; Proposition 10.17	Proposition / proved	Scalar probes are incomplete in S2
ak-hpdst:21.0.0:main:ch10:theorem-10.18	DOC-MAIN p.296; Theorem 10.18	Theorem / proved	S2 enriched generator reconstruction
ak-hpdst:21.0.0:main:ch10:definition-10.19	DOC-MAIN p.296; Definition 10.19	Definition / operational	S2 validation verdict
ak-hpdst:21.0.0:main:ch10:theorem-10.20	DOC-MAIN p.296; Theorem 10.20	Theorem / proved	S3 object-level descent
ak-hpdst:21.0.0:main:ch10:proposition-10.21	DOC-MAIN p.296; Proposition 10.21	Proposition / proved	S3 strict derived descent
ak-hpdst:21.0.0:main:ch10:definition-10.22	DOC-MAIN p.297; Definition 10.22	Definition / operational	S3 validation verdict
ak-hpdst:21.0.0:main:ch10:theorem-10.23	DOC-MAIN p.297; Theorem 10.23	Theorem / proved	S4 finite-state inverse-limit Return
ak-hpdst:21.0.0:main:ch10:definition-10.24	DOC-MAIN p.297; Definition 10.24	Definition / operational	S4 validation verdict
ak-hpdst:21.0.0:main:ch10:proposition-10.25	DOC-MAIN p.298; Proposition 10.25	Proposition / proved	S5 source persistence
ak-hpdst:21.0.0:main:ch10:definition-10.26	DOC-MAIN p.298; Definition 10.26	Definition / operational	S5 external Return predicate
ak-hpdst:21.0.0:main:ch10:theorem-10.27	DOC-MAIN p.298; Theorem 10.27	Theorem / proved	S5 complete CoreP + CoreDG Return
ak-hpdst:21.0.0:main:ch10:proposition-10.28	DOC-MAIN p.298; Proposition 10.28	Proposition / proved	S5 Return ceiling
ak-hpdst:21.0.0:main:ch10:definition-10.29	DOC-MAIN p.298; Definition 10.29	Definition / operational	S5 validation verdict
ak-hpdst:21.0.0:main:ch10:theorem-10.30	DOC-MAIN p.299; Theorem 10.30	Theorem / proved	Exact query complexity of rank-table reconstruction
ak-hpdst:21.0.0:main:ch10:corollary-10.31	DOC-MAIN p.299; Corollary 10.31	Corollary / proved	Model-relative certificate and verifier cost
ak-hpdst:21.0.0:main:ch10:definition-10.33	DOC-MAIN p.300; Definition 10.33	Definition / operational	Two-track validation cycle
ak-hpdst:21.0.0:main:ch10:definition-10.34	DOC-MAIN p.300; Definition 10.34	Definition / operational	Challenge stopping rule
ak-hpdst:21.0.0:main:ch10:proposition-10.35	DOC-MAIN p.300; Proposition 10.35	Proposition / proved	Stable fallback is informative but noncompensating
ak-hpdst:21.0.0:main:ch10:definition-10.36	DOC-MAIN p.300; Definition 10.36	Definition / operational	Initial v22 Challenge portfolio
ak-hpdst:21.0.0:main:ch10:theorem-10.43	DOC-MAIN p.301; Theorem 10.43	Theorem / proved	Negative validation closure
ak-hpdst:21.0.0:main:ch10:definition-10.44	DOC-MAIN p.301; Definition 10.44	Definition / operational	v21 Main validation gates
ak-hpdst:21.0.0:main:ch10:proposition-10.45	DOC-MAIN p.302; Proposition 10.45	Proposition / proved	v21 Main validation gate results
ak-hpdst:21.0.0:main:ch10:theorem-10.46	DOC-MAIN p.302; Theorem 10.46	Theorem / proved	Required v21 validation portfolio closure
ak-hpdst:21.0.0:main:ch10:definition-10.48	DOC-MAIN p.303; Definition 10.48	Definition / operational	Successor classification
ak-hpdst:21.0.0:main:ch10:proposition-10.49	DOC-MAIN p.303; Proposition 10.49	Proposition / proved	v22 Core promotion candidates
ak-hpdst:21.0.0:main:ch10:proposition-10.50	DOC-MAIN p.303; Proposition 10.50	Proposition / proved	Technical Extension candidates
ak-hpdst:21.0.0:main:ch10:proposition-10.51	DOC-MAIN p.303; Proposition 10.51	Proposition / proved	Challenge-continuation registry
ak-hpdst:21.0.0:main:ch10:proposition-10.52	DOC-MAIN p.303; Proposition 10.52	Proposition / proved	Permanent no-go registry
ak-hpdst:21.0.0:main:ch10:theorem-10.53	DOC-MAIN p.304; Theorem 10.53	Theorem / proved	Validated v21 Main-body portfolio architecture

9.2 Foundation Appendices A-G (140)

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:foundation:appendix-a:lemma-1.16	DOC-FOUNDATION p.6; Lemma 1.16	Lemma / proved	Infinite-bar threshold invariance
ak-hpdst:21.0.0:foundation:appendix-a:proposition-1.18	DOC-FOUNDATION p.6; Proposition 1.18	Proposition / proved	Positive-threshold short-bar objects are not Serre in general
ak-hpdst:21.0.0:foundation:appendix-a:corollary-1.19	DOC-FOUNDATION p.7; Corollary 1.19	Corollary / proved	No global exact reflector claim
ak-hpdst:21.0.0:foundation:appendix-a:proposition-1.23	DOC-FOUNDATION p.7; Proposition 1.23	Proposition / proved	Basic properties of hard threshold deletion
ak-hpdst:21.0.0:foundation:appendix-a:proposition-1.24	DOC-FOUNDATION p.7; Proposition 1.24	Proposition / proved	Threshold composition
ak-hpdst:21.0.0:foundation:appendix-a:corollary-1.27	DOC-FOUNDATION p.8; Corollary 1.27	Corollary / proved	Rejected global non-expansiveness claim
ak-hpdst:21.0.0:foundation:appendix-a:corollary-1.40	DOC-FOUNDATION p.12; Corollary 1.40	Corollary / proved	Scalarized-ledger-safe threshold classification
ak-hpdst:21.0.0:foundation:appendix-a:lemma-1.43	DOC-FOUNDATION p.12; Lemma 1.43	Lemma / proved	Distance-to-zero formula
ak-hpdst:21.0.0:foundation:appendix-a:proposition-1.46	DOC-FOUNDATION p.13; Proposition 1.46	Proposition / proved	Exact zero transport
ak-hpdst:21.0.0:foundation:appendix-a:corollary-1.48	DOC-FOUNDATION p.14; Corollary 1.48	Corollary / proved	Scalarized metric-safe repaired-zero transport
ak-hpdst:21.0.0:foundation:appendix-a:corollary-1.51	DOC-FOUNDATION p.15; Corollary 1.51	Corollary / proved	Detector-certified metric zero transport
ak-hpdst:21.0.0:foundation:appendix-a:proposition-1.59	DOC-FOUNDATION p.16; Proposition 1.59	Proposition / proved	No unconditional crop-collapse commutation
ak-hpdst:21.0.0:foundation:appendix-a:corollary-1.90	DOC-FOUNDATION p.23; Corollary 1.90	Corollary / proved	Overlap repaired-zero equivalence
ak-hpdst:21.0.0:foundation:appendix-a:proposition-1.120	DOC-FOUNDATION p.31; Proposition 1.120	Proposition / proved	Appendix A contribution ceiling
ak-hpdst:21.0.0:foundation:appendix-b:proposition-2.19	DOC-FOUNDATION p.41; Proposition 2.19	Proposition / proved	Comparison modes do not self-promote
ak-hpdst:21.0.0:foundation:appendix-b:policy-2.28	DOC-FOUNDATION p.42; Policy 2.28	Policy / operational	Exact-zero use of randomized or approximate access
ak-hpdst:21.0.0:foundation:appendix-b:lemma-2.52	DOC-FOUNDATION p.47; Lemma 2.52	Lemma / proved	Augmented-mesh hitting lemma
ak-hpdst:21.0.0:foundation:appendix-b:corollary-2.59	DOC-FOUNDATION p.50; Corollary 2.59	Corollary / proved	One-way transport of a detector-certified zero
ak-hpdst:21.0.0:foundation:appendix-b:theorem-2.77	DOC-FOUNDATION p.53; Theorem 2.77	Theorem / proved	Common-model exact query complexity benchmark
ak-hpdst:21.0.0:foundation:appendix-b:theorem-2.79	DOC-FOUNDATION p.54; Theorem 2.79	Theorem / proved	Uniform finite-detector no-go on an unbounded birth family
ak-hpdst:21.0.0:foundation:appendix-b:proposition-2.85	DOC-FOUNDATION p.55; Proposition 2.85	Proposition / proved	Replayable shared-query composition benchmark
ak-hpdst:21.0.0:foundation:appendix-b:lemma-2.93	DOC-FOUNDATION p.57; Lemma 2.93	Lemma / proved	Persistence of an elementary interval complex
ak-hpdst:21.0.0:foundation:appendix-b:theorem-2.95	DOC-FOUNDATION p.57; Theorem 2.95	Theorem / proved	Finite barcode family realization
ak-hpdst:21.0.0:foundation:appendix-b:proposition-2.99	DOC-FOUNDATION p.58; Proposition 2.99	Proposition / proved	Compatibility is exactly the representative condition
ak-hpdst:21.0.0:foundation:appendix-b:axiom-2.105	DOC-FOUNDATION p.59; Axiom 2.105	Axiom / proved	Structure-compatible representative availability
ak-hpdst:21.0.0:foundation:appendix-b:proposition-2.107	DOC-FOUNDATION p.59; Proposition 2.107	Proposition / proved	Non-uniqueness of filtered representatives
ak-hpdst:21.0.0:foundation:appendix-b:proposition-2.109	DOC-FOUNDATION p.60; Proposition 2.109	Proposition / proved	Representative invariance of persistence-level clauses
ak-hpdst:21.0.0:foundation:appendix-b:proposition-2.112	DOC-FOUNDATION p.60; Proposition 2.112	Proposition / proved	Detector preservation by an exact stage-compatible isomorphism
ak-hpdst:21.0.0:foundation:appendix-b:proposition-2.117	DOC-FOUNDATION p.61; Proposition 2.117	Proposition / proved	Restriction under window compatibility
ak-hpdst:21.0.0:foundation:appendix-b:proposition-2.120	DOC-FOUNDATION p.61; Proposition 2.120	Proposition / proved	Windowed representative idempotence
ak-hpdst:21.0.0:foundation:appendix-b:proposition-2.122	DOC-FOUNDATION p.62; Proposition 2.122	Proposition / proved	Windowed threshold composition
ak-hpdst:21.0.0:foundation:appendix-b:proposition-2.127	DOC-FOUNDATION p.63; Proposition 2.127	Proposition / proved	Weak functoriality is conditional
ak-hpdst:21.0.0:foundation:appendix-b:proposition-2.130	DOC-FOUNDATION p.63; Proposition 2.130	Proposition / proved	Realization invariance under stronger equivalence
ak-hpdst:21.0.0:foundation:appendix-c:policy-3.6	DOC-FOUNDATION p.73; Policy 3.6	Policy / operational	No verdict-conditioned realization
ak-hpdst:21.0.0:foundation:appendix-c:lemma-3.40	DOC-FOUNDATION p.79; Lemma 3.40	Lemma / proved	Semisimplicity over a field
ak-hpdst:21.0.0:foundation:appendix-c:corollary-3.44	DOC-FOUNDATION p.80; Corollary 3.44	Corollary / proved	Higher edge vanishing criterion
ak-hpdst:21.0.0:foundation:appendix-c:corollary-3.45	DOC-FOUNDATION p.80; Corollary 3.45	Corollary / proved	Two-term splitting
ak-hpdst:21.0.0:foundation:appendix-c:proposition-3.82	DOC-FOUNDATION p.86; Proposition 3.82	Proposition / proved	Repaired degree-n vanishing implies edge vanishing
ak-hpdst:21.0.0:foundation:appendix-c:proposition-3.84	DOC-FOUNDATION p.87; Proposition 3.84	Proposition / proved	Repaired degree-one vanishing implies edge vanishing
ak-hpdst:21.0.0:foundation:appendix-c:theorem-3.97	DOC-FOUNDATION p.89; Theorem 3.97	Theorem / proved	Non-synthetic family-mesh zero-reflection
ak-hpdst:21.0.0:foundation:appendix-c:proposition-3.108	DOC-FOUNDATION p.92; Proposition 3.108	Proposition / proved	Reverse implication fails in the inherited degree-one regime
ak-hpdst:21.0.0:foundation:appendix-c:theorem-3.113	DOC-FOUNDATION p.93; Theorem 3.113	Theorem / proved	No synthetic promotion
ak-hpdst:21.0.0:foundation:appendix-c:proposition-3.122	DOC-FOUNDATION p.95; Proposition 3.122	Proposition / proved	Eligibility is invariant under admissible representative equivalence
ak-hpdst:21.0.0:foundation:appendix-c:proposition-3.126	DOC-FOUNDATION p.95; Proposition 3.126	Proposition / proved	Representative independence of the higher discharge
ak-hpdst:21.0.0:foundation:appendix-d:proposition-4.62	DOC-FOUNDATION p.113; Proposition 4.62	Proposition / proved	Diagnostics count terminal-cofinal defects
ak-hpdst:21.0.0:foundation:appendix-d:proposition-4.72	DOC-FOUNDATION p.115; Proposition 4.72	Proposition / proved	Zero diagnostics plus no interior defects imply isomorphism
ak-hpdst:21.0.0:foundation:appendix-d:proposition-4.94	DOC-FOUNDATION p.119; Proposition 4.94	Proposition / proved	Finite admissibility does not imply monitored tower admissibility
ak-hpdst:21.0.0:foundation:appendix-d:corollary-4.99	DOC-FOUNDATION p.120; Corollary 4.99	Corollary / proved	Inherited v20 exact comparison consequence
ak-hpdst:21.0.0:foundation:appendix-d:proposition-4.123	DOC-FOUNDATION p.125; Proposition 4.123	Proposition / proved	Degreewise stable bands exclude degree-wise Type IV
ak-hpdst:21.0.0:foundation:appendix-d:proposition-4.128	DOC-FOUNDATION p.125; Proposition 4.128	Proposition / proved	Full-isomorphism band implies stable band
ak-hpdst:21.0.0:foundation:appendix-d:proposition-4.134	DOC-FOUNDATION p.126; Proposition 4.134	Proposition / proved	Inherited degree-one eligible reduction of tower-clean admissibility
ak-hpdst:21.0.0:foundation:appendix-d:proposition-4.135	DOC-FOUNDATION p.126; Proposition 4.135	Proposition / proved	Inherited v20 monitored eligible reduction
ak-hpdst:21.0.0:foundation:appendix-d:proposition-4.141	DOC-FOUNDATION p.127; Proposition 4.141	Proposition / proved	Invariance of terminal and transient diagnostics
ak-hpdst:21.0.0:foundation:appendix-e:proposition-5.35	DOC-FOUNDATION p.137; Proposition 5.35	Proposition / proved	Overlap Gate preserves pairwise audit readability
ak-hpdst:21.0.0:foundation:appendix-e:theorem-5.65	DOC-FOUNDATION p.141; Theorem 5.65	Theorem / proved	Countable definable-cover gluing
ak-hpdst:21.0.0:foundation:appendix-e:proposition-5.70	DOC-FOUNDATION p.142; Proposition 5.70	Proposition / proved	Budget auditability under summability
ak-hpdst:21.0.0:foundation:appendix-e:proposition-5.73	DOC-FOUNDATION p.142; Proposition 5.73	Proposition / proved	Overlap cleanliness is required for gluing
ak-hpdst:21.0.0:foundation:appendix-e:proposition-5.76	DOC-FOUNDATION p.143; Proposition 5.76	Proposition / proved	Certificate gluing does not imply object gluing
ak-hpdst:21.0.0:foundation:appendix-e:proposition-5.81	DOC-FOUNDATION p.144; Proposition 5.81	Proposition / proved	Certificate-atlas and object-compatibility separation
ak-hpdst:21.0.0:foundation:appendix-e:proposition-5.84	DOC-FOUNDATION p.144; Proposition 5.84	Proposition / proved	Inverse-audit participation

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:foundation:appendix-e:proposition-5.99	DOC-FOUNDATION p.150; Proposition 5.99	Proposition / proved	Pairwise bound through a metric simplex reference
ak-hpdst:21.0.0:foundation:appendix-e:proposition-5.135	DOC-FOUNDATION p.158; Proposition 5.135	Proposition / proved	Transient-clean participation rule
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.16	DOC-FOUNDATION p.165; Proposition 6.16	Proposition / proved	No scalar manufacture of non-scalar evidence
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.32	DOC-FOUNDATION p.167; Proposition 6.32	Proposition / proved	Preflight failure blocks Core readiness
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.36	DOC-FOUNDATION p.168; Proposition 6.36	Proposition / proved	Pass is downward closed
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.39	DOC-FOUNDATION p.168; Proposition 6.39	Proposition / proved	Dependency blocker propagation
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.41	DOC-FOUNDATION p.169; Proposition 6.41	Proposition / proved	Status-algebra laws
ak-hpdst:21.0.0:foundation:appendix-f:theorem-6.43	DOC-FOUNDATION p.169; Theorem 6.43	Theorem / proved	Non-compensation
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.44	DOC-FOUNDATION p.169; Proposition 6.44	Proposition / proved	Partition invariance of hierarchical aggregation
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.48	DOC-FOUNDATION p.170; Proposition 6.48	Proposition / proved	Atomic status and failure route are different types
ak-hpdst:21.0.0:foundation:appendix-f:theorem-6.55	DOC-FOUNDATION p.171; Theorem 6.55	Theorem / proved	Certificate-readiness criterion
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.57	DOC-FOUNDATION p.172; Proposition 6.57	Proposition / proved	Complete Restart verdict
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.58	DOC-FOUNDATION p.172; Proposition 6.58	Proposition / proved	Atlas composition rule
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.61	DOC-FOUNDATION p.173; Proposition 6.61	Proposition / proved	Finite-prefix checks do not certify countable readiness
ak-hpdst:21.0.0:foundation:appendix-f:theorem-6.62	DOC-FOUNDATION p.173; Theorem 6.62	Theorem / proved	Countable aggregate soundness
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.64	DOC-FOUNDATION p.173; Proposition 6.64	Proposition / proved	No retroactive verdict mutation
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.67	DOC-FOUNDATION p.174; Proposition 6.67	Proposition / proved	Schema conformance preserves semantics
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.76	DOC-FOUNDATION p.175; Proposition 6.76	Proposition / proved	No cross-sort token substitution
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.85	DOC-FOUNDATION p.178; Proposition 6.85	Proposition / proved	Layered non-compensation
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.93	DOC-FOUNDATION p.180; Proposition 6.93	Proposition / proved	Descent-stage noncompensation
ak-hpdst:21.0.0:foundation:appendix-f:theorem-6.95	DOC-FOUNDATION p.181; Theorem 6.95	Theorem / proved	v21 compatibility readiness criterion
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.98	DOC-FOUNDATION p.181; Proposition 6.98	Proposition / proved	No lossy migration of theorem evidence
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.100	DOC-FOUNDATION p.182; Proposition 6.100	Proposition / proved	Replay invariance under fixed inputs
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.123	DOC-FOUNDATION p.186; Proposition 6.123	Proposition / proved	No reversal of weakening
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.128	DOC-FOUNDATION p.186; Proposition 6.128	Proposition / proved	Unlicensed cycle blocks certification
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.133	DOC-FOUNDATION p.187; Proposition 6.133	Proposition / proved	Typing precedes obstruction
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.138	DOC-FOUNDATION p.188; Proposition 6.138	Proposition / proved	Source storage is not theorem recovery
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.140	DOC-FOUNDATION p.188; Proposition 6.140	Proposition / proved	Hash identity supports identity, not truth
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.145	DOC-FOUNDATION p.189; Proposition 6.145	Proposition / proved	No lossy migration
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.148	DOC-FOUNDATION p.189; Proposition 6.148	Proposition / proved	Agent assertion does not establish a Core node
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.150	DOC-FOUNDATION p.189; Proposition 6.150	Proposition / proved	Formal verification is not silent interpretation
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.152	DOC-FOUNDATION p.189; Proposition 6.152	Proposition / proved	Contamination blocks status inheritance
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.157	DOC-FOUNDATION p.190; Proposition 6.157	Proposition / proved	Replay disagreement blocks silent reuse
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.160	DOC-FOUNDATION p.190; Proposition 6.160	Proposition / proved	No automatic status inheritance
ak-hpdst:21.0.0:foundation:appendix-f:proposition-6.167	DOC-FOUNDATION p.191; Proposition 6.167	Proposition / proved	No conclusion beyond route intersection
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.7	DOC-FOUNDATION p.196; Proposition 7.7	Proposition / proved	Appendix F semantics are authoritative
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.12	DOC-FOUNDATION p.197; Proposition 7.12	Proposition / proved	Identity collision is manifest inconsistency
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.23	DOC-FOUNDATION p.198; Proposition 7.23	Proposition / proved	No silent default theorem
ak-hpdst:21.0.0:foundation:appendix-g:axiom-7.24	DOC-FOUNDATION p.198; Axiom 7.24	Axiom / proved	Minimal Manifest Axiom
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.26	DOC-FOUNDATION p.199; Proposition 7.26	Proposition / proved	Necessity of manifest data
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.31	DOC-FOUNDATION p.199; Proposition 7.31	Proposition / proved	Finite prefix cannot certify a countable registry
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.36	DOC-FOUNDATION p.200; Proposition 7.36	Proposition / proved	Status/classification separation
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.41	DOC-FOUNDATION p.201; Proposition 7.41	Proposition / proved	Window data are audit-critical
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.47	DOC-FOUNDATION p.201; Proposition 7.47	Proposition / proved	Threshold and collapse data are audit-critical
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.52	DOC-FOUNDATION p.202; Proposition 7.52	Proposition / proved	Evaluation records are audit-critical
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.57	DOC-FOUNDATION p.202; Proposition 7.57	Proposition / proved	Representative and eligibility records are audit-critical when invoked
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.64	DOC-FOUNDATION p.203; Proposition 7.64	Proposition / proved	Ledger data are audit-critical
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.71	DOC-FOUNDATION p.204; Proposition 7.71	Proposition / proved	Diagnostic and comparison-artifact data are audit-critical
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.76	DOC-FOUNDATION p.204; Proposition 7.76	Proposition / proved	Pairwise records do not imply higher coherence
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.82	DOC-FOUNDATION p.205; Proposition 7.82	Proposition / proved	Gate verdicts are audit-critical
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.87	DOC-FOUNDATION p.205; Proposition 7.87	Proposition / proved	Reference data are audit-critical when invoked
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.94	DOC-FOUNDATION p.206; Proposition 7.94	Proposition / proved	Artifact references are audit-critical
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.97	DOC-FOUNDATION p.206; Proposition 7.97	Proposition / proved	No retroactive verdict mutation
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.103	DOC-FOUNDATION p.207; Proposition 7.103	Proposition / proved	Minimal Manifest implies audit readability
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.104	DOC-FOUNDATION p.207; Proposition 7.104	Proposition / proved	Audit readability enables proposer-independent evaluation
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.108	DOC-FOUNDATION p.207; Proposition 7.108	Proposition / proved	Only verifier-side data determine Core verdicts
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.114	DOC-FOUNDATION p.208; Proposition 7.114	Proposition / proved	Concrete execution schemas must refine the abstract schema
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.117	DOC-FOUNDATION p.208; Proposition 7.117	Proposition / proved	Canonical manifest conformance
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.119	DOC-FOUNDATION p.209; Proposition 7.119	Proposition / proved	Schema sufficiency for Core audit
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.124	DOC-FOUNDATION p.209; Proposition 7.124	Proposition / proved	Failure is preferable to incompleteness
ak-hpdst:21.0.0:foundation:appendix-g:theorem-7.125	DOC-FOUNDATION p.209; Theorem 7.125	Theorem / proved	Appendix G manifest package
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.128	DOC-FOUNDATION p.210; Proposition 7.128	Proposition / proved	Cross-sort inconsistency is detectable
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.144	DOC-FOUNDATION p.215; Proposition 7.144	Proposition / proved	Semantic replay is stronger than byte reproduction

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.147	DOC-FOUNDATION p.215; Proposition 7.147	Proposition / proved	Lossy migration is non-conformant
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.152	DOC-FOUNDATION p.216; Proposition 7.152	Proposition / proved	Document completeness and theorem strength are independent
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.155	DOC-FOUNDATION p.217; Proposition 7.155	Proposition / proved	v20 schema sufficiency for audit reconstruction
ak-hpdst:21.0.0:foundation:appendix-g:theorem-7.156	DOC-FOUNDATION p.217; Theorem 7.156	Theorem / proved	Appendix G v20 manifest and replay package
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.164	DOC-FOUNDATION p.219; Proposition 7.164	Proposition / proved	No manifest-axis promotion
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.173	DOC-FOUNDATION p.220; Proposition 7.173	Proposition / proved	Proof-DAG record completeness is not discharge
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.178	DOC-FOUNDATION p.221; Proposition 7.178	Proposition / proved	Numerical records cannot manufacture semantics
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.184	DOC-FOUNDATION p.223; Proposition 7.184	Proposition / proved	Lossless descent-stage serialization
ak-hpdst:21.0.0:foundation:appendix-g:theorem-7.190	DOC-FOUNDATION p.224; Theorem 7.190	Theorem / proved	Quarantine serialization soundness
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.192	DOC-FOUNDATION p.224; Proposition 7.192	Proposition / proved	Artifact identity is not mathematical truth
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.195	DOC-FOUNDATION p.224; Proposition 7.195	Proposition / proved	Dangling or ambiguous artifacts block pass
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.200	DOC-FOUNDATION p.225; Proposition 7.200	Proposition / proved	Consensus and formal build are non-compensating
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.204	DOC-FOUNDATION p.226; Proposition 7.204	Proposition / proved	Build and render success have bounded authority
ak-hpdst:21.0.0:foundation:appendix-g:theorem-7.209	DOC-FOUNDATION p.226; Theorem 7.209	Theorem / proved	Immutable manifest history
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.213	DOC-FOUNDATION p.227; Proposition 7.213	Proposition / proved	Handoffis not a Bridge theorem
ak-hpdst:21.0.0:foundation:appendix-g:proposition-7.220	DOC-FOUNDATION p.228; Proposition 7.220	Proposition / proved	Schema validation reconstructs but does not create pass
ak-hpdst:21.0.0:foundation:appendix-g:theorem-7.221	DOC-FOUNDATION p.228; Theorem 7.221	Theorem / proved	Appendix G v21 canonical manifest and replay package

9.3 Appendix DG (20)

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:dg:dg:policy-DG.1.3	DOC-DG p.4; Policy DG.1.3	Policy / operational	Conservative-evolution and proof-ownership policy
ak-hpdst:21.0.0:dg:dg:lemma-DG.2.4	DOC-DG p.6; Lemma DG.2.4	Lemma / proved	Presentation completeness is reconstruction data
ak-hpdst:21.0.0:dg:dg:policy-DG.6.3	DOC-DG p.10; Policy DG.6.3	Policy / operational	Concrete cone artifact policy
ak-hpdst:21.0.0:dg:dg:lemma-DG.7.2	DOC-DG p.10; Lemma DG.7.2	Lemma / proved	Category-algebra identification
ak-hpdst:21.0.0:dg:dg:lemma-DG.7.5	DOC-DG p.11; Lemma DG.7.5	Lemma / proved	Explicit reconstruction from a left module
ak-hpdst:21.0.0:dg:dg:lemma-DG.8.5	DOC-DG p.13; Lemma DG.8.5	Lemma / proved	Information lower bound from independent arrows
ak-hpdst:21.0.0:dg:dg:lemma-DG.10.4	DOC-DG p.15; Lemma DG.10.4	Lemma / proved	Appendix-local: descent of morphisms, chain homotopies, and equations
ak-hpdst:21.0.0:dg:dg:theorem-DG.12.2	DOC-DG p.16; Theorem DG.12.2	Theorem / proved	DescentInput lane-selection theorem
ak-hpdst:21.0.0:dg:dg:policy-DG.15.4	DOC-DG p.20; Policy DG.15.4	Policy / operational	Evidence reuse and no-free-advice policy
ak-hpdst:21.0.0:dg:dg:proposition-DG.16.7	DOC-DG p.21; Proposition DG.16.7	Proposition / proved	Counterexample coverage theorem
ak-hpdst:21.0.0:dg:dg:policy-DG.17.4	DOC-DG p.22; Policy DG.17.4	Policy / operational	Chapter 9 non-overreach policy
ak-hpdst:21.0.0:dg:dg:theorem-DG.18.1	DOC-DG p.22; Theorem DG.18.1	Theorem / proved	Core-P support-record boundary
ak-hpdst:21.0.0:dg:dg:theorem-DG.18.2	DOC-DG p.23; Theorem DG.18.2	Theorem / proved	P2DG consumer audit theorem
ak-hpdst:21.0.0:dg:dg:policy-DG.18.3	DOC-DG p.23; Policy DG.18.3	Policy / operational	Finite-to-infinite and tower consumption
ak-hpdst:21.0.0:dg:dg:theorem-DG.18.4	DOC-DG p.24; Theorem DG.18.4	Theorem / proved	Object-descent handoff consumer audit
ak-hpdst:21.0.0:dg:dg:proposition-DG.18.6	DOC-DG p.24; Proposition DG.18.6	Proposition / proved	DG-to-F status preservation
ak-hpdst:21.0.0:dg:dg:proposition-DG.18.10	DOC-DG p.26; Proposition DG.18.10	Proposition / proved	Consumer-chain closure
ak-hpdst:21.0.0:dg:dg:proposition-DG.18.8	DOC-DG p.26; Proposition DG.18.8	Proposition / proved	DG-to-G semantic closure
ak-hpdst:21.0.0:dg:dg:policy-DG.18.9	DOC-DG p.26; Policy DG.18.9	Policy / operational	Downstream-only validation
ak-hpdst:21.0.0:dg:dg:proposition-DG.19.1	DOC-DG p.27; Proposition DG.19.1	Proposition / proved	Identity closure audit

9.4 Appendix HT (23)

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:ht:ht-v21:axiom-1.2	DOC-HT p.9; Axiom 1.2	Axiom / proved	Immutable v20 predecessor boundary
ak-hpdst:21.0.0:ht:ht-v21:proposition-1.8	DOC-HT p.11; Proposition 1.8	Proposition / proved	No duplicate mathematical authority
ak-hpdst:21.0.0:ht:ht-v21:proposition-2.2	DOC-HT p.11; Proposition 2.2	Proposition / proved	Handoff non-promotion
ak-hpdst:21.0.0:ht:ht-v21:theorem-3.3	DOC-HT p.13; Theorem 3.3	Theorem / proved	Expanded ownership consequence of Main Theorem 9.6
ak-hpdst:21.0.0:ht:ht-v21:proposition-4.3	DOC-HT p.13; Proposition 4.3	Proposition / proved	No unrelated partial-Return comparison
ak-hpdst:21.0.0:ht:ht-v21:theorem-4.4	DOC-HT p.14; Theorem 4.4	Theorem / proved	Assumption and strength non-erasure
ak-hpdst:21.0.0:ht:ht-v21:theorem-5.3	DOC-HT p.14; Theorem 5.3	Theorem / proved	HT-local specialization of component composition
ak-hpdst:21.0.0:ht:ht-v21:proposition-5.4	DOC-HT p.15; Proposition 5.4	Proposition / proved	Owner partition is non-overlapping at theorem identity
ak-hpdst:21.0.0:ht:ht-v21:proposition-6.6	DOC-HT p.17; Proposition 6.6	Proposition / proved	Dual-Core and descent-stage non-identification
ak-hpdst:21.0.0:ht:ht-v21:proposition-8.2	DOC-HT p.18; Proposition 8.2	Proposition / proved	HC conclusion ceiling
ak-hpdst:21.0.0:ht:ht-v21:proposition-8.11	DOC-HT p.19; Proposition 8.11	Proposition / proved	No universal apex or universal inversion
ak-hpdst:21.0.0:ht:ht-v21:theorem-8.7	DOC-HT p.19; Theorem 8.7	Theorem / proved	SS non-collapse firewall
ak-hpdst:21.0.0:ht:ht-v21:proposition-8.9	DOC-HT p.19; Proposition 8.9	Proposition / proved	Three-way RL separation
ak-hpdst:21.0.0:ht:ht-v21:proposition-10.4	DOC-HT p.21; Proposition 10.4	Proposition / proved	Noncompensation
ak-hpdst:21.0.0:ht:ht-v21:proposition-10.5	DOC-HT p.21; Proposition 10.5	Proposition / proved	Repair precedence
ak-hpdst:21.0.0:ht:ht-v21:theorem-11.2	DOC-HT p.21; Theorem 11.2	Theorem / proved	Expanded research-spine proof record for Main Theorem 9.47
ak-hpdst:21.0.0:ht:ht-v21:corollary-11.3	DOC-HT p.22; Corollary 11.3	Corollary / proved	HT selected-sub-DAG composition
ak-hpdst:21.0.0:ht:ht-v21:theorem-11.4	DOC-HT p.22; Theorem 11.4	Theorem / proved	Non-composition firewall
ak-hpdst:21.0.0:ht:ht-v21:theorem-11.5	DOC-HT p.22; Theorem 11.5	Theorem / proved	Return ceiling and complete failure localization
ak-hpdst:21.0.0:ht:ht-v21:proposition-13.2	DOC-HT p.24; Proposition 13.2	Proposition / proved	Lossless HT serialization
ak-hpdst:21.0.0:ht:ht-v21:theorem-15.1	DOC-HT p.25; Theorem 15.1	Theorem / proved	Counterexample coverage
ak-hpdst:21.0.0:ht:ht-v21:theorem-16.1	DOC-HT p.26; Theorem 16.1	Theorem / proved	Cross-document impact-isolation theorem
ak-hpdst:21.0.0:ht:ht-v21:theorem-16.2	DOC-HT p.26; Theorem 16.2	Theorem / proved	Appendix HT v21 research-spine closure

9.5 Appendix UB (21)

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:ub:ub-v21:axiom-1.2	DOC-UB p.11; Axiom 1.2	Axiom / proved	Immutable predecessor boundary
ak-hpdst:21.0.0:ub:ub-v21:theorem-2.4	DOC-UB p.13; Theorem 2.4	Theorem / proved	UB implementation compatibility
ak-hpdst:21.0.0:ub:ub-v21:proposition-3.3	DOC-UB p.13; Proposition 3.3	Proposition / proved	No assumption erasure
ak-hpdst:21.0.0:ub:ub-v21:proposition-4.1	DOC-UB p.14; Proposition 4.1	Proposition / proved	Owner partition is non-overlapping
ak-hpdst:21.0.0:ub:ub-v21:proposition-7.3	DOC-UB p.16; Proposition 7.3	Proposition / proved	Noncompensation
ak-hpdst:21.0.0:ub:ub-v21:axiom-8.2	DOC-UB p.17; Axiom 8.2	Axiom / proved	Chapter 10 leaf and benchmark quarantine
ak-hpdst:21.0.0:ub:ub-v21:proposition-9.2	DOC-UB p.18; Proposition 9.2	Proposition / proved	Replay hierarchy
ak-hpdst:21.0.0:ub:ub-v21:theorem-11.1	DOC-UB p.19; Theorem 11.1	Theorem / proved	Appendix UB v21 synchronization closure
ak-hpdst:21.0.0:ub:ub-v21:proposition-13.2	DOC-UB p.21; Proposition 13.2	Proposition / proved	Interface equivalence is a compositional congruence
ak-hpdst:21.0.0:ub:ub-UB-v21-DER-01	DOC-UB p.21; Theorem 13.3	Theorem / proved	UB-v21-DER-01: certified implementation paracategory
ak-hpdst:21.0.0:ub:ub-UB-v21-DER-02	DOC-UB p.22; Corollary 13.4	Corollary / proved	UB-v21-DER-02: implementation substitution invariance
ak-hpdst:21.0.0:ub:ub-UB-v21-DER-03	DOC-UB p.22; Theorem 14.2	Theorem / proved	UB-v21-DER-03: finite Pareto strength-frontier theorem
ak-hpdst:21.0.0:ub:ub-UB-v21-DER-04	DOC-UB p.23; Theorem 14.4	Theorem / proved	UB-v21-DER-04: unique minimal failure-antichain theorem
ak-hpdst:21.0.0:ub:ub-UB-v21-DER-05	DOC-UB p.23; Theorem 14.5	Theorem / proved	UB-v21-DER-05: independent target-product Return
ak-hpdst:21.0.0:ub:ub-UB-v21-DER-06	DOC-UB p.24; Theorem 15.2	Theorem / proved	UB-v21-DER-06: strict descent of finite-state derived inverse limits
ak-hpdst:21.0.0:ub:ub-UB-v21-DER-07	DOC-UB p.25; Corollary 15.3	Corollary / proved	UB-v21-DER-07: local-to-global finite-state quasi-isomorphism criterion
ak-hpdst:21.0.0:ub:ub-UB-v21-DER-08	DOC-UB p.25; Theorem 15.4	Theorem / proved	UB-v21-DER-08: field-extension invariance of finite-state derived limits

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:ub:UB-v21-DER-09	DOC-UB p.27; Theorem 16.2	Theorem / proved	UB-v21-DER-09: exact mixed regular–singular finite-layer theorem
ak-hpdst:21.0.0:ub:UB-v21-DER-10	DOC-UB p.27; Corollary 16.3	Corollary / proved	UB-v21-DER-10: arithmetic mixed-layer Return
ak-hpdst:21.0.0:ub:UB-v21-DER-11	DOC-UB p.27; Theorem 16.4	Theorem / proved	UB-v21-DER-11: branch-complete decision theorem on the exact one-variable class
ak-hpdst:21.0.0:ub:UB-v21-DER-12	DOC-UB p.29; Theorem 17.5	Theorem / proved	UB-v21-DER-12: derived-theorem-pack closure

9.6 Appendix CM (19)

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:cm:CM-v21-OWN-01	DOC-CM p.4; Policy 2	Policy / operational	CM-v21-OWN-01: immutable predecessor and new-identity policy
ak-hpdst:21.0.0:cm:CM-v21-TB-01	DOC-CM p.6; Policy 10	Policy / operational	CM-v21-TB-01: TB consumer handoff
ak-hpdst:21.0.0:cm:CM-v21-TB-02	DOC-CM p.6; Policy 11	Policy / operational	CM-v21-TB-02: dangling v20 source-reference quarantine
ak-hpdst:21.0.0:cm:CM-v21-IF-01	DOC-CM p.6; Policy 8	Policy / operational	CM-v21-IF-01: P2DG and Bridge nonpromotion
ak-hpdst:21.0.0:cm:CM-v21-SRC-01	DOC-CM p.7; Remark 12	Remark / provenance_only	CM-v21-SRC-01: imported single-morphism source-binding audit
ak-hpdst:21.0.0:cm:CM-v21-DER-01	DOC-CM p.8; Theorem 13	Theorem / proved	CM-v21-DER-01: threshold-deletion band action
ak-hpdst:21.0.0:cm:CM-v21-DER-02	DOC-CM p.8; Theorem 14	Theorem / proved	CM-v21-DER-02: window-restriction meet action
ak-hpdst:21.0.0:cm:CM-v21-DER-03	DOC-CM p.8; Theorem 15	Theorem / proved	CM-v21-DER-03: upward closure of complete stencils
ak-hpdst:21.0.0:cm:CM-v21-DER-04	DOC-CM p.9; Theorem 16	Theorem / proved	CM-v21-DER-04: family-union query bound in one fixed stencil model
ak-hpdst:21.0.0:cm:CM-v21-DER-05	DOC-CM p.9; Theorem 17	Theorem / proved	CM-v21-DER-05: canonical minimal augmented-mesh stencil and refinement
ak-hpdst:21.0.0:cm:CM-v21-DER-06	DOC-CM p.10; Theorem 18	Theorem / proved	CM-v21-DER-06: associative Restart summary calculus
ak-hpdst:21.0.0:cm:CM-v21-DER-07	DOC-CM p.10; Theorem 19	Theorem / proved	CM-v21-DER-07: terminal-clean two-out-of-three
ak-hpdst:21.0.0:cm:CM-v21-DER-08	DOC-CM p.11; Theorem 20	Theorem / proved	CM-v21-DER-08: terminal short-five principle
ak-hpdst:21.0.0:cm:CM-v21-DER-09	DOC-CM p.11; Theorem 21	Theorem / proved	CM-v21-DER-09: finite-chain accumulation of Type IV defects
ak-hpdst:21.0.0:cm:CM-v21-DER-10	DOC-CM p.11; Theorem 22	Theorem / proved	CM-v21-DER-10: stabilization-witness independence
ak-hpdst:21.0.0:cm:CM-v21-DER-11	DOC-CM p.12; Theorem 23	Theorem / proved	CM-v21-DER-11: object comparison does not determine morphism defects
ak-hpdst:21.0.0:cm:CM-v21-DER-12	DOC-CM p.12; Theorem 24	Theorem / proved	CM-v21-DER-12: local-zero/global-nonzero window witness
ak-hpdst:21.0.0:cm:CM-v21-GOV-01	DOC-CM p.12; Policy 25	Policy / operational	CM-v21-GOV-01: lossless serialization and package closure
ak-hpdst:21.0.0:cm:CM-v21-CLS-01	DOC-CM p.13; Theorem 26	Theorem / conditional	CM-v21-CLS-01: conservative CM-v21 closure

9.7 Appendix TB (19)

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:tb:TB-v21-OWN-01	DOC-TB p.4; Policy 2	Policy / operational	TB-v21-OWN-01: no third theorem authority
ak-hpdst:21.0.0:tb:TB-v21-TYPE-01	DOC-TB p.4; Policy 4	Policy / operational	TB-v21-TYPE-01: status and result-sort separation
ak-hpdst:21.0.0:tb:TB-v21-CM-01	DOC-TB p.5; Policy 5	Policy / operational	TB-v21-CM-01: repaired CM consumer bindings
ak-hpdst:21.0.0:tb:TB-v21-DG-01	DOC-TB p.6; Policy 6	Policy / operational	TB-v21-DG-01: one active descent lane and no local-to-global shortcut
ak-hpdst:21.0.0:tb:TB-v21-VAL-01	DOC-TB p.6; Policy 7	Policy / operational	TB-v21-VAL-01: validated demonstration and no benchmark backflow
ak-hpdst:21.0.0:tb:TB-v21-DER-01	DOC-TB p.7; Theorem 8	Theorem / proved	TB-v21-DER-01: strict-transition budget for injective finite-dimensional towers
ak-hpdst:21.0.0:tb:TB-v21-DER-02	DOC-TB p.7; Theorem 9	Theorem / proved	TB-v21-DER-02: strict-transition budget for finite-length module towers
ak-hpdst:21.0.0:tb:TB-v21-DER-03	DOC-TB p.8; Theorem 10	Theorem / conditional	TB-v21-DER-03: finite direct-sum closure of exact toy towers
ak-hpdst:21.0.0:tb:TB-v21-DER-04	DOC-TB p.8; Theorem 11	Theorem / proved	TB-v21-DER-04: finite-chain accumulation of terminal-generic defects
ak-hpdst:21.0.0:tb:TB-v21-DER-05	DOC-TB p.9; Theorem 12	Theorem / proved	TB-v21-DER-05: terminal-clean two-out-of-three
ak-hpdst:21.0.0:tb:TB-v21-DER-06	DOC-TB p.9; Theorem 13	Theorem / conditional	TB-v21-DER-06: terminal short-five principle
ak-hpdst:21.0.0:tb:TB-v21-DER-07	DOC-TB p.10; Theorem 14	Theorem / conditional	TB-v21-DER-07: corrected effective-domain family mesh
ak-hpdst:21.0.0:tb:TB-v21-DER-08	DOC-TB p.10; Theorem 15	Theorem / proved	TB-v21-DER-08: instance-relative finite retained-bar witness extraction
ak-hpdst:21.0.0:tb:TB-v21-DER-09	DOC-TB p.11; Theorem 16	Theorem / conditional	TB-v21-DER-09: strict finite-diagram exact globalization
ak-hpdst:21.0.0:tb:TB-v21-DER-10	DOC-TB p.11; Theorem 17	Theorem / conditional	TB-v21-DER-10: strict descent commutes with eventual-constancy validation
ak-hpdst:21.0.0:tb:TB-v21-DER-11	DOC-TB p.12; Theorem 18	Theorem / conditional	TB-v21-DER-11: independent target-product toy Return
ak-hpdst:21.0.0:tb:TB-v21-DER-12	DOC-TB p.12; Theorem 19	Theorem / proved	TB-v21-DER-12: objectwise isomorphism does not manufacture naturality
ak-hpdst:21.0.0:tb:TB-v21-GOV-01	DOC-TB p.13; Policy 21	Policy / operational	TB-v21-GOV-01: lossless register and replay closure
ak-hpdst:21.0.0:tb:TB-v21-CLS-01	DOC-TB p.14; v21 Theorem 22; local UID TB-v21-CLS-01	Theorem / conditional	TB-v21-CLS-01: v21 toy portfolio closure

9.8 Technical Extension Appendices H-N (70)

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:te:h:policy-2	DOC-TE p.10; Policy 2	Policy / operational	Immutable predecessor and current-reading overlay
ak-hpdst:21.0.0:te:H-v21-DER-01	DOC-TE p.12; Theorem 6	Theorem / proved	H-v21-DER-01: stale closure and unique minimal advisory recomputation frontier
ak-hpdst:21.0.0:te:H-v21-DER-02	DOC-TE p.13; Theorem 7	Theorem / proved	H-v21-DER-02: jointly compatible advisory-attachment merge
ak-hpdst:21.0.0:te:H-v21-DER-03	DOC-TE p.13; Theorem 9	Theorem / proved	H-v21-DER-03: re-entry factorization, route admissibility, and nonmutation
ak-hpdst:21.0.0:te:H-v21-IF-01	DOC-TE p.14; Policy 10	Policy / operational	H-v21-IF-01: no direct Core-DG, descent, or Bridge handoff
ak-hpdst:21.0.0:te:H-v21-DER-04	DOC-TE p.15; Theorem 12	Theorem / proved	H-v21-DER-04: diagram-summary invariance and sharp nonconservativity
ak-hpdst:21.0.0:te:H-v21-DER-05	DOC-TE p.15; Theorem 13	Theorem / proved	H-v21-DER-05: exact finite threshold-cell reduction
ak-hpdst:21.0.0:te:H-v21-DER-06	DOC-TE p.16; Theorem 14	Theorem / proved	H-v21-DER-06: matching-sensitive global gap-free stability of excess mass
ak-hpdst:21.0.0:te:H-v21-DER-07	DOC-TE p.16; Theorem 15	Theorem / proved	H-v21-DER-07: finite exact moment reconstruction of the length spectrum
ak-hpdst:21.0.0:te:H-v21-DER-08	DOC-TE p.17; Theorem 16	Theorem / proved	H-v21-DER-08: right-derivative and distributional recovery of the finite length spectrum
ak-hpdst:21.0.0:te:H-v21-DER-09	DOC-TE p.17; Theorem 17	Theorem / proved	H-v21-DER-09: exact $\blacklozenge$ 1 tail distance for finite length spectra
ak-hpdst:21.0.0:te:h:policy-18	DOC-TE p.18; Policy 18	Policy / operational	H-v21-BRIDGE-01: target-relative output ceiling
ak-hpdst:21.0.0:te:H-v21-VAL-01	DOC-TE p.18; Policy 19	Policy / operational	H-v21-VAL-01: validation-leaf and benchmark-quarantine rule
ak-hpdst:21.0.0:te:h:theorem-21	DOC-TE p.19; Theorem 21	Theorem / proved	H-v21 package closure and contribution ceiling
ak-hpdst:21.0.0:te:i:policy-3	DOC-TE p.21; Policy 3	Policy / operational	Discretization route separation
ak-hpdst:21.0.0:te:I-v21-DER-01	DOC-TE p.22; Theorem 4	Theorem / proved	I-v21-DER-01: finite-family common refinement
ak-hpdst:21.0.0:te:I-v21-DER-02	DOC-TE p.22; Theorem 5	Theorem / proved	I-v21-DER-02: monotone readiness frontier on a finite refinement poset
ak-hpdst:21.0.0:te:I-v21-DER-03	DOC-TE p.22; Theorem 6	Theorem / proved	I-v21-DER-03: natural discretization conjugacy preserves diagram defects
ak-hpdst:21.0.0:te:I-v21-DER-04	DOC-TE p.22; Theorem 7	Theorem / proved	I-v21-DER-04: strict descent of a compatible discretization isomorphism
ak-hpdst:21.0.0:te:I-v21-DER-07	DOC-TE p.23; Theorem 10	Theorem / proved	I-v21-DER-07: coherent common-refinement certificate transport
ak-hpdst:21.0.0:te:i:policy-11	DOC-TE p.23; Policy 11	Policy / operational	I-v21 target and Return ceiling
ak-hpdst:21.0.0:te:I-v21-DER-05	DOC-TE p.23; Theorem 8	Theorem / proved	I-v21-DER-05: finite monitored-certificate union bound
ak-hpdst:21.0.0:te:I-v21-DER-06	DOC-TE p.23; Theorem 9	Theorem / proved	I-v21-DER-06: finite simultaneous risk allocation
ak-hpdst:21.0.0:te:j:policy-2	DOC-TE p.25; Policy 2	Policy / operational	Measurement target separation
ak-hpdst:21.0.0:te:J-v21-DER-01	DOC-TE p.25; Theorem 3	Theorem / proved	J-v21-DER-01: threshold-mass sweep exactly recovers the finite length multiset
ak-hpdst:21.0.0:te:J-v21-DER-02	DOC-TE p.26; Theorem 4	Theorem / proved	J-v21-DER-02: equivalence of exact finite length-spectrum presentations
ak-hpdst:21.0.0:te:J-v21-DER-03	DOC-TE p.26; Theorem 5	Theorem / proved	J-v21-DER-03: universal translation obstruction for length-factorizing measurements
ak-hpdst:21.0.0:te:J-v21-DER-04	DOC-TE p.26; Theorem 6	Theorem / proved	J-v21-DER-04: objectwise measurement summaries do not reconstruct a finite diagram
ak-hpdst:21.0.0:te:J-v21-DER-05	DOC-TE p.26; Theorem 7	Theorem / proved	J-v21-DER-05: independent product-measurement target
ak-hpdst:21.0.0:te:J-v21-DER-06	DOC-TE p.26; Theorem 8	Theorem / proved	J-v21-DER-06: finite measurement-certificate subadditivity
ak-hpdst:21.0.0:te:j:policy-9	DOC-TE p.27; Policy 9	Policy / operational	J-v21 re-entry and Return ceiling
ak-hpdst:21.0.0:te:k:policy-2	DOC-TE p.28; Policy 2	Policy / operational	Exact, controlled, and advisory separation
ak-hpdst:21.0.0:te:K-v21-DER-01	DOC-TE p.28; Theorem 3	Theorem / proved	K-v21-DER-01: controlled implementations form a partial composition category
ak-hpdst:21.0.0:te:K-v21-DER-02	DOC-TE p.29; Theorem 4	Theorem / proved	K-v21-DER-02: interface-equivalent substitution invariance
ak-hpdst:21.0.0:te:K-v21-DER-03	DOC-TE p.29; Theorem 5	Theorem / proved	K-v21-DER-03: single-owner finite pasting bound
ak-hpdst:21.0.0:te:K-v21-DER-04	DOC-TE p.29; Theorem 6	Theorem / proved	K-v21-DER-04: independent product defect component
ak-hpdst:21.0.0:te:K-v21-DER-05	DOC-TE p.29; Theorem 7	Theorem / proved	K-v21-DER-05: exact natural conjugacy transports cones and strict descent comparisons
ak-hpdst:21.0.0:te:K-v21-DER-06	DOC-TE p.30; Theorem 8	Theorem / proved	K-v21-DER-06: AWFS lifting remains target-local
ak-hpdst:21.0.0:te:k:policy-9	DOC-TE p.30; Policy 9	Policy / operational	K-v21 result ceiling
ak-hpdst:21.0.0:te:l:policy-2	DOC-TE p.31; Policy 2	Policy / operational	Opposite-category and terminal-policy firewall
ak-hpdst:21.0.0:te:L-v21-DER-01	DOC-TE p.31; Theorem 3	Theorem / proved	L-v21-DER-01: contravariant diagram transport changes the indexing category

9.9 Problem Interface Appendices MS/NS (73)

9.10 Search/Platform Appendices U-Z (128)

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:sp:SP-v21-DEF-01	DOC-SP p.3; Definition 1	Definition / operational	SP-v21-DEF-01: Search / Platform ownership boundary
ak-hpdst:21.0.0:sp:SP-v21-DEF-02	DOC-SP p.3; Definition 2	Definition / operational	SP-v21-DEF-02: Independent status and result sorts
ak-hpdst:21.0.0:sp:SP-v21-DEF-03	DOC-SP p.3; Definition 3	Definition / operational	SP-v21-DEF-03: Comparison modes and detector stages
ak-hpdst:21.0.0:sp:SP-v21-DEF-04	DOC-SP p.3; Definition 4	Definition / operational	SP-v21-DEF-04: Canonical handoff projections
ak-hpdst:21.0.0:sp:SP-v21-DEF-01	DOC-SP p.4; Proposition 5	Proposition / proved	SP-v21-DEF-01: Active-lane isolation
ak-hpdst:21.0.0:sp:SP-v21-DEF-05	DOC-SP p.4; Definition 6	Definition / operational	SP-v21-DEF-05: Lossless platform packet
ak-hpdst:21.0.0:sp:SP-v21-DEF-06	DOC-SP p.4; Definition 7	Definition / operational	SP-v21-DEF-06: Packet adapter, semantic field map, and consumer contract
ak-hpdst:21.0.0:sp:SP-v21-DEF-02	DOC-SP p.4; Theorem 8	Theorem / proved	SP-v21-DEF-02: Consumer-projection determinacy
ak-hpdst:21.0.0:sp:SP-v21-DEF-03	DOC-SP p.4; Theorem 9	Theorem / proved	SP-v21-DEF-03: Typed composition of preservation certificates
ak-hpdst:21.0.0:sp:SP-v21-DEF-04	DOC-SP p.5; Proposition 10	Proposition / proved	SP-v21-DEF-04: Readiness is domain membership, not theorem evaluation
ak-hpdst:21.0.0:sp:SP-v21-DEF-05	DOC-SP p.5; Theorem 11	Theorem / proved	SP-v21-DEF-05: Infrastructure non-compensation
ak-hpdst:21.0.0:sp:SP-v21-DEF-07	DOC-SP p.5; Definition 12	Definition / operational	SP-v21-DEF-07: Pipeline-width vector
ak-hpdst:21.0.0:sp:SP-v21-DEF-06	DOC-SP p.5; Proposition 13	Proposition / proved	SP-v21-DEF-06: Width and theorem strength are independent
ak-hpdst:21.0.0:sp:U-v21-DEF-01	DOC-SP p.5; Definition 14	Definition / operational	U-v21-DEF-01: Capability-qualified agent action
ak-hpdst:21.0.0:sp:U-v21-DEF-02	DOC-SP p.5; Definition 15	Definition / operational	U-v21-DEF-02: Dependency-separated review
ak-hpdst:21.0.0:sp:U-v21-DEF-01	DOC-SP p.5; Proposition 16	Proposition / proved	U-v21-DEF-01: Role product does not merge authority
ak-hpdst:21.0.0:sp:U-v21-DEF-03	DOC-SP p.6; Definition 17	Definition / operational	U-v21-DEF-03: Verifier projection
ak-hpdst:21.0.0:sp:U-v21-DEF-02	DOC-SP p.6; Theorem 18	Theorem / proved	U-v21-DEF-02: Idempotence of verifier projection
ak-hpdst:21.0.0:sp:U-v21-DEF-03	DOC-SP p.6; Theorem 19	Theorem / proved	U-v21-DEF-03: Re-entry substitution invariance
ak-hpdst:21.0.0:sp:U-v21-DEF-04	DOC-SP p.6; Proposition 20	Proposition / proved	U-v21-DEF-04: Consensus cannot create a missing proof path
ak-hpdst:21.0.0:sp:U-v21-DEF-05	DOC-SP p.6; Theorem 21	Theorem / proved	U-v21-DEF-05: Semantic transformation chain
ak-hpdst:21.0.0:sp:U-v21-DEF-06	DOC-SP p.6; Proposition 22	Proposition / proved	U-v21-DEF-06: Agent-side fan-out is verdict-local
ak-hpdst:21.0.0:sp:U-v21-DEF-07	DOC-SP p.6; Theorem 23	Theorem / proved	U-v21-DEF-07: Dependency-separated double checking
ak-hpdst:21.0.0:sp:V-v21-DEF-01	DOC-SP p.7; Definition 24	Definition / operational	V-v21-DEF-01: Typed search portfolio
ak-hpdst:21.0.0:sp:V-v21-DEF-02	DOC-SP p.7; Definition 25	Definition / operational	V-v21-DEF-02: Search capability set
ak-hpdst:21.0.0:sp:V-v21-DEF-01	DOC-SP p.7; Theorem 26	Theorem / proved	V-v21-DEF-01: Portfolio coverage union
ak-hpdst:21.0.0:sp:V-v21-DEF-03	DOC-SP p.7; Definition 27	Definition / operational	V-v21-DEF-03: Finite Pareto search order
ak-hpdst:21.0.0:sp:V-v21-DEF-02	DOC-SP p.7; Theorem 28	Theorem / proved	V-v21-DEF-02: Finite Pareto-frontier completeness
ak-hpdst:21.0.0:sp:V-v21-DEF-03	DOC-SP p.7; Theorem 29	Theorem / proved	V-v21-DEF-03: Exhaustive finite-search closure
ak-hpdst:21.0.0:sp:V-v21-DEF-04	DOC-SP p.7; Theorem 30	Theorem / proved	V-v21-DEF-04: Verified counterexample closure
ak-hpdst:21.0.0:sp:V-v21-DEF-05	DOC-SP p.7; Proposition 31	Proposition / proved	V-v21-DEF-05: Search-space refinement monotonicity
ak-hpdst:21.0.0:sp:V-v21-DEF-04	DOC-SP p.8; Definition 32	Definition / operational	V-v21-DEF-04: Source-adapter bus
ak-hpdst:21.0.0:sp:V-v21-DEF-06	DOC-SP p.8; Theorem 33	Theorem / proved	V-v21-DEF-06: Immutable adapter fan-out
ak-hpdst:21.0.0:sp:V-v21-DEF-07	DOC-SP p.8; Proposition 34	Proposition / proved	V-v21-DEF-07: Detector-search stage safety
ak-hpdst:21.0.0:sp:V-v21-DEF-08	DOC-SP p.8; Theorem 35	Theorem / proved	V-v21-DEF-08: Typed fallback program correctness
ak-hpdst:21.0.0:sp:W-v21-DEF-01	DOC-SP p.8; Definition 36	Definition / operational	W-v21-DEF-01: v21 Bridge-program record
ak-hpdst:21.0.0:sp:W-v21-DEF-02	DOC-SP p.8; Definition 37	Definition / operational	W-v21-DEF-02: Complete route and alternate portfolio
ak-hpdst:21.0.0:sp:W-v21-DEF-01	DOC-SP p.8; Theorem 38	Theorem / proved	W-v21-DEF-01: Typed serial composition
ak-hpdst:21.0.0:sp:W-v21-DEF-02	DOC-SP p.9; Theorem 39	Theorem / proved	W-v21-DEF-02: Alternative-route capability union
ak-hpdst:21.0.0:sp:W-v21-DEF-03	DOC-SP p.9; Theorem 40	Theorem / proved	W-v21-DEF-03: Independent target product
ak-hpdst:21.0.0:sp:W-v21-DEF-04	DOC-SP p.9; Theorem 41	Theorem / proved	W-v21-DEF-04: Route-local failure isolation
ak-hpdst:21.0.0:sp:W-v21-DEF-03	DOC-SP p.9; Definition 42	Definition / operational	W-v21-DEF-03: Target cut
ak-hpdst:21.0.0:sp:W-v21-DEF-05	DOC-SP p.9; Theorem 43	Theorem / proved	W-v21-DEF-05: Minimal target cuts need not be antichains
ak-hpdst:21.0.0:sp:W-v21-DEF-06	DOC-SP p.10; Theorem 45	Theorem / proved	W-v21-DEF-06: Component substitution invariance
ak-hpdst:21.0.0:sp:W-v21-DEF-07	DOC-SP p.10; Proposition 46	Proposition / proved	W-v21-DEF-07: No cross-route borrowing
ak-hpdst:21.0.0:sp:W-v21-DEF-08	DOC-SP p.10; Theorem 47	Theorem / proved	W-v21-DEF-08: Targetwise partial Return
ak-hpdst:21.0.0:sp:W-v21-DEF-09	DOC-SP p.10; Theorem 48	Theorem / proved	W-v21-DEF-09: Identity-node normalization
ak-hpdst:21.0.0:sp:W-v21-DEF-10	DOC-SP p.10; Proposition 49	Proposition / proved	W-v21-DEF-10: Shared-component reuse does not multiply evidence
ak-hpdst:21.0.0:sp:X-v21-DEF-01	DOC-SP p.10; Definition 50	Definition / operational	X-v21-DEF-01: Typed proof DAG
ak-hpdst:21.0.0:sp:X-v21-DEF-02	DOC-SP p.10; Definition 51	Definition / operational	X-v21-DEF-02: Target ancestor slice
ak-hpdst:21.0.0:sp:X-v21-DEF-01	DOC-SP p.10; Theorem 52	Theorem / proved	X-v21-DEF-01: Target-slice sufficiency
ak-hpdst:21.0.0:sp:X-v21-DEF-02	DOC-SP p.11; Theorem 53	Theorem / proved	X-v21-DEF-02: Finite topological execution order
ak-hpdst:21.0.0:sp:X-v21-DEF-03	DOC-SP p.11; Definition 54	Definition / operational	X-v21-DEF-03: Level schedule
ak-hpdst:21.0.0:sp:X-v21-DEF-03	DOC-SP p.11; Theorem 55	Theorem / proved	X-v21-DEF-03: Antichain parallelization
ak-hpdst:21.0.0:sp:X-v21-DEF-04	DOC-SP p.11; Definition 56	Definition / operational	X-v21-DEF-04: Stale closure
ak-hpdst:21.0.0:sp:X-v21-DEF-04	DOC-SP p.11; Theorem 57	Theorem / proved	X-v21-DEF-04: Structural stale propagation
ak-hpdst:21.0.0:sp:X-v21-DEF-05	DOC-SP p.11; Definition 58	Definition / operational	X-v21-DEF-05: Minimal recomputation frontier
ak-hpdst:21.0.0:sp:X-v21-DEF-05	DOC-SP p.11; Theorem 59	Theorem / proved	X-v21-DEF-05: Frontier reconstruction
ak-hpdst:21.0.0:sp:X-v21-DEF-06	DOC-SP p.12; Theorem 60	Theorem / proved	X-v21-DEF-06: Target-cut characterization
ak-hpdst:21.0.0:sp:X-v21-DEF-07	DOC-SP p.12; Theorem 61	Theorem / proved	X-v21-DEF-07: Incremental target evaluation
ak-hpdst:21.0.0:sp:X-v21-DEF-08	DOC-SP p.12; Proposition 62	Proposition / proved	X-v21-DEF-08: Alias-node contraction
ak-hpdst:21.0.0:sp:X-v21-DEF-09	DOC-SP p.12; Theorem 63	Theorem / proved	X-v21-DEF-09: Product-target fan-out
ak-hpdst:21.0.0:sp:X-v21-DEF-10	DOC-SP p.12; Theorem 64	Theorem / proved	X-v21-DEF-10: Certificate closure criterion
ak-hpdst:21.0.0:sp:Y-v21-DEF-01	DOC-SP p.12; Definition 65	Definition / operational	Y-v21-DEF-01: Content-addressed proof object
ak-hpdst:21.0.0:sp:Y-v21-DEF-02	DOC-SP p.13; Definition 66	Definition / operational	Y-v21-DEF-02: Store dependency closure
ak-hpdst:21.0.0:sp:Y-v21-DEF-01	DOC-SP p.13; Theorem 67	Theorem / proved	Y-v21-DEF-01: Exact content deduplication
ak-hpdst:21.0.0:sp:Y-v21-DEF-02	DOC-SP p.13; Theorem 68	Theorem / proved	Y-v21-DEF-02: Live-root garbage-collection safety
ak-hpdst:21.0.0:sp:Y-v21-DEF-03	DOC-SP p.13; Definition 69	Definition / operational	Y-v21-DEF-03: Immutable version lineage
ak-hpdst:21.0.0:sp:Y-v21-DEF-03	DOC-SP p.13; Theorem 70	Theorem / proved	Y-v21-DEF-03: Acyclicity of frozen lineage
ak-hpdst:21.0.0:sp:Y-v21-DEF-04	DOC-SP p.13; Theorem 71	Theorem / proved	Y-v21-DEF-04: Mirror extensionality
ak-hpdst:21.0.0:sp:Y-v21-DEF-04	DOC-SP p.13; Definition 72	Definition / operational	Y-v21-DEF-04: Query contract
ak-hpdst:21.0.0:sp:Y-v21-DEF-05	DOC-SP p.13; Theorem 73	Theorem / proved	Y-v21-DEF-05: Sound positive query and conditional negative query
ak-hpdst:21.0.0:sp:Y-v21-DEF-05	DOC-SP p.13; Definition 74	Definition / operational	Y-v21-DEF-05: Schema migration
ak-hpdst:21.0.0:sp:Y-v21-DEF-06	DOC-SP p.14; Theorem 75	Theorem / proved	Y-v21-DEF-06: Migration composition
ak-hpdst:21.0.0:sp:Y-v21-DEF-07	DOC-SP p.14; Proposition 76	Proposition / proved	Y-v21-DEF-07: Store fan-out is non-promotional
ak-hpdst:21.0.0:sp:Y-v21-DEF-08	DOC-SP p.14; Theorem 77	Theorem / proved	Y-v21-DEF-08: Proof-bundle closure portability
ak-hpdst:21.0.0:sp:Y-v21-DEF-09	DOC-SP p.14; Proposition 78	Proposition / proved	Y-v21-DEF-09: Hash identity is not semantic identity
ak-hpdst:21.0.0:sp:Z-v21-DEF-01	DOC-SP p.14; Definition 79	Definition / operational	Z-v21-DEF-01: Hermetic run envelope
ak-hpdst:21.0.0:sp:Z-v21-DEF-01	DOC-SP p.14; Theorem 80	Theorem / proved	Z-v21-DEF-01: Deterministic hermetic replay
ak-hpdst:21.0.0:sp:Z-v21-DEF-02	DOC-SP p.14; Definition 81	Definition / operational	Z-v21-DEF-02: Incremental build graph
ak-hpdst:21.0.0:sp:Z-v21-DEF-02	DOC-SP p.15; Theorem 82	Theorem / proved	Z-v21-DEF-02: Structurally complete conservative rebuild set
ak-hpdst:21.0.0:sp:Z-v21-DEF-03	DOC-SP p.15; Theorem 83	Theorem / proved	Z-v21-DEF-03: Cache-key soundness
ak-hpdst:21.0.0:sp:Z-v21-DEF-03	DOC-SP p.15; Definition 84	Definition / operational	Z-v21-DEF-03: Seeded and distributional replay
ak-hpdst:21.0.0:sp:Z-v21-DEF-04	DOC-SP p.15; Theorem 85	Theorem / proved	Z-v21-DEF-04: Seeded pseudo-random replay
ak-hpdst:21.0.0:sp:Z-v21-DEF-05	DOC-SP p.15; Proposition 86	Proposition / proved	Z-v21-DEF-05: Distributional replay is not byte replay
ak-hpdst:21.0.0:sp:Z-v21-DEF-06	DOC-SP p.15; Definition 87	Definition / operational	Z-v21-DEF-06: Reproducibility vector
ak-hpdst:21.0.0:sp:Z-v21-DEF-06	DOC-SP p.15; Theorem 88	Theorem / proved	Z-v21-DEF-06: Product order of reproducibility guarantees
ak-hpdst:21.0.0:sp:Z-v21-DEF-05	DOC-SP p.15; Definition 89	Definition / operational	Z-v21-DEF-05: Semantic replay morphism
ak-hpdst:21.0.0:sp:Z-v21-DEF-07	DOC-SP p.16; Theorem 90	Theorem / proved	Z-v21-DEF-07: Composition of semantic replay
ak-hpdst:21.0.0:sp:Z-v21-DEF-06	DOC-SP p.16; Definition 91	Definition / operational	Z-v21-DEF-06: Extensional backend certificate
ak-hpdst:21.0.0:sp:Z-v21-DEF-08	DOC-SP p.16; Theorem 92	Theorem / proved	Z-v21-DEF-08: Backend substitution
ak-hpdst:21.0.0:sp:Z-v21-DEF-09	DOC-SP p.16; Theorem 93	Theorem / proved	Z-v21-DEF-09: Parallel execution by proof-DAG levels
ak-hpdst:21.0.0:sp:Z-v21-DEF-10	DOC-SP p.16; Theorem 94	Theorem / proved	Z-v21-DEF-10: Supply-chain closure criterion
ak-hpdst:21.0.0:sp:SP-v21-DEF-08	DOC-SP p.16; Definition 95	Definition / operational	SP-v21-DEF-08: Consumer-contract refinement
ak-hpdst:21.0.0:sp:V-v21-DEF-09	DOC-SP p.17; Proposition 100	Proposition / proved	V-v21-DEF-09: Search/validation separation
ak-hpdst:21.0.0:sp:V-v21-DEF-10	DOC-SP p.17; Theorem 101	Theorem / proved	V-v21-DEF-10: Consumer-relative Pareto pruning
ak-hpdst:21.0.0:sp:SP-v21-DEF-07	DOC-SP p.17; Theorem 96	Theorem / proved	SP-v21-DEF-07: Safe contract weakening
ak-hpdst:21.0.0:sp:SP-v21-DEF-09	DOC-SP p.17; Definition 97	Definition / operational	SP-v21-DEF-09: Jointly compatible packet family
ak-hpdst:21.0.0:sp:SP-v21-DEF-08	DOC-SP p.17; Theorem 98	Theorem / proved	SP-v21-DEF-08: Finite compatible packet amalgamation
ak-hpdst:21.0.0:sp:V-v21-DEF-05	DOC-SP p.17; Definition 99	Definition / operational	V-v21-DEF-05: Adaptive-search validation firewall
ak-hpdst:21.0.0:sp:W-v21-DEF-04	DOC-SP p.18; Definition 102	Definition / operational	W-v21-DEF-04: Internal route resilience
ak-hpdst:21.0.0:sp:W-v21-DEF-11	DOC-SP p.18; Theorem 103	Theorem / proved	W-v21-DEF-11: Disjoint-route failure tolerance
ak-hpdst:21.0.0:sp:W-v21-DEF-12	DOC-SP p.18; Proposition 104	Proposition / proved	W-v21-DEF-12: Shared critical node defeats route-count resilience
ak-hpdst:21.0.0:sp:W-v21-DEF-13	DOC-SP p.18; Theorem 105	Theorem / proved	W-v21-DEF-13: Fallback availability after failures
ak-hpdst:21.0.0:sp:X-v21-DEF-06	DOC-SP p.18; Definition 106	Definition / operational	X-v21-DEF-06: Target-complete pure proof DAG
ak-hpdst:21.0.0:sp:X-v21-DEF-11	DOC-SP p.18; Theorem 107	Theorem / proved	X-v21-DEF-11: Least graph-only state set
ak-hpdst:21.0.0:sp:X-v21-DEF-12	DOC-SP p.18; Theorem 108	Theorem / proved	X-v21-DEF-12: Multi-target incremental union
ak-hpdst:21.0.0:sp:X-v21-DEF-07	DOC-SP p.19; Definition 109	Definition / operational	X-v21-DEF-07: Certified DAG congruence
ak-hpdst:21.0.0:sp:X-v21-DEF-13	DOC-SP p.19; Theorem 110	Theorem / proved	X-v21-DEF-13: Congruence-quotient normalization
ak-hpdst:21.0.0:sp:Y-v21-DEF-06	DOC-SP p.19; Definition 111	Definition / operational	Y-v21-DEF-06: Merkle closure manifest
ak-hpdst:21.0.0:sp:Y-v21-DEF-10	DOC-SP p.19; Theorem 112	Theorem / proved	Y-v21-DEF-10: Root commitment to a finite proof closure
ak-hpdst:21.0.0:sp:Y-v21-DEF-07	DOC-SP p.19; Definition 113	Definition / operational	Y-v21-DEF-07: Snapshot epoch and deletion barrier
ak-hpdst:21.0.0:sp:Y-v21-DEF-11	DOC-SP p.19; Theorem 114	Theorem / proved	Y-v21-DEF-11: Snapshot-safe garbage collection
ak-hpdst:21.0.0:sp:Y-v21-DEF-12	DOC-SP p.19; Theorem 115	Theorem / proved	Y-v21-DEF-12: Lossless schema-migration round trip
ak-hpdst:21.0.0:sp:Y-v21-DEF-13	DOC-SP p.20; Theorem 116	Theorem / proved	Y-v21-DEF-13: Certified migration-diamond confluence
ak-hpdst:21.0.0:sp:Z-v21-DEF-07	DOC-SP p.20; Definition 117	Definition / operational	Z-v21-DEF-07: Captured external input
ak-hpdst:21.0.0:sp:Z-v21-DEF-11	DOC-SP p.20; Theorem 118	Theorem / proved	Z-v21-DEF-11: External-response capture
ak-hpdst:21.0.0:sp:Z-v21-DEF-12	DOC-SP p.20; Theorem 119	Theorem / proved	Z-v21-DEF-12: Worst-case minimality of structural rebuild
ak-hpdst:21.0.0:sp:Z-v21-DEF-13	DOC-SP p.20; Theorem 120	Theorem / proved	Z-v21-DEF-13: Schedule-independent parallel determinism
ak-hpdst:21.0.0:sp:Z-v21-DEF-08	DOC-SP p.20; Definition 121	Definition / operational	Z-v21-DEF-08: Cache-admission certificate
ak-hpdst:21.0.0:sp:Z-v21-DEF-14	DOC-SP p.20; Theorem 122	Theorem / proved	Z-v21-DEF-14: Verified cache admission
ak-hpdst:21.0.0:sp:C-v21-DEF-04	DOC-SP p.21; Theorem 123	Theorem / proved	C-v21-DEF-04: Reproducibly auditable target closure
ak-hpdst:21.0.0:sp:C-v21-DEF-05	DOC-SP p.21; Proposition 124	Proposition / proved	C-v21-DEF-05: Horizontal scaling is non-promotional
ak-hpdst:21.0.0:sp:C-v21-DEF-01	DOC-SP p.21; Definition 125	Definition / operational	C-v21-DEF-01: Application lane
ak-hpdst:21.0.0:sp:C-v21-DEF-01	DOC-SP p.21; Theorem 126	Theorem / proved	C-v21-DEF-01: End-to-end pipeline soundness

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:sp:C-v21-DER-02	DOC-SP p.22; Theorem 127	Theorem / proved	C-v21-DER-02: Safe widening by adapters and fallbacks
ak-hpdst:21.0.0:sp:C-v21-DER-03	DOC-SP p.22; Theorem 128	Theorem / proved	C-v21-DER-03: Part V v21 closure principle
ak-hpdst:21.0.0:sp:R-v21-DER-01	DOC-SP p.22; Proposition 130	Proposition / proved	R-v21-DER-01: Conservative predecessor embedding

9.11 Companion (78)

UID	Source locator	Kind / status	Title
ak-hpdst:21.0.0:comp:C-v21-DEF-01	DOC-COMP p.2; Definition 1.1	Definition / operational	C-v21-DEF-01: Companion projection
ak-hpdst:21.0.0:comp:C-v21-DEF-01	DOC-COMP p.2; Proposition 1.2	Proposition / proved	C-v21-DEF-01: Projection is non-promotional
ak-hpdst:21.0.0:comp:C-v21-DEF-02	DOC-COMP p.2; Definition 1.3	Definition / operational	C-v21-DEF-02: Independent status and result sorts
ak-hpdst:21.0.0:comp:C-v21-DEF-02	DOC-COMP p.3; Proposition 1.4	Proposition / proved	C-v21-DEF-02: Status-axis noninterference
ak-hpdst:21.0.0:comp:C-v21-DEF-03	DOC-COMP p.3; Definition 1.5	Definition / operational	C-v21-DEF-03: Lossless Companion packet
ak-hpdst:21.0.0:comp:C-v21-DEF-04	DOC-COMP p.3; Definition 1.6	Definition / operational	C-v21-DEF-04: Consumer projection
ak-hpdst:21.0.0:comp:C-v21-DEF-04	DOC-COMP p.3; Proposition 1.7	Proposition / proved	C-v21-DEF-04: Consumer-projection invariance
ak-hpdst:21.0.0:comp:C-v21-DEF-05	DOC-COMP p.3; Definition 1.8	Definition / operational	C-v21-DEF-05: Semantic field map
ak-hpdst:21.0.0:comp:C-v21-DEF-05	DOC-COMP p.3; Proposition 1.9	Proposition / proved	C-v21-DEF-05: Preservation certificates compose
ak-hpdst:21.0.0:comp:C-v21-DEF-06	DOC-COMP p.4; Definition 1.10	Definition / operational	C-v21-DEF-06: Compatible packet amalgamation
ak-hpdst:21.0.0:comp:C-v21-DEF-05	DOC-COMP p.4; Proposition 1.11	Proposition / proved	C-v21-DEF-05: Algebra of compatible amalgamation
ak-hpdst:21.0.0:comp:C-v21-DEF-07	DOC-COMP p.4; Definition 1.12	Definition / operational	C-v21-DEF-07: Target-relative evidence support profile
ak-hpdst:21.0.0:comp:C-v21-DEF-10	DOC-COMP p.4; Proposition 1.13	Proposition / proved	C-v21-DEF-12: Alternate-route support union
ak-hpdst:21.0.0:comp:C-v21-DEF-06	DOC-COMP p.4; Proposition 1.14	Proposition / proved	C-v21-DEF-06: Infrastructure noncompensation
ak-hpdst:21.0.0:comp:C-v21-DEF-08	DOC-COMP p.4; Definition 1.15	Definition / operational	C-v21-DEF-08: Companion route portfolio
ak-hpdst:21.0.0:comp:C-v21-DEF-09	DOC-COMP p.4; Definition 1.16	Definition / operational	C-v21-DEF-09: Target slice and target cut
ak-hpdst:21.0.0:comp:C-v21-DEF-07	DOC-COMP p.5; Proposition 1.17	Proposition / proved	C-v21-DEF-07: Target-slice sufficiency
ak-hpdst:21.0.0:comp:C-v21-DEF-08	DOC-COMP p.5; Proposition 1.18	Proposition / proved	C-v21-DEF-08: Complete fallback and failure isolation
ak-hpdst:21.0.0:comp:C-v21-DEF-08	DOC-COMP p.5; Definition 1.19	Definition / operational	C-v21-DEF-10: Adaptive validation firewall
ak-hpdst:21.0.0:comp:C-v21-DEF-09	DOC-COMP p.5; Proposition 1.20	Proposition / proved	C-v21-DEF-09: Validation reuse boundary
ak-hpdst:21.0.0:comp:C-v21-DEF-13	DOC-COMP p.5; Definition 1.21	Definition / operational	C-v21-DEF-13: Legacy-schema current-reading map
ak-hpdst:21.0.0:comp:C-v21-DEF-09	DOC-COMP p.5; Proposition 1.22	Proposition / proved	C-v21-DEF-13: Legacy favorable tokens are nonpromotional
ak-hpdst:21.0.0:comp:C1-v21-DEF-01	DOC-COMP p.6; Definition 2.1	Definition / operational	C1-v21-DEF-01: Problem-Interface calibration packet
ak-hpdst:21.0.0:comp:C1-v21-DEF-02	DOC-COMP p.6; Definition 2.2	Definition / operational	C1-v21-DEF-02: Source-adaptor registry
ak-hpdst:21.0.0:comp:C1-v21-DEF-01	DOC-COMP p.6; Proposition 2.3	Proposition / proved	C1-v21-DEF-01: Conservative adaptor extension
ak-hpdst:21.0.0:comp:C1-v21-DEF-03	DOC-COMP p.6; Definition 2.4	Definition / operational	C1-v21-DEF-03: MS arithmetic lane record
ak-hpdst:21.0.0:comp:C1-v21-DEF-04	DOC-COMP p.6; Definition 2.5	Definition / operational	C1-v21-DEF-04: NS analytic lane record
ak-hpdst:21.0.0:comp:C1-v21-DEF-02	DOC-COMP p.6; Proposition 2.6	Proposition / proved	C1-v21-DEF-02: Source-specific lane noninterchangeability
ak-hpdst:21.0.0:comp:C1-v21-DEF-05	DOC-COMP p.6; Definition 2.7	Definition / operational	C1-v21-DEF-05: Calibration target portfolio
ak-hpdst:21.0.0:comp:C1-v21-DEF-04	DOC-COMP p.7; Proposition 2.10	Proposition / proved	C1-v21-DEF-04: Equivalent arithmetic calibrations
ak-hpdst:21.0.0:comp:C1-v21-DEF-07	DOC-COMP p.7; Definition 2.11	Definition / operational	C1-v21-DEF-07: Recurrence calibration packet
ak-hpdst:21.0.0:comp:C1-v21-DEF-05	DOC-COMP p.7; Proposition 2.12	Proposition / proved	C1-v21-DEF-05: Certified finite tail replay
ak-hpdst:21.0.0:comp:C1-v21-DEF-08	DOC-COMP p.7; Definition 2.13	Definition / operational	C1-v21-DEF-08: Branch calibration packet
ak-hpdst:21.0.0:comp:C1-v21-DEF-06	DOC-COMP p.7; Proposition 2.14	Proposition / proved	C1-v21-DEF-06: Branch portfolio completeness
ak-hpdst:21.0.0:comp:C1-v21-DEF-03	DOC-COMP p.7; Proposition 2.8	Proposition / proved	C1-v21-DEF-03: Immutable calibration fan-out
ak-hpdst:21.0.0:comp:C1-v21-DEF-06	DOC-COMP p.7; Definition 2.9	Definition / operational	C1-v21-DEF-06: Finite-length module calibration packet
ak-hpdst:21.0.0:comp:C1-v21-DEF-09	DOC-COMP p.8; Definition 2.15	Definition / operational	C1-v21-DEF-09: Mixed-norm certificate packet
ak-hpdst:21.0.0:comp:C1-v21-DEF-10	DOC-COMP p.8; Definition 2.16	Definition / operational	C1-v21-DEF-10: Local cylinder portfolio
ak-hpdst:21.0.0:comp:C1-v21-DEF-07	DOC-COMP p.8; Proposition 2.17	Proposition / proved	C1-v21-DEF-07: Local-to-global certificate assembly
ak-hpdst:21.0.0:comp:C1-v21-DEF-11	DOC-COMP p.8; Definition 2.18	Definition / operational	C1-v21-DEF-11: Approximation-limit packet
ak-hpdst:21.0.0:comp:C1-v21-DEF-08	DOC-COMP p.8; Proposition 2.19	Proposition / proved	C1-v21-DEF-08: Limit transport is topology-scoped
ak-hpdst:21.0.0:comp:C1-v21-DEF-12	DOC-COMP p.8; Definition 2.20	Definition / operational	C1-v21-DEF-12: Heterogeneous criterion stitching record
ak-hpdst:21.0.0:comp:C1-v21-DEF-09	DOC-COMP p.8; Proposition 2.21	Proposition / proved	C1-v21-DEF-09: Criterion stitching ceiling
ak-hpdst:21.0.0:comp:C1-v21-DEF-13	DOC-COMP p.8; Definition 2.22	Definition / operational	C1-v21-DEF-13: Calibration-family coverage certificate
ak-hpdst:21.0.0:comp:C1-v21-DEF-10	DOC-COMP p.8; Proposition 2.23	Proposition / proved	C1-v21-DEF-10: No implicit continuum coverage
ak-hpdst:21.0.0:comp:C2-v21-DEF-01	DOC-COMP p.9; Definition 3.1	Definition / operational	C2-v21-DEF-01: Hermetic execution envelope
ak-hpdst:21.0.0:comp:C2-v21-DEF-02	DOC-COMP p.9; Definition 3.2	Definition / operational	C2-v21-DEF-02: Execution packet
ak-hpdst:21.0.0:comp:C2-v21-DEF-01	DOC-COMP p.10; Proposition 3.3	Proposition / proved	C2-v21-DEF-01: Hermetic determinism
ak-hpdst:21.0.0:comp:C2-v21-DEF-03	DOC-COMP p.10; Definition 3.4	Definition / operational	C2-v21-DEF-03: Merkle proof closure manifest
ak-hpdst:21.0.0:comp:C2-v21-DEF-02	DOC-COMP p.10; Proposition 3.5	Proposition / proved	C2-v21-DEF-02: Closure commitment
ak-hpdst:21.0.0:comp:C2-v21-DEF-04	DOC-COMP p.10; Definition 3.6	Definition / operational	C2-v21-DEF-04: Cache-admission certificate
ak-hpdst:21.0.0:comp:C2-v21-DEF-03	DOC-COMP p.10; Proposition 3.7	Proposition / proved	C2-v21-DEF-03: Verified cache substitution
ak-hpdst:21.0.0:comp:C2-v21-DEF-05	DOC-COMP p.10; Definition 3.8	Definition / operational	C2-v21-DEF-05: Structural state set and rebuild plan
ak-hpdst:21.0.0:comp:C2-v21-DEF-04	DOC-COMP p.10; Proposition 3.9	Proposition / proved	C2-v21-DEF-04: Multi-target incremental sufficiency
ak-hpdst:21.0.0:comp:C2-v21-DEF-06	DOC-COMP p.11; Definition 3.10	Definition / operational	C2-v21-DEF-06: Deterministic parallel schedule
ak-hpdst:21.0.0:comp:C2-v21-DEF-05	DOC-COMP p.11; Proposition 3.11	Proposition / proved	C2-v21-DEF-05: Schedule independence
ak-hpdst:21.0.0:comp:C2-v21-DEF-07	DOC-COMP p.11; Definition 3.12	Definition / operational	C2-v21-DEF-07: Typed schema migration
ak-hpdst:21.0.0:comp:C2-v21-DEF-06	DOC-COMP p.11; Proposition 3.13	Proposition / proved	C2-v21-DEF-06: Lossless round-trip equivalence
ak-hpdst:21.0.0:comp:C2-v21-DEF-08	DOC-COMP p.11; Definition 3.14	Definition / operational	C2-v21-DEF-08: Migration confluence certificate
ak-hpdst:21.0.0:comp:C2-v21-DEF-07	DOC-COMP p.11; Proposition 3.15	Proposition / proved	C2-v21-DEF-07: Confluent migration invariance
ak-hpdst:21.0.0:comp:C2-v21-DEF-09	DOC-COMP p.11; Definition 3.16	Definition / operational	C2-v21-DEF-09: Semantic replay morphism
ak-hpdst:21.0.0:comp:C2-v21-DEF-08	DOC-COMP p.11; Proposition 3.17	Proposition / proved	C2-v21-DEF-08: Semantic replay composes
ak-hpdst:21.0.0:comp:C2-v21-DEF-10	DOC-COMP p.11; Definition 3.18	Definition / operational	C2-v21-DEF-10: Snapshot-safe garbage collection
ak-hpdst:21.0.0:comp:C2-v21-DEF-09	DOC-COMP p.11; Proposition 3.19	Proposition / proved	C2-v21-DEF-09: Live-proof preservation
ak-hpdst:21.0.0:comp:C2-v21-DEF-11	DOC-COMP p.12; Definition 3.20	Definition / operational	C2-v21-DEF-11: Reproducibility vector
ak-hpdst:21.0.0:comp:C2-v21-DEF-14	DOC-COMP p.12; Definition 3.21	Definition / operational	C2-v21-DEF-14: External-input identity and freshness split
ak-hpdst:21.0.0:comp:C2-v21-DEF-12	DOC-COMP p.12; Proposition 3.22	Proposition / proved	C2-v21-DEF-12: Freshness is not replay identity
ak-hpdst:21.0.0:comp:C2-v21-DEF-15	DOC-COMP p.12; Definition 3.23	Definition / operational	C2-v21-DEF-15: Target-required reproducibility mask
ak-hpdst:21.0.0:comp:C2-v21-DEF-13	DOC-COMP p.12; Proposition 3.24	Proposition / proved	C2-v21-DEF-13: Componentwise reproducibility acceptance
ak-hpdst:21.0.0:comp:C2-v21-DEF-12	DOC-COMP p.12; Definition 3.25	Definition / operational	C2-v21-DEF-12: Reproducibly auditable target closure
ak-hpdst:21.0.0:comp:C2-v21-DEF-10	DOC-COMP p.12; Proposition 3.26	Proposition / proved	C2-v21-DEF-10: Independent reconstruction
ak-hpdst:21.0.0:comp:C2-v21-DEF-13	DOC-COMP p.13; Definition 3.27	Definition / operational	C2-v21-DEF-13: Release handoff packet
ak-hpdst:21.0.0:comp:C2-v21-DEF-11	DOC-COMP p.13; Proposition 3.28	Proposition / proved	C2-v21-DEF-11: Release handoff is non-promotional
ak-hpdst:21.0.0:comp:C-v21-DEF-11	DOC-COMP p.14; Definition 4.1	Definition / operational	C-v21-DEF-11: Companion closure-state vector
ak-hpdst:21.0.0:comp:C-v21-DEF-14	DOC-COMP p.14; Proposition 4.2	Proposition / proved	C-v21-DEF-14: Release-axis noncollapse
ak-hpdst:21.0.0:comp:C-v21-DEF-12	DOC-COMP p.14; Definition 4.3	Definition / operational	C-v21-DEF-12: Companion migration record
ak-hpdst:21.0.0:comp:C-v21-DEF-10	DOC-COMP p.14; Proposition 4.4	Proposition / proved	C-v21-DEF-10: Conservative v21 migration
ak-hpdst:21.0.0:comp:C-v21-DEF-11	DOC-COMP p.14; Proposition 4.5	Proposition / proved	C-v21-DEF-11: Companion closure theorem

10 Preserved v20 lower-baseline epoch

The v20 compact register contributes its complete registered epoch without renumbering. The machine files preserve every historical UID, source artifact, locator, type/status/use digest and title. The semantic migration status is `inherited`; exact successor equality is not inferred. Current ownership is resolved by source family and the v21 canonical-owner rules.

Historical preservation result

The preserved v20 epoch contains 1950 registered identities. No historical UID is reused for a v21 strengthening or repair. The nine nonexistent CM-v20 tokens quoted by frozen TB-v20 are not registered historical identities; they remain quarantined repair ancestry.

11 Source-binding and Known-Theorem Recovery closure

11.1 Iwasawa track

The authoritative local statement-binding artifact has SHA-256 `a5f0a65a73c802492167676aba024b01651cd855bbdc00bc3d3c48d46cd6f0d7`. The legacy registry records a stale different hash and is retained only as repair ancestry. `1W-KTR-GROWTH` is registered as exact recovery of the Iwasawa 1959 Theorem 4

characteristic-growth clause relative to the permitted precursor package. No Iwasawa main conjecture, class-number specialization, or unrestricted module-structure theorem is registered.

11.2 Serrin track

The authoritative source-statement artifact has SHA-256 608216fda60bd3fdc055243c0169bdb31b53c337bb99d9b7c9cb43a215996423. NS-KTR-SERRIN-L6 is registered as the exact one-way fixed-exponent strict-subcritical criterion: the frozen $L_t^6 L_x^6$ hypothesis implies the frozen interior-regularity conclusion. No new analytic proof or unconditional three-dimensional regularity is registered.

12 Release-state vector and final nonconfusion

Closure-state vector

claim_register=registered_final_v21; source_identity=closed; uid_uniqueness=pass; canonical_ownership=pass; migration_epoch=closed; source_binding=pass_with_r
ecorded_IW_hash_repair; package_integrity=pass;
package_release=not_asserted; publication=not_asserted.

The final Claim Register governs identity and admissible use. It does not restate or strengthen the source mathematics. Every ambiguity is resolved by the weaker source-supported, target-typed reading. Package publication, repository publication, DOI/Zenodo deposition and public release are separate acts.